Supplementary Material

Soil-Water Modelling and Decision Support through a Graphical User Interface for the Daisy Model

Derivations of the equations, the mechanisms behind the results, the data, and the origin of every table and figure

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S1. What is new in this article

Table S1: What this article adds, item by item: what was established before, what is new here, and the evidence in this article.

#	Established before	New in this article	Evidence here
1	Daisy is run from text set-up files (Hansen et al., 2012); field, design, satellite and model tools are separate programs.	A graphical workbench that checks every design against a catalogue of what Daisy can vary, keeps an exact run budget, keeps every run and explains every failure by its root cause.	End-to-end Daisy 7.1.14 runs of the library scenarios, deficit irrigation and the soil-hydraulic factors; a defect that stopped every run on a sample horizon found and removed.
2	Decision tools show numbers without their origin.	Two auditable rules: every number leads to an equation with its origin, and a Daisy layer shows a Daisy result only from a Daisy run.	All layers opened without a page error; Daisy layers show nothing before a run.
3	Shared model functions are trusted once published.	A check of a shared function against stored Daisy results that found a defect.	The denitrification function corrected after comparison with 792 Daisy results.
4	Sentinel-2 indices, SoilGrids and elevation models are used separately (Drusch et al., 2012; Poggio et al., 2021).	One offline chain from a field outlined on a satellite image to indices, water sources, terrain, soil loss, retention maps, a design and a schedule.	Run on a real 5.85 ha field near Zaria, Nigeria.
5	Sensitivity to hydraulic parameters is studied with dedicated codes (Šimůnek et al., 2016).	Factors that scale the van Genuchten–Mualem parameters of every horizon of a Daisy run inside the workbench.	One season's matrix percolation rose from 177 to 222 mm and runoff vanished with the scaled parameters.
6	–	A manual made with the program and opened at the page in use (Halilu, 2026).	Every control, view and equation documented; page links from each screen.

S2. How this supplement is organised

Each of the following sections gathers everything about one result of the article under one heading: the problem and the author's contribution with its pinpointed equation, the worked numbers that can be checked on a calculator or in the workbook, graphs that mark the exact points, the figures of the article with their numbered labels, the derivations, and why it happens. The terms, the constants, the tables, the path from the data to every result, the data themselves and the origin of every item follow at the end. The derivations of this supplement are of established methods (spectral indices, FAO-56, the Richards equation, budgets, D8 flow and wetness, RUSLE, Latin-hypercube sampling), given so that each computation of the software can be followed. They are not claimed as new; the contribution of the article is the integration, the two rules of origin, and the verification that found defects (Table S1).

S3. The rule of origin and the bridge to Daisy

How every number shown can be traced to a Daisy run.

S3.1 Contribution 1: the rule of origin

The problem

Decision tools that couple a process model with fast approximations can show a number that looks like a model result but was not produced by the model. How can a user be sure that a "Daisy" number came from Daisy?

How the author developed it

Every value passed by the bridge carries the log file and line it was read from; before drawing, a layer tests that origin and draws nothing if there is no run, stating which run is needed.

The result

How it is derived, line by line

1. Every value carries its origin:

$$v \mapsto \text{source}(v) = (\text{run}, \text{file}, \text{line})$$

2. Draw only values with a Daisy origin:

$$\text{show}(v) \iff \text{source}(v) \in \text{Daisy runs}$$

The result, with its parts marked

$$\underbrace{\text{show}(v)}_{(1)} \iff \underbrace{\text{source}(v) \in \{\text{Daisy runs of this project}\}}_{(2)}$$

$$\underbrace{\text{no run} \Rightarrow \text{empty layer with the reason}}_{(3)}$$

developed by the author

Pinpoints: what each circled mark shows and why

(1) A value of a Daisy layer is drawn only if the condition on the right holds.

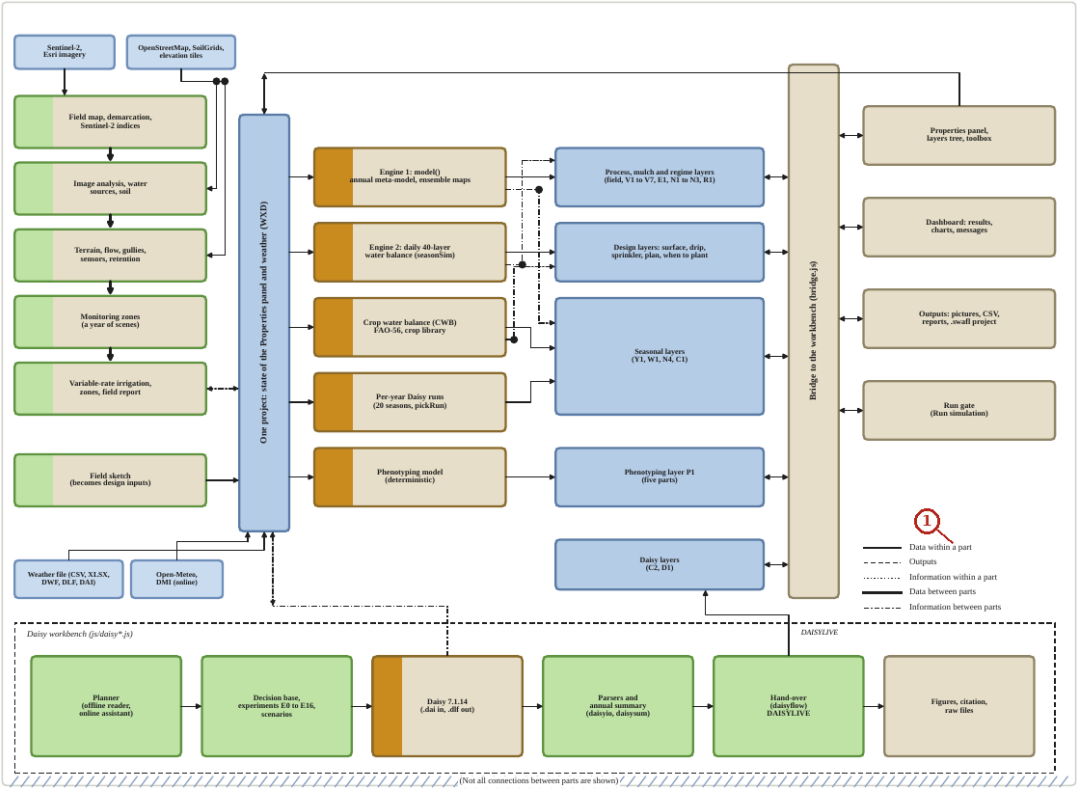
(2) The source of every value is the log file and line of a run; the bridge carries it to the layer.

(3) Before any run the layer says what it needs and shows no example numbers, which was checked for every layer. The rule makes the origin auditable instead of trusted.

S3.2 Worked numbers, part 8: the rule of origin

A decision layer may show a Daisy result only if a Daisy run made it. Before a run, the layer therefore says that it needs a run and shows nothing; after a run it shows the run's numbers with their file and line.

S3.3 Reading Figure 1 of the article



What the numbered labels point to.

(1) Panel (a): see the caption above.

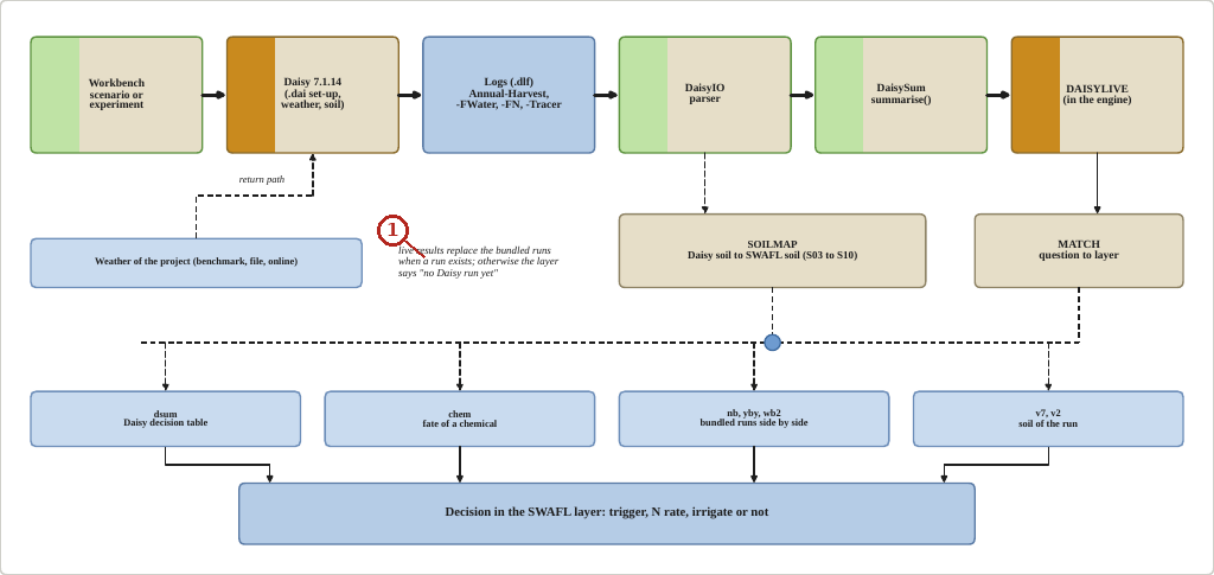
What it shows. Architecture of SWAFL Desktop 1.1.0 in the style of the process map of Daisy. Every arrow is a dependency found in the code: the field tools (green) call the engines and write to the project; the project (blue bar) carries the state of the Properties panel and the weather to the five computing parts (brown); these feed the layers (blue); the layers and the workbench shell talk through one bridge; the Daisy workbench (dashed frame) hands its runs to the Daisy layers.

How to read the labels, symbols and colours. Architecture: green field tools, blue project and layers, brown computing parts, dashed frame the Daisy workbench; every arrow a dependency in the code.

What affects it, and why it happens. Everything passes through the project and one bridge, which is what makes the origin of every number traceable.

How it is made. Figure drawn by the script from the input data. Script: `Manuscripts/reproducibility/p456/p456_figs_architecture.py`. Input: `Data/daisy_ensemble.csv.gz`.

S3.4 Reading Figure 12 of the article



What the numbered labels point to.

(1) Panel (a): see the caption above.

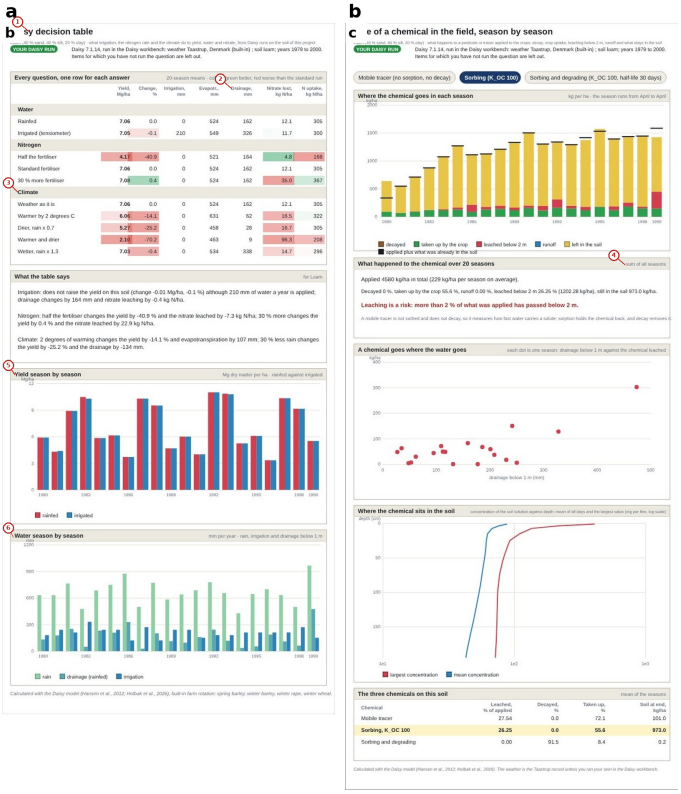
What it shows. The bridge from a Daisy run to a decision. Heavy arrows are the path of a run, dashed arrows are outputs and the dashed arrow at the left is the return path of the weather of the project to Daisy.

How to read the labels, symbols and colours. Heavy arrows: path of a run; dashed: outputs; left dashed arrow: the project's weather back to Daisy.

What affects it, and why it happens. One bridge carries the values with their origin, which enforces the rule of origin.

How it is made. Figure drawn by the script from the input data. Script: `Manuscripts/reproducibility/p456/p456_figs_schematics.py`. Input: `Data/daisy_ensemble.csv.gz`.

S3.5 Reading Figure 17 of the article



What the numbered labels point to.

1. "y decision table": a heading or control of this screen; the caption above says which layer it belongs to.
2. "Drainage": rain, from the weather of the project.
3. "Climate": a heading or control of this screen; the caption above says which layer it belongs to.
4. "sum of all seasons": a heading or control of this screen; the caption above says which layer it belongs to.
5. "Yield season by season": a heading or control of this screen; the caption above says which layer it belongs to.
6. "Water season by season": a heading or control of this screen; the caption above says which layer it belongs to.

What it shows. Results of the hand-over. (a) The Daisy decision table with the labels of the run (rainfed and irrigated, half and 30 % more fertiliser, warmer and drier climates). (b) The fate of a chemical for a mobile, a sorbing and a sorbing and degrading tracer.

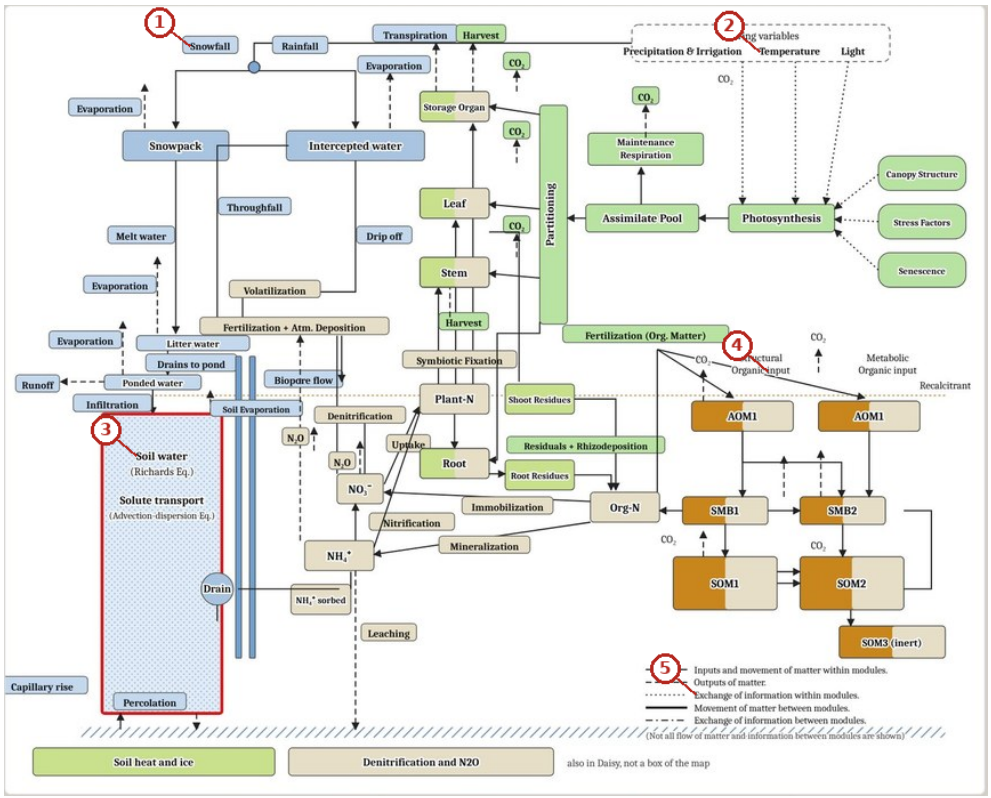
How to read the labels, symbols and colours. (a) The decision table of the hand-over with

run labels; (b) fate of three tracers.

What affects it, and why it happens. A mobile tracer leaches, a sorbing one stays, a sorbing and degrading one disappears in the soil.

How it is made. Screen capture of SWAFL Desktop 1.1.0 (not regenerated).

S3.6 Reading Figure 8 of the article



What the numbered labels point to.

1. "Snowfall": a heading or control of this screen; the caption above says which layer it belongs to.
2. "Temperature": a heading or control of this screen; the caption above says which layer it belongs to.
3. "Soil water": the soil of the project, which every layer uses.
4. "input": a heading or control of this screen; the caption above says which layer it belongs to.
5. "Exchange of information within modules": a heading or control of this screen; the caption above says which layer it belongs to.

What it shows. The process map of Daisy. Each process is a button that opens the figures it

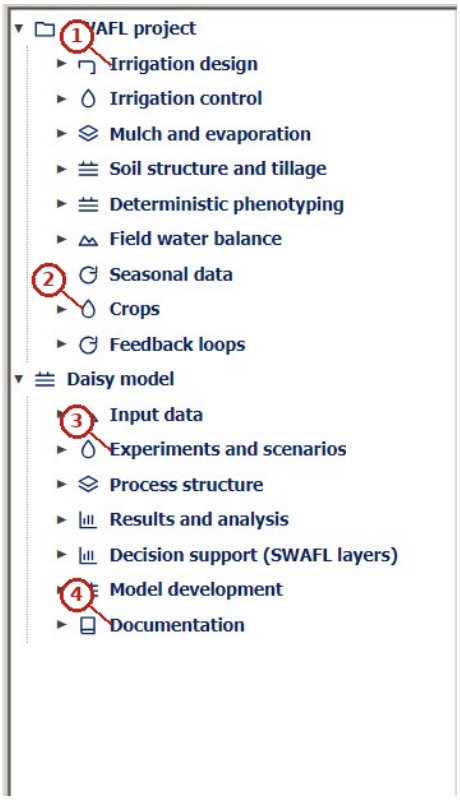
produces from a run.

How to read the labels, symbols and colours. The process map of Daisy; each box is a button.

What affects it, and why it happens. It shows which process produces which output, so a user finds the figure of a process.

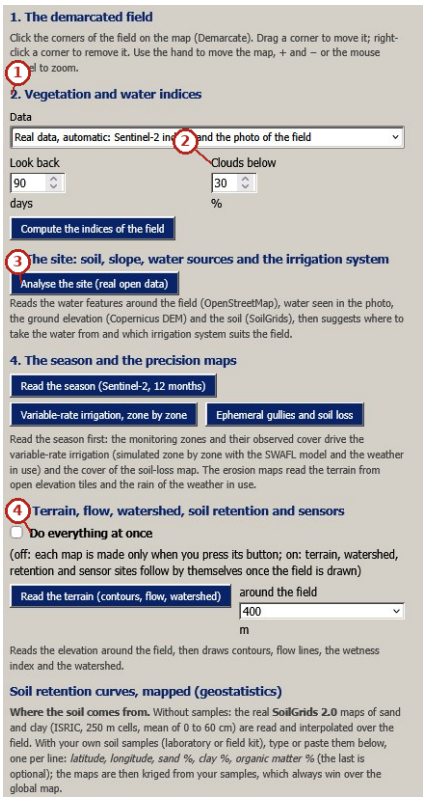
How it is made. Screen capture of SWAFL Desktop 1.1.0 (not regenerated).

S3.7 Reading Figure 18 of the article



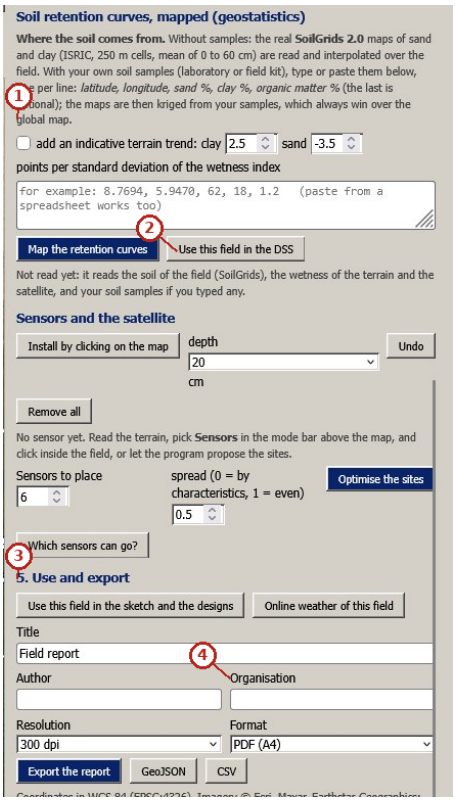
What the numbered labels point to (picture 1 of 3: The Layers panel: the nine groups of the SWAFL project and the seven of the Daisy model.).

1. "Irrigation design": the irrigation-design group of the Layers tree and its layers.
2. "Crops": the crop of the project and its stage.
3. "Experiments and scenarios": a heading or control of this screen; the caption above says which layer it belongs to.
4. "Documentation": a heading or control of this screen; the caption above says which layer it belongs to.



What the numbered labels point to (picture 2 of 3: The panel of the real field, sections 1 to 4b.).

1. "2. Vegetation and water indices": satellite indices of the vegetation computed from the Sentinel-2 bands.
2. "Clouds below": a heading or control of this screen; the caption above says which layer it belongs to.
3. "Analyse the site (real open data)": a heading or control of this screen; the caption above says which layer it belongs to.
4. "Do everything at once": a heading or control of this screen; the caption above says which layer it belongs to.



What the numbered labels point to (picture 3 of 3: Soil retention mapped, sensors, and export.).

1. "global map": a heading or control of this screen; the caption above says which layer it belongs to.
2. "Use this field in the DSS": a heading or control of this screen; the caption above says which layer it belongs to.
3. "5. Use and export": a heading or control of this screen; the caption above says which layer it belongs to.
4. "Organisation": a heading or control of this screen; the caption above says which layer it belongs to.

What it shows. The interface of release 1.1.0 as it appears to the user (screen captures). Panels: (a) The Layers panel: the nine groups of the SWAFL project and the seven of the Daisy model. (b) The panel of the real field, sections 1 to 4b. (c) Soil retention mapped, sensors, and export.

How to read the labels, symbols and colours. Screens of the Layers panel and the field panel.

What affects it, and why it happens. The layers are grouped by question, so the user finds

the layer that answers it. The groups organise the layers by question.

How it is made. Screen capture of SWAFL Desktop 1.1.0 (not regenerated).

S3.8 Why it happens

Why every number can be traced. Each value shown is computed by a function that carries the number of its equation and a label of origin (literature, fitted to data, engineered); the interface shows both, so the origin is a property of the code, not of the documentation.

Why a Daisy layer draws nothing before a Daisy run. The rule of origin forbids a layer to show a Daisy result from any other source; without a run there is no result, so the layer says so rather than showing an example.

Why the routes from Daisy to a decision differ in time by orders of magnitude. Running Daisy solves the Richards equation and a crop for every day of every run; the meta-model looks up and combines stored outcomes, which takes microseconds.

S4. Experiments within a run budget

Designs that test every level of every factor within the budget the user sets.

S4.1 Contribution 2: an exact run budget that still tests every level

The problem

A design of a few factors can need hundreds of Daisy runs of minutes each. How can the user set a budget and still test every level of every factor?

How the author developed it

The full factorial is counted; if it exceeds the budget B, a Latin hypercube of B runs is drawn in which the levels of each factor are used as evenly as possible.

The result

How it is derived, line by line

1. **Full factorial:**

   $\prod_i n_i$

2. **Never above the budget:**

   $N = \min(\prod_i n_i, B)$

3. **Latin hypercube, levels balanced:**

   each level used $\lfloor N/n_i \rfloor$ or $\lceil N/n_i \rceil$ times

The result, with its parts marked

$$\underbrace{N}_{①} = \min\Big(\underbrace{\textstyle\prod_i n_i}_{②},\ \underbrace{B}_{③}\Big)$$

$$\underbrace{\text{each level of factor } i \text{ used } \lfloor N/n_i \rfloor \text{ or } \lceil N/n_i \rceil \text{ times}}_{④}$$

developed by the author

Pinpoints: what each circled mark shows and why

① The number of runs actually made, never above the budget.

② The full factorial (for 4, 10 and 5 levels: 200 runs).

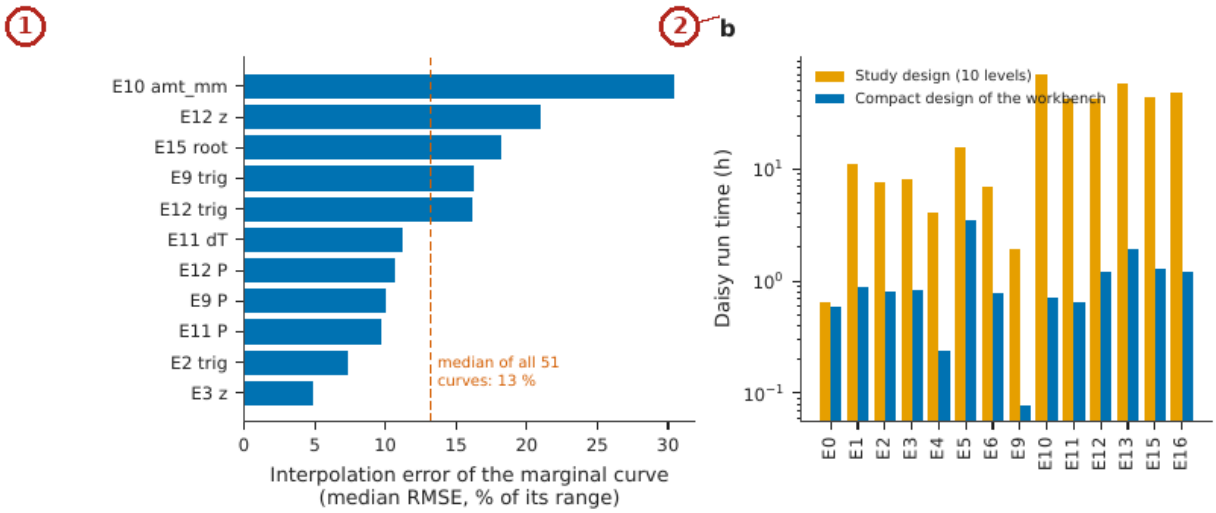
③ The budget set by the user (for example 60).

④ The Latin-hypercube choice spreads the runs so that every level appears almost equally often (each of the 10 levels 6 times), so main effects remain estimable.

S4.2 Worked numbers, part 6: keeping an experiment within its budget

Three factors with 4, 10 and 5 levels make 200 combinations; with a budget of 60 runs a Latin-hypercube choice makes 60 runs in which each level of the second factor appears 6 times, so every level is still tested.

S4.3 Reading Figure 13 of the article



What the numbered labels point to.

1. Panel (a): Error of the marginal curves rebuilt from the levels of the compact designs (median over five responses, per experiment and factor). *Why:* Fewer levels miss the curvature between them, so curves rebuilt from compact designs err more where the response is curved.
2. Panel (b): Daisy run time of the study designs and of the compact designs, by experiment (logarithmic scale). *Why:* Compact designs need orders of magnitude fewer runs, so far less Daisy time.

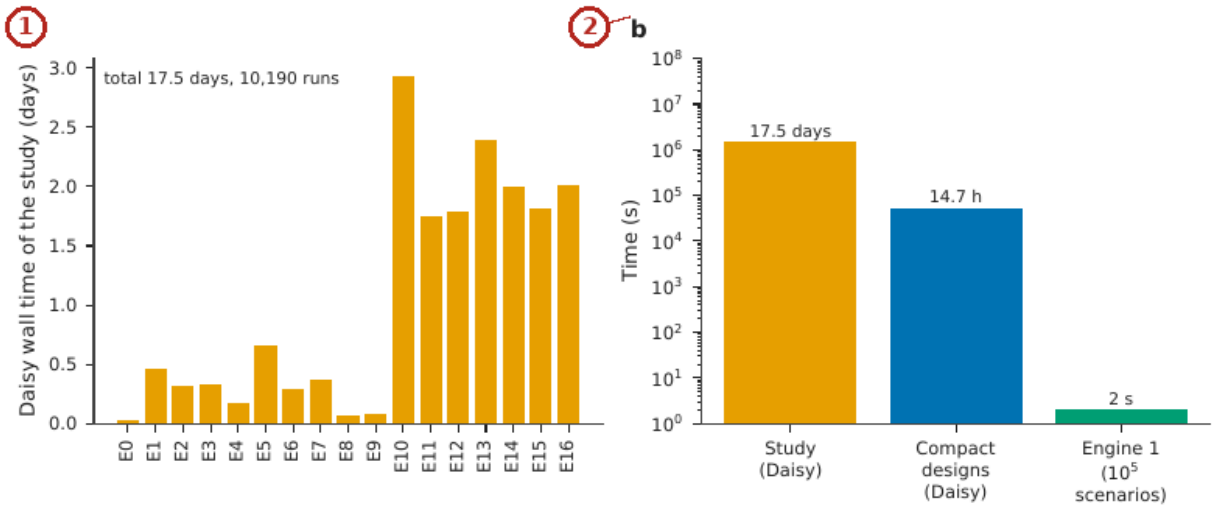
What it shows. (a) Error of the marginal curves rebuilt from the levels of the compact designs (median over five responses, per experiment and factor). (b) Daisy run time of the study designs and of the compact designs, by experiment (logarithmic scale).

How to read the labels, symbols and colours. (a) Error of the marginal curves of the compact designs by experiment and factor; (b) Daisy time of the study and compact designs (log scale).

What affects it, and why it happens. Fewer levels miss curvature between them (error) but save orders of magnitude of time.

How it is made. Figure drawn by the script from the input data. Script: `Manuscripts/reproducibility/p456/p456_figs_data.py`. Input: `Data/daisy_ensemble.csv.gz`.

S4.4 Reading Figure 16 of the article



What the numbered labels point to.

1. Panel (a): Daisy wall-clock time of the 17 experiments of the study (total 17.5 days). *Why:* Experiments with more runs or longer runs take longer.
2. Panel (b): Time of the study, of the compact designs of the workbench and of Engine 1 for 10^5 scenarios (logarithmic scale). *Why:* Daisy solves every day of every run; the meta-model looks up stored results, so it is millions of times faster.

What it shows. (a) Daisy wall-clock time of the 17 experiments of the study (total 17.5 days). (b) Time of the study, of the compact designs of the workbench and of Engine 1 for 10^5 scenarios (logarithmic scale).

How to read the labels, symbols and colours. (a) Daisy time of each experiment; (b) time of the study, compact designs and Engine 1 (log scale).

What affects it, and why it happens. Each route trades fidelity for speed by orders of magnitude.

How it is made. Figure drawn by the script from the input data. Script: `Manuscripts/reproducibility/p456/p456_figs_data.py`. Input: `Data/daisy_ensemble.csv.gz`.

S4.5 Derivation: Run budget of an experiment

For k factors with $n_1, \ldots, n_k$ levels the full design has $\prod n_i$ runs. When this exceeds the budget B set by the user, a Latin-hypercube choice of B runs is made: each factor's levels are used as

evenly as possible (B/n_i times each), so every level of every factor appears and the main effects remain estimable while the number of runs never exceeds B.

S4.6 Why it happens

Why the compact designs lose some shape of a curve. A design with fewer levels samples a curve at fewer points; the marginal curve is then drawn through those points and misses curvature between them, which is the loss measured.

S5. Root causes of failed runs

The classes of Daisy failures and what to do about each.

S5.1 Contribution 3: the root cause of a failed run

The result

Daisy message contains	Class	What to do
(1) Unknown '…' model	a name not defined	take only the needed definitions
(2) did not converge, NaN	numerical	bring extreme values closer to typical
(3) out of range, must be	range	change the level that writes the value
(4) cannot open, not found	missing file	rerun; check the weather and soil

Pinpoints: what each circled mark shows and why

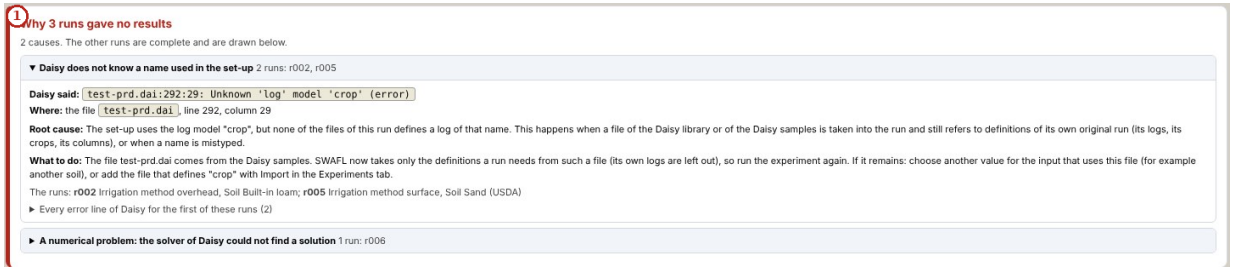
(1) This class found a real defect: a file of the Daisy samples referred to logs of its own run; taking only the definitions a run needs removed it.

(2) Numerical failures come from extreme hydraulic values; the message names the factor and the typical range.

(3) The line and column of the set-up file are shown, so the user sees which input wrote the value.

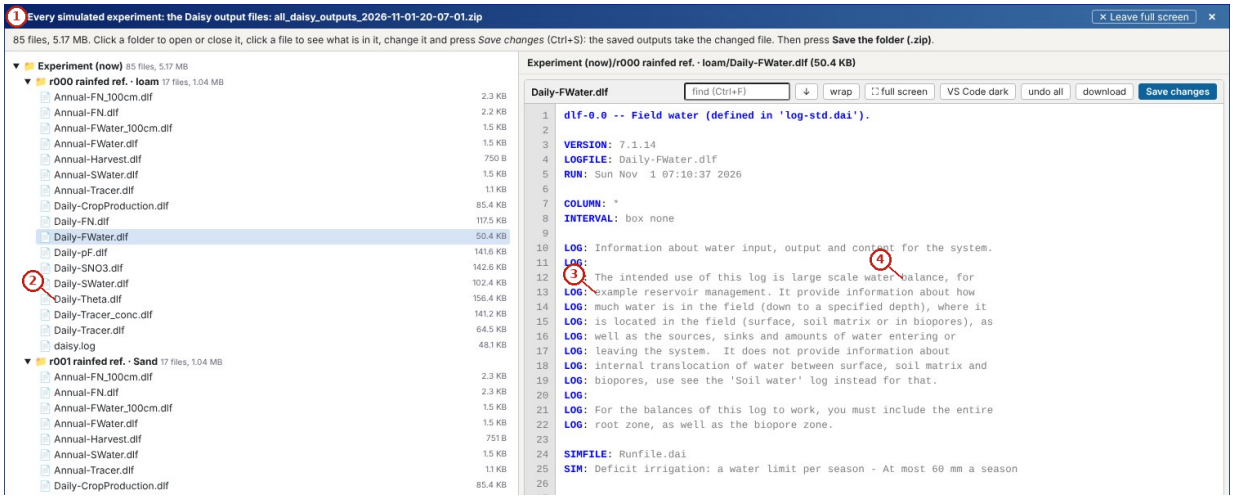
(4) Failed runs are grouped by cause, so one explanation covers all runs that failed for the same reason.

S5.2 Reading Figure 20 of the article



What the numbered labels point to (picture 1 of 2: The root cause of runs that gave no results, grouped by cause, with the place in the set-up and what to do.).

(1) "Why 3 runs gave no results": a heading or control of this screen; the caption above says which layer it belongs to.



What the numbered labels point to (picture 2 of 2: Every file Daisy wrote, experiment by experiment and run by run, in a full-screen viewer.).

(1) "Every simulated": a heading or control of this screen; the caption above says which layer it belongs to.

(2) "Daily-Theta.dlf": a heading or control of this screen; the caption above says which layer it belongs to.

(3) "example reservoir": a heading or control of this screen; the caption above says which layer it belongs to.

(4) "balance": a heading or control of this screen; the caption above says which layer it belongs to.

What it shows. Diagnosis and provenance of the runs (screen captures). Panels: (a) The root

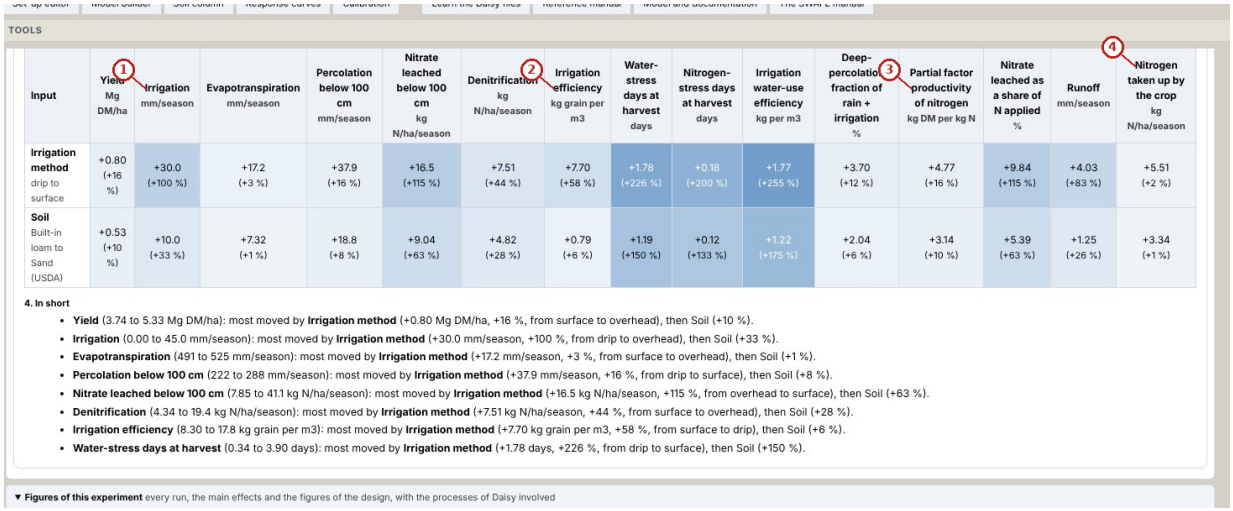
cause of runs that gave no results, grouped by cause, with the place in the set-up and what to do. (b) Every file Daisy wrote, experiment by experiment and run by run, in a full-screen viewer.

How to read the labels, symbols and colours. Diagnosis of failed runs and the viewer of every file.

What affects it, and why it happens. Failed runs are grouped by the cause in Daisy's message, so one correction fixes every run that failed the same way. Errors are grouped by cause, so one fix addresses all runs that failed the same way.

How it is made. Screen capture of SWAFL Desktop 1.1.0 (not regenerated).

S5.3 Reading Figure 19 of the article


TOOLS

Input	Yield Mg DM/ha	Irrigation mm/season	Evapotranspiration mm/season	Percolation below 100 cm mm/season	Nitrate leached below 100 cm kg N/ha/season	Denitrification kg N/ha/season	Irrigation efficiency kg grain per m3	Water-stress days at harvest days	Nitrogen-stress days at harvest days	Irrigation water-use efficiency kg per m3	Deep-percolation fraction of rain + irrigation %	Partial factor productivity of nitrogen kg DM per kg N	Nitrate leached as a share of N applied %	Runoff mm/season	Nitrogen taken up by the crop kg N/ha/season
Irrigation method drip to surface	+0.80 (+16 %)	+30.0 (+100 %)	+17.2 (+3 %)	+37.9 (+16 %)	+16.5 (+115 %)	+7.51 (+44 %)	+7.70 (+58 %)	+1.78 (+226 %)	+0.18 (+200 %)	+1.77 (+255 %)	+3.70 (+12 %)	+4.77 (+16 %)	+9.84 (+115 %)	+4.03 (+83 %)	+5.51 (+2 %)
Soil Built-in loam to Sand (USDA)	+0.53 (+10 %)	+10.0 (+33 %)	+7.32 (+1 %)	+18.8 (+8 %)	+9.04 (+63 %)	+4.82 (+28 %)	+0.79 (+6 %)	+1.19 (+150 %)	+0.12 (+133 %)	+1.22 (+175 %)	+2.04 (+6 %)	+3.14 (+10 %)	+5.39 (+63 %)	+1.25 (+26 %)	+3.34 (+1 %)

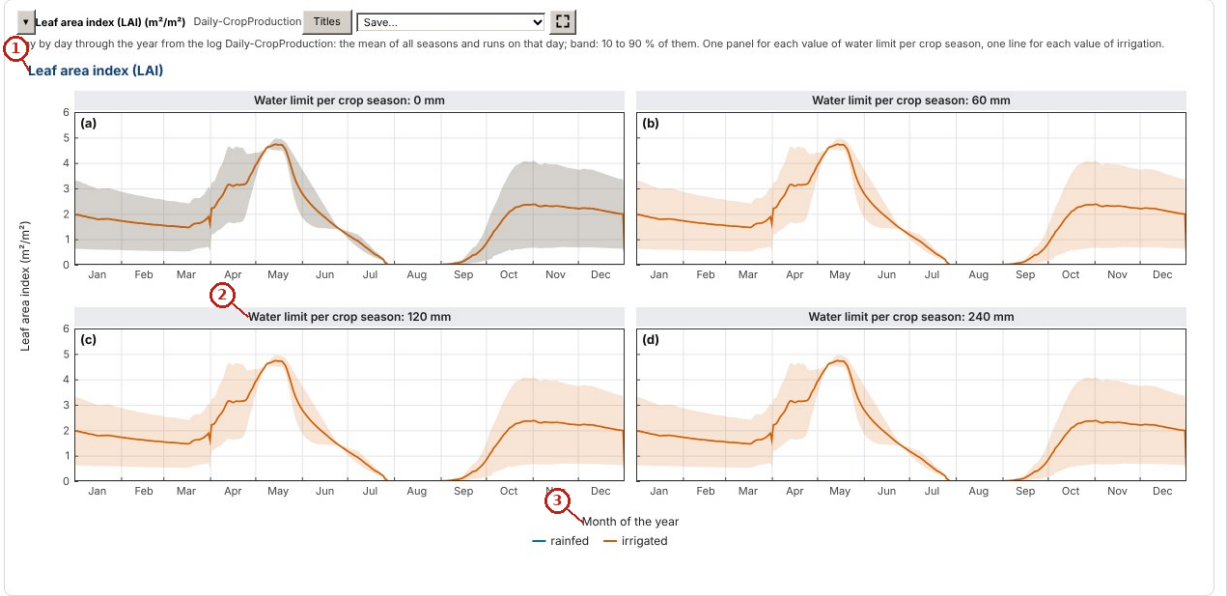
4. In short

- **Yield** (3.74 to 5.33 Mg DM/ha): most moved by **Irrigation method** (+0.80 Mg DM/ha, +16 %, from surface to overhead), then Soil (+10 %).
- **Irrigation** (0.00 to 45.0 mm/season): most moved by **Irrigation method** (+30.0 mm/season, +100 %, from drip to overhead), then Soil (+33 %).
- **Evapotranspiration** (491 to 525 mm/season): most moved by **Irrigation method** (+17.2 mm/season, +3 %, from surface to overhead), then Soil (+1 %).
- **Percolation below 100 cm** (222 to 288 mm/season): most moved by **Irrigation method** (+37.9 mm/season, +16 %, from drip to surface), then Soil (+8 %).
- **Nitrate leached below 100 cm** (7.85 to 41.1 kg N/ha/season): most moved by **Irrigation method** (+16.5 kg N/ha/season, +115 %, from overhead to surface), then Soil (+63 %).
- **Denitrification** (4.34 to 19.4 kg N/ha/season): most moved by **Irrigation method** (+7.51 kg N/ha/season, +44 %, from surface to overhead), then Soil (+28 %).
- **Irrigation efficiency** (8.30 to 17.8 kg grain per m3): most moved by **Irrigation method** (+7.70 kg grain per m3, +58 %, from surface to drip), then Soil (+6 %).
- **Water-stress days at harvest** (0.34 to 3.90 days): most moved by **Irrigation method** (+1.78 days, +226 %, from drip to surface), then Soil (+150 %).

▾ **Figures of this experiment** every run, the main effects and the figures of the design, with the processes of Daisy involved

What the numbered labels point to (picture 1 of 2: The summary of an experiment: the effect of every input on every result (blue rises, red falls; in brackets the share of the mean) and the summary in words.).

1. "Irrigation": a heading or control of this screen; the caption above says which layer it belongs to.
2. "efficiency": a heading or control of this screen; the caption above says which layer it belongs to.
3. "productivity": a heading or control of this screen; the caption above says which layer it belongs to.
4. "Nitrogen": a heading or control of this screen; the caption above says which layer it belongs to.



What the numbered labels point to (picture 2 of 2: One of the figures of every log: leaf area index through the year, one panel per water limit, rainfed and irrigated, the mean of the seasons with 10 to 90 % as the band.).

1. "Leaf area index (LAI)": a heading or control of this screen; the caption above says which layer it belongs to.
2. "Water limit per crop season: 120": the crop of the project and its stage.
3. "Month of the year": a heading or control of this screen; the caption above says which layer it belongs to.

What it shows. Outputs of the experiment engine drawn from Daisy runs (screen captures of the workbench). Panels: (a) The summary of an experiment: the effect of every input on every result (blue rises, red falls; in brackets the share of the mean) and the summary in words. (b) One of the figures of every log: leaf area index through the year, one panel per water limit, rainfed and irrigated, the mean of the seasons with 10 to 90 % as the band.

How to read the labels, symbols and colours. Summary of an experiment (blue rises, red falls) and a figure of one log.

What affects it, and why it happens. The effect of each input is the change of each result across its levels, so the largest bars are the inputs that matter most. Each input's effect on each result is its range across the runs, shown with sign.

How it is made. Screen capture of SWAFL Desktop 1.1.0 (not regenerated).

S5.4 Why it happens

Why runs on a horizon of the Daisy samples failed. The sample file also defined logs that referred to a model of its original run; Daisy stopped on the unknown name. Taking only the definitions a run needs removed the error.

S6. The soil column and its hydraulic factors

The Richards equation of the column and the factors that rewrite every horizon.

S6.1 Contribution 4: soil-hydraulic factors for every horizon

How the author developed it

Each factor rewrites the hydraulic block of every horizon of the set-up file before Daisy runs; $n - 1$ rather than n is scaled so that $n > 1$ for any factor; one run of the loam was made with Daisy 7.1.14 with and without the factors.

The result

How it is derived, line by line

1. **Keep $n > 1$ for any factor by scaling $n - 1$:**

$$n' = 1 + f_n(n - 1)$$

2. **Scale the other parameters of every horizon:**

$$K'_s = f_k K_s,\ \alpha' = f_a \alpha,\ \theta'_s = f_t \theta_s,\ \ell' = \ell + \delta\ell$$

The result, with its parts marked

$$\underbrace{K_s \to f_k K_s}_{①},\quad \underbrace{\alpha \to f_a \alpha}_{②},\quad \underbrace{n - 1 \to f_n(n - 1)}_{③},\quad \theta_s \to f_t \theta_s,\quad \ell \to \ell + \delta\ell$$

$$\underbrace{\text{percolation } 177 \to 222 \text{ mm}}_{④}$$

developed by the author

Pinpoints: what each circled mark shows and why

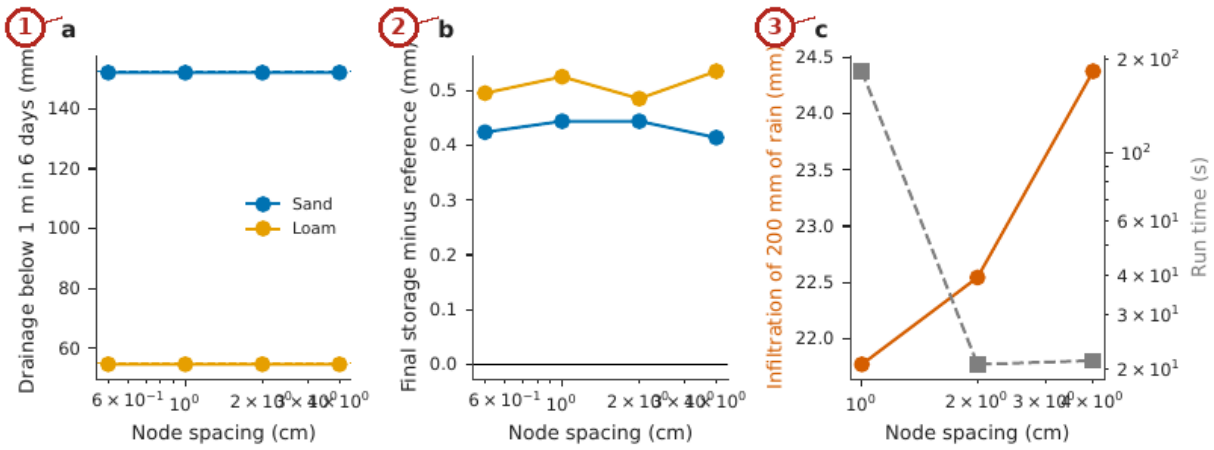
(1) Scaling the saturated conductivity of every horizon at once keeps the layering of the soil.

(2) Scaling α moves the air-entry suction.

(3) Scaling $n - 1$ (not n) keeps $n > 1$, so the van Genuchten curve stays valid for any factor.

(4) A faster-draining parameter set, run end to end with Daisy, raised one season's matrix percolation from 177 to 222 mm and removed runoff, the direction expected from the conductivity.

S6.2 Reading Figure 14 of the article



What the numbered labels point to.

(1) Panel (a): Drainage below 1 m with the node spacing; dashed lines are the reference. *Why:* Coarser node spacing smooths wetting fronts, so drainage departs from the fine-grid reference.

(2) Panel (b): Final storage minus the reference. *Why:* Storage errors follow the drainage errors because water must be conserved.

(3) Panel (c): Infiltration of 200 mm of rain into a ponded silty clay loam (left axis) and run time (right axis), showing the lack of convergence. *Why:* A ponded fine soil has very steep suction gradients at the surface and tiny conductivity, so the solver needs tiny steps and does not converge in reasonable time.

What it shows. Soil column. (a) Drainage below 1 m with the node spacing; dashed lines

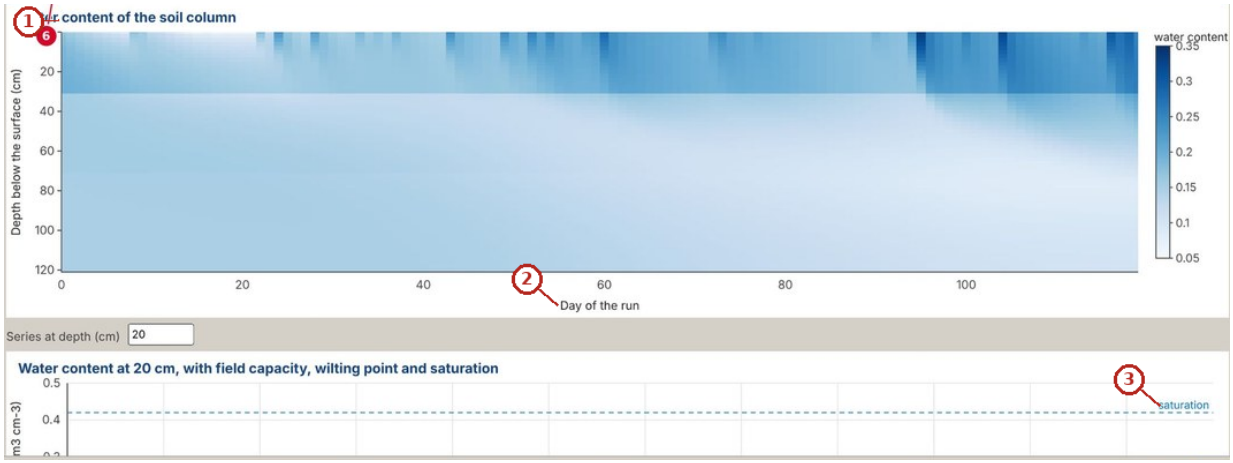
are the reference. (b) Final storage minus the reference. (c) Infiltration of 200 mm of rain into a ponded silty clay loam (left axis) and run time (right axis), showing the lack of convergence.

How to read the labels, symbols and colours. (a) Drainage below 1 m against node spacing, dashed reference; (b) storage error; (c) infiltration into a ponded fine soil and run time.

What affects it, and why it happens. Coarser grids smooth the fronts; a ponded fine soil has steep gradients and tiny conductivity, so the solver needs small steps and does not converge in reasonable time.

How it is made. Figure drawn by the script from the input data. Script: `Manuscripts/reproducibility/p456/p456_figs_data.py`. Input: `Data/daisy_ensemble.csv.gz`.

S6.3 Reading Figure 9 of the article



What the numbered labels point to.

1. "content of the soil column": the soil of the project, which every layer uses.
2. "Day of the run": a heading or control of this screen; the caption above says which layer it belongs to.
3. "saturation": a heading or control of this screen; the caption above says which layer it belongs to.

What it shows. Water content of a soil column through time, computed by the Richards solver of the workbench.

How to read the labels, symbols and colours. Water content through depth and time from the column solver; colour is water content.

What affects it, and why it happens. Wetting fronts move down after rain and the profile dries from the top by evaporation and uptake.

How it is made. Screen capture of SWAFL Desktop 1.1.0 (not regenerated).

S6.4 Derivation: The Richards equation of the soil column

Mass conservation of water in a layer of thickness dz gives $\partial\theta/\partial t = -\partial q/\partial z - S$, with S the root uptake. The Darcy–Buckingham flux with z positive downward and h the suction head is $q = K(h)(1 - \partial h/\partial z)$: gravity drives water down and a suction gradient drives it towards the drier soil. With $\theta(h)$ and $K(h)$ from the van Genuchten–Mualem functions the equation is closed and is solved implicitly with Picard iteration in the column of the workbench.

S6.5 Why it happens

Why the soil column is slow for a ponded fine soil. With ponding the top boundary is a fixed head and the fine soil has a very small conductivity and a steep head gradient at the surface; the implicit solver then needs small time steps and many iterations to converge.

S7. Nitrogen and the denitrification split

The nitrogen budget and the defect found in a shared function.

S7.1 Contribution 5: a defect found in a shared function

The problem

The denitrification function is shared with Daisy's Python tools. Is the copy used by the software right?

How the author developed it

The function was evaluated on the stored inputs of the 792 cases distributed with Daisy's tools and its outputs compared case by case with Daisy's stored results.

The result

How it is derived, line by line

1. **Run the function on the stored inputs:**

   $D_j^{\text{SWAFL}},\ j = 1..792$

2. **Compare case by case:**

   $\max_j |D_j^{\text{SWAFL}} - D_j^{\text{Daisy}}|$

The result, with its parts marked

$$\underbrace{\max_{j=1..792} \left| D_j^{\text{SWAFL}} - D_j^{\text{Daisy}} \right|}_{①} \quad \Rightarrow \quad \underbrace{\text{one branch wrong, corrected}}_{②}$$

developed by the author

Pinpoints: what each circled mark shows and why

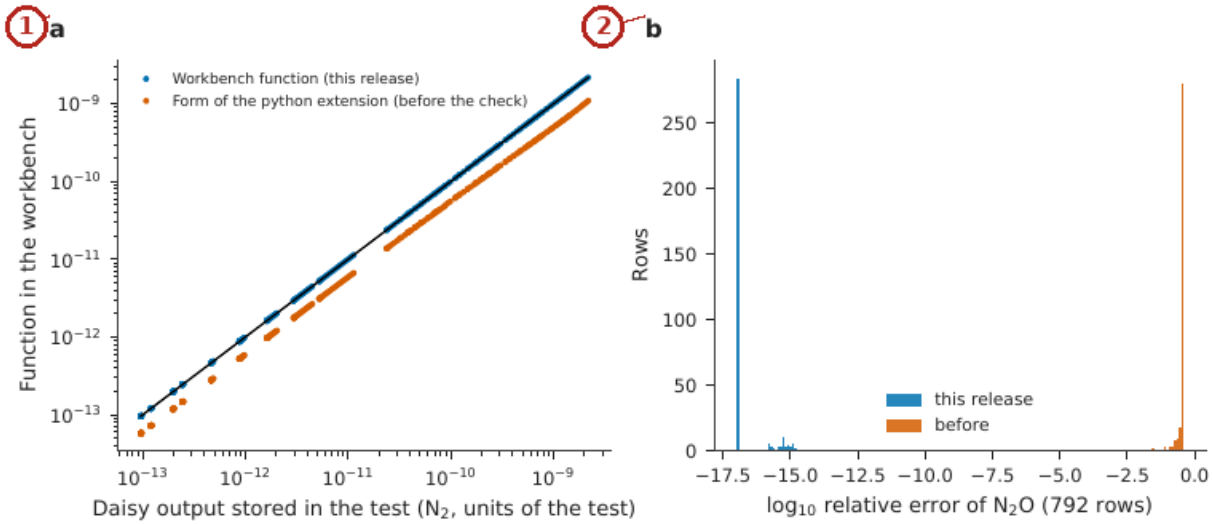
① The function was evaluated on the 792 cases for which Daisy's own results are stored, and compared case by case.

② One class of cases disagreed; the branch responsible was found and corrected, after which the cases agree. A shared function was tested rather than trusted.

S7.2 Worked numbers, part 7: splitting denitrification

With $12\,\text{kg N ha}^{-1}$ denitrified and an N_2:N_2O ratio of 4, $N_2O = 12/5 = 2.4$ and $N_2 = 9.6\,\text{kg N ha}^{-1}$.

S7.3 Reading Figure 15 of the article



What the numbered labels point to.

(1) Panel (a): N_2 loss; the 1:1 line is black. *Why:* After the correction the function and Daisy agree, so points lie on the 1:1 line.

(2) Panel (b): Histogram of the relative error of the N_2O loss before and after the correction. *Why:* Before the correction one class of cases had large errors; after it the errors collapse to zero.

What it shows. Denitrification function against the 792 stored Daisy outputs. (a) N_2 loss; the 1:1 line is black. (b) Histogram of the relative error of the N_2O loss before and after the correction.

How to read the labels, symbols and colours. (a) N_2 loss of the function against 792 stored Daisy outputs, 1:1 line; (b) histogram of the N_2O error before and after the correction.

What affects it, and why it happens. Before the correction one class of cases was off; after it, the errors collapse to zero.

How it is made. Figure drawn by the script from the input data. Script: `Manuscripts/reproducibility/p456/p456_figs_data.py`. Input: `Data/daisy_ensemble.csv.gz`.

S7.4 Derivation: Nitrogen and chemical budgets

For each season, the change of the mineral nitrogen store equals inputs minus outputs: fertiliser, deposition and mineralisation in; uptake, leaching, denitrification and volatilisation out. For a chemical, $\Delta M = A - U - L - R - X$ (applied, taken up, leached by every route, run off,

transformed); the bridge reads each term from the Daisy logs and the closure of this budget is its test.

S7.5 Derivation: Splitting denitrification into N_2O and N_2

Let r be the ratio of N_2 to N_2O in the gaseous product of denitrification D. Then $D_{N_2O} + D_{N_2} = D$ and $D_{N_2} = r\, D_{N_2O}$, so $D_{N_2O}(1 + r) = D$, giving $D_{N_2O} = D/(1 + r)$ and $D_{N_2} = rD/(1 + r) = D/(1 + 1/r)$, the two expressions of the paper. The ratio is the smaller of a nitrate and a carbon-dioxide limited ratio, scaled by a water factor, so wetter, more active soils give relatively more N_2.

S7.6 Why it happens

Why the denitrification defect was found. The check compared the shared function with the stored Daisy results over 792 cases; one case class disagreed, which pointed to a wrong branch in the function, corrected in the release.

S8. Maps of the field

Vegetation indices, crop coefficient, water balance, terrain, wetness and soil loss of every map cell.

S8.1 Worked numbers, part 1: a vegetation index from a satellite pixel

Green leaves reflect near-infrared light and absorb red light. *Example:* near-infrared reflectance 0.28 and red 0.11 give NDVI $= (0.28 - 0.11)/(0.28 + 0.11) = 0.44$, a moderately green crop.

S8.2 Worked numbers, part 2: the crop coefficient of the day

FAO-56 describes the crop's water use relative to a reference grass with four straight lines. *Example:* with stages of 30, 40, 50 and 30 days and K_c of 0.40, 1.15 and 0.40, day 50 lies in the development stage: $K_c = 0.40 + (1.15 - 0.40) \times (50 - 30)/40 = 0.78$; day 70 starts the mid-season, $K_c = 1.15$.

S8.3 Worked numbers, part 3: the daily water balance of a map cell

Each 10 m cell keeps a root-zone depletion: yesterday's depletion minus rain and irrigation plus crop water use; when it passes the readily available water, the cell needs irrigation of that depth.

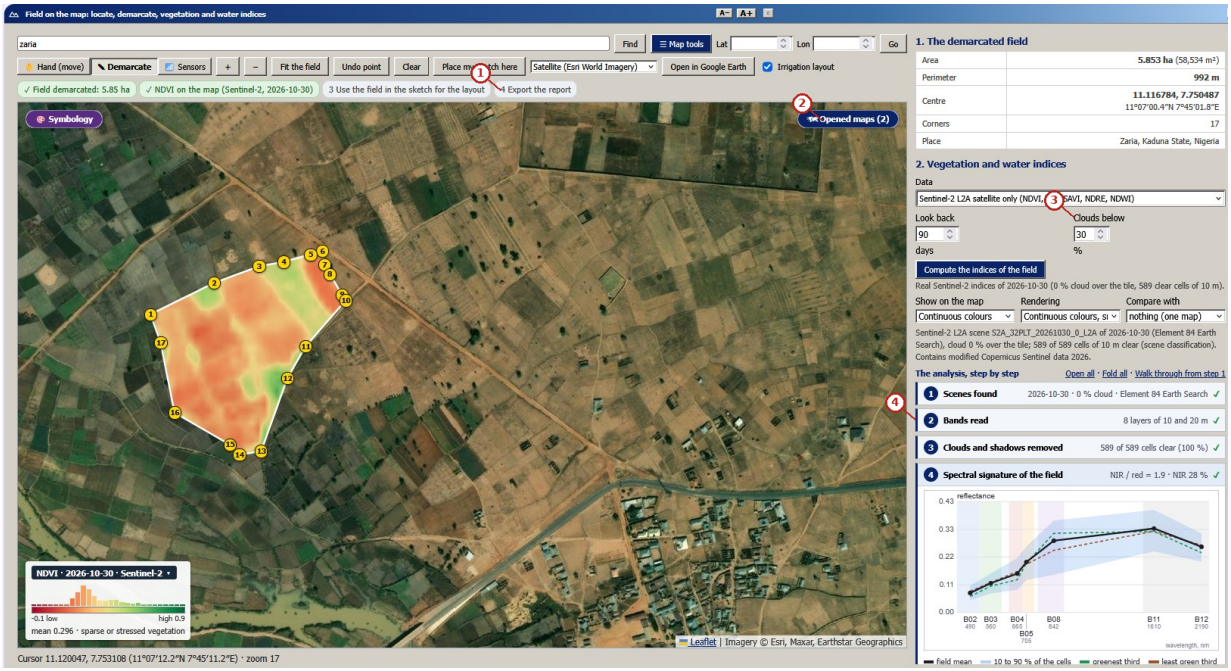
S8.4 Worked numbers, part 4: soil loss

$A = R\,K\,LS\,C\,P$. *Example:* $450 \times 0.03 \times 0.4 \times 0.3 \times 1 = 1.6\,\mathrm{t\,ha^{-1}\,yr^{-1}}$, below a tolerable 2 t.

S8.5 Worked numbers, part 5: where the soil stays wet

The wetness index is $\ln(a/\tan\beta)$. *Example:* 250 m of upslope area per metre of contour on a slope of 1.9° gives $\ln(250/0.0332) = 8.9$, a wet position.

S8.6 Reading Figure 2 of the article

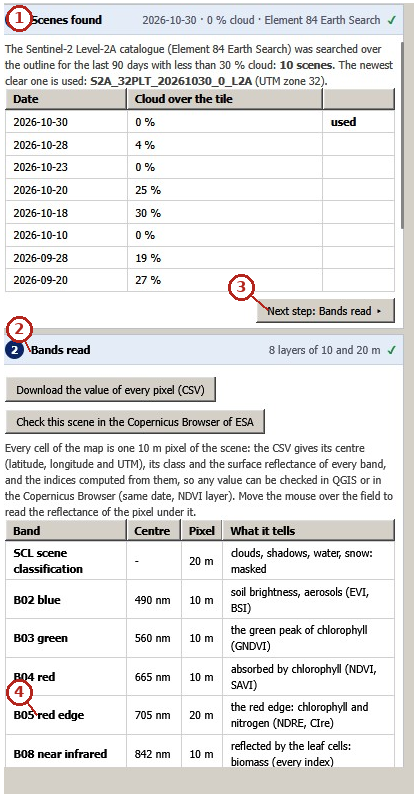


What the numbered labels point to (picture 1 of 3: The demarcated field (5.85 ha) with the NDVI of the Sentinel-2 scene of 30 October 2026 cut to its outline, the legend with the histogram of the field, and the analysis step by step with the spectral signature of the field.).

1. "Export the report": export of the field report and its maps.
2. "Opened maps (2)": a heading or control of this screen; the caption above says which layer it belongs to.

3 “Clouds below”: a heading or control of this screen; the caption above says which layer it belongs to.

4 “Bands read”: satellite indices of the vegetation computed from the Sentinel-2 bands.



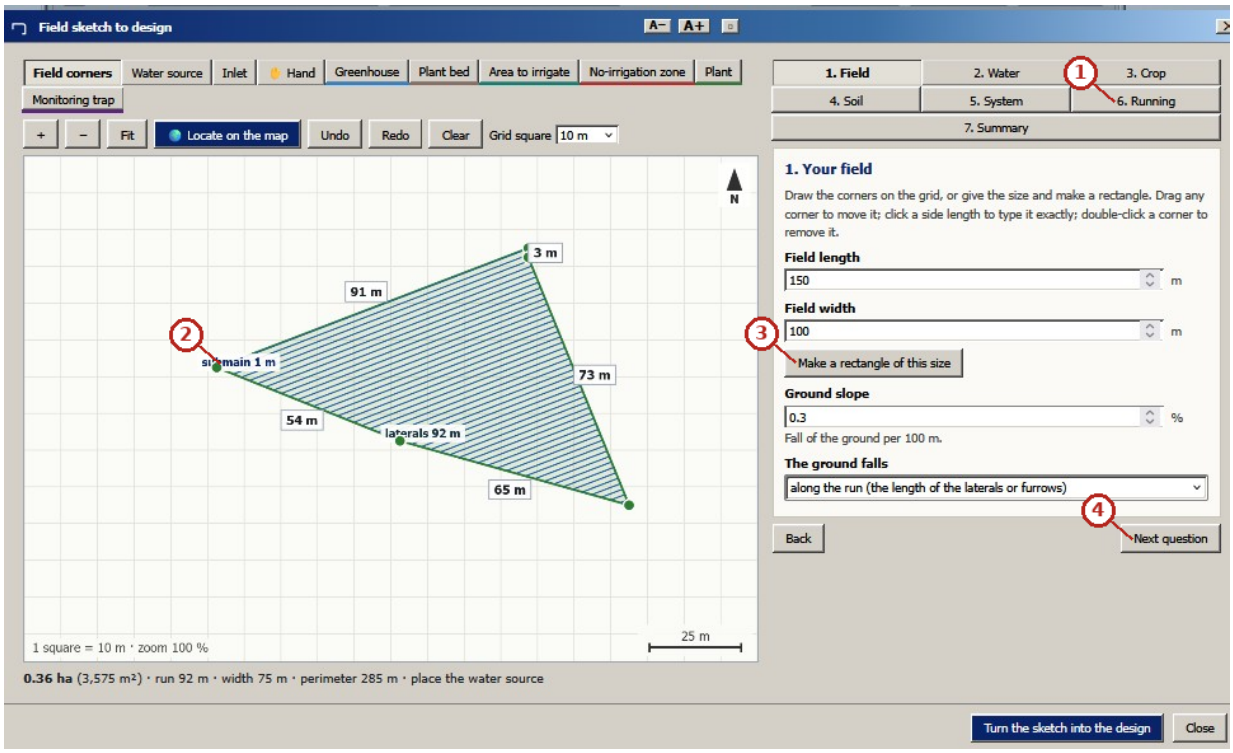
What the numbered labels point to (picture 2 of 3: The scenes found and the bands read.).

1 “Scenes found”: a heading or control of this screen; the caption above says which layer it belongs to.

2 “Bands read”: satellite indices of the vegetation computed from the Sentinel-2 bands.

3 “Next step”: moves to the next or previous step of the guided path.

4 “red edge”: a heading or control of this screen; the caption above says which layer it belongs to.



What the numbered labels point to (picture 3 of 3: The field sketch that carries a field into the designs.).

1. "6. Running": a heading or control of this screen; the caption above says which layer it belongs to.
2. "main 1m": a heading or control of this screen; the caption above says which layer it belongs to.
3. "Make a rectangle of this size": a heading or control of this screen; the caption above says which layer it belongs to.
4. "Next question": moves to the next or previous step of the guided path.

What it shows. A real field near Zaria, Nigeria: demarcation and satellite indices (screen captures of release 1.1.0). Panels: (a) The demarcated field (5.85 ha) with the NDVI of the Sentinel-2 scene of 30 October 2026 cut to its outline, the legend with the histogram of the field, and the analysis step by step with the spectral signature of the field. (b) The scenes found and the bands read. (c) The field sketch that carries a field into the designs.

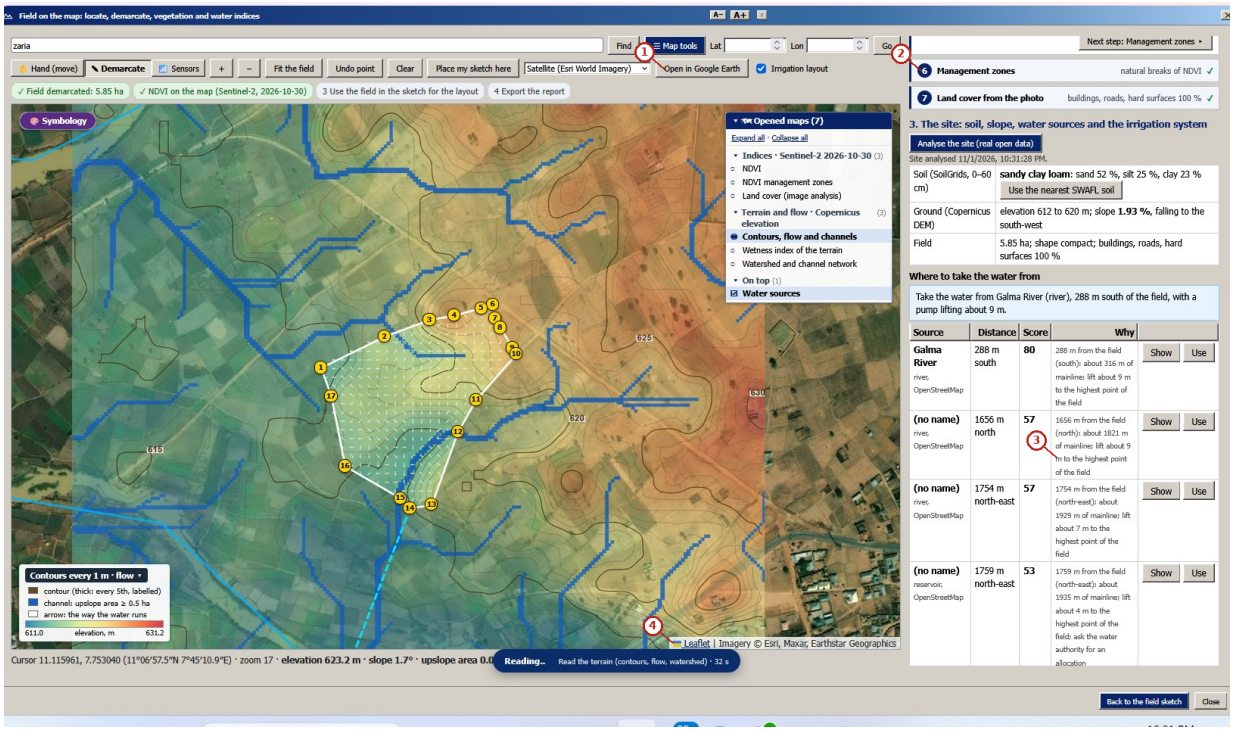
How to read the labels, symbols and colours. Screens of the real field: NDVI map with its histogram and spectral signature, the scenes and bands, and the sketch of the field.

What affects it, and why it happens. Greener cells reflect more near-infrared and less red light, so NDVI is higher where the crop is denser. Greener cells (higher NDVI) have more active

vegetation; the scene is the clearest of the last 90 days.

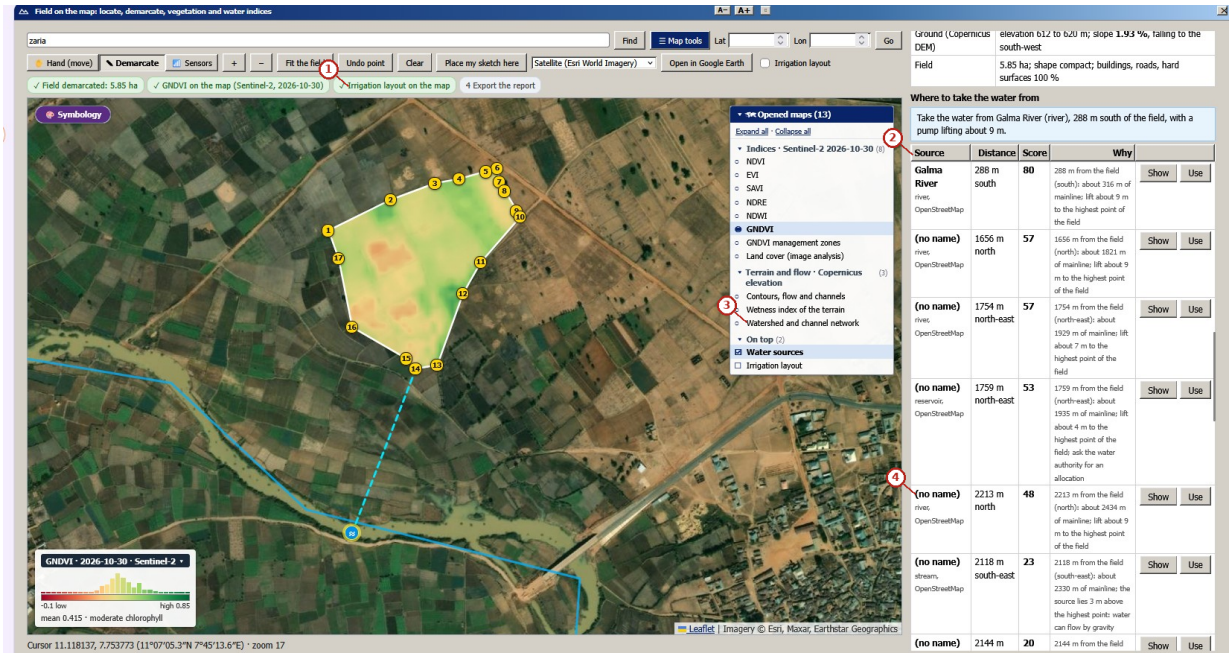
How it is made. Screen capture of SWAFL Desktop 1.1.0 (not regenerated).

S8.7 Reading Figure 3 of the article



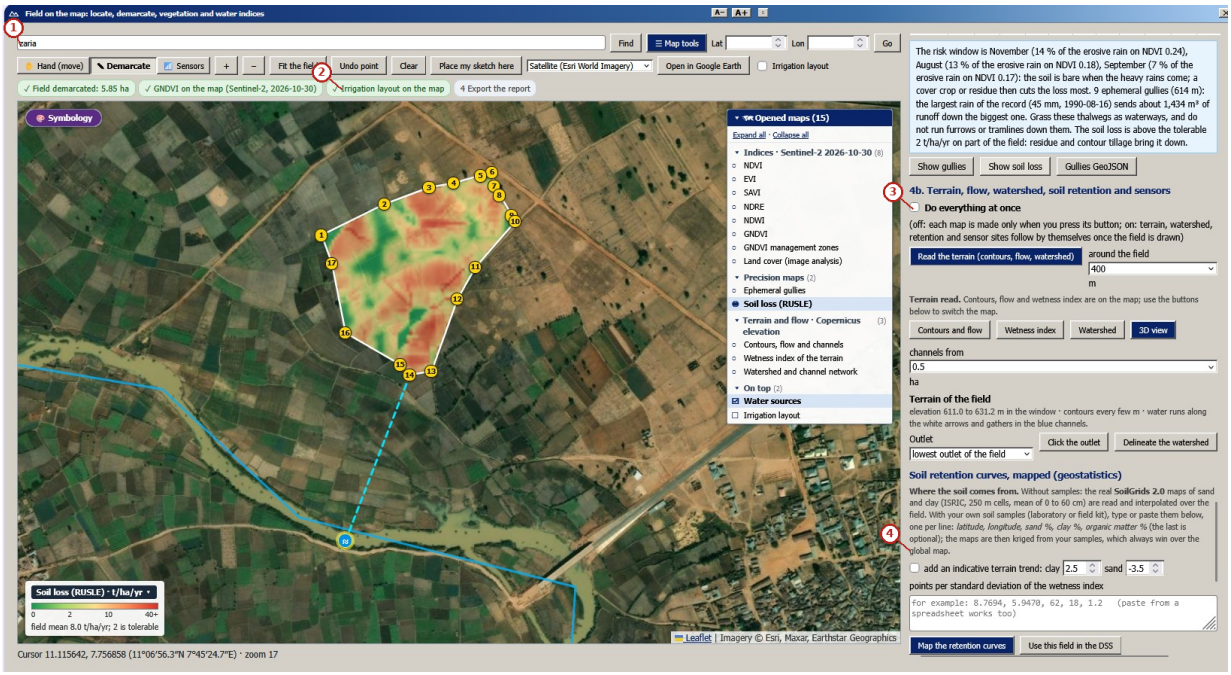
What the numbered labels point to (picture 1 of 4: Contours every metre, flow arrows and channels, and the ranked water sources.).

1. "Open in Google Earth": a heading or control of this screen; the caption above says which layer it belongs to.
2. "zones": a heading or control of this screen; the caption above says which layer it belongs to.
3. "m to the highest point": a heading or control of this screen; the caption above says which layer it belongs to.
4. "Imagery ©": a heading or control of this screen; the caption above says which layer it belongs to.



What the numbered labels point to (picture 2 of 4: GNDVI and the line to the best water source, the Galma River (288 m).).

1. "Irrigation layout on the map": a heading or control of this screen; the caption above says which layer it belongs to.
2. "Source Score": a heading or control of this screen; the caption above says which layer it belongs to.
3. "Watershed and channel network": a heading or control of this screen; the caption above says which layer it belongs to.
4. "(no name)": a heading or control of this screen; the caption above says which layer it belongs to.



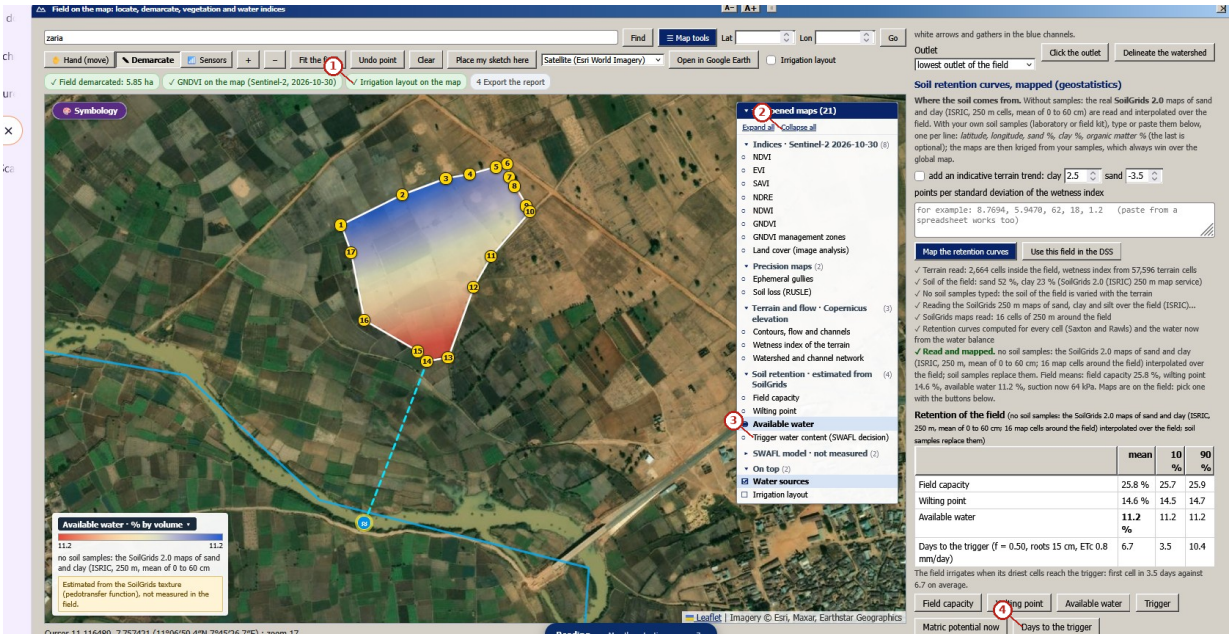
What the numbered labels point to (picture 3 of 4: Soil loss (RUSLE), field mean 8.0 t ha^{-1} yr^{-1}, and the months of risk.).

(1) "Zaria": a heading or control of this screen; the caption above says which layer it belongs to.

(2) "Irrigation layout on the map": a heading or control of this screen; the caption above says which layer it belongs to.

(3) "Do everything at once": a heading or control of this screen; the caption above says which layer it belongs to.

(4) "global map": a heading or control of this screen; the caption above says which layer it belongs to.



What the numbered labels point to (picture 4 of 4: Available water from the retention curves mapped from SoilGrids (11.2 % by volume) and the days to the trigger.).

1. "layout on the map": a heading or control of this screen; the caption above says which layer it belongs to.
2. "Collapse all": a heading or control of this screen; the caption above says which layer it belongs to.
3. "Trigger water content (SWAFL decision)": a heading or control of this screen; the caption above says which layer it belongs to.
4. "Days to the trigger": the suction (or depletion) at which irrigation starts.

What it shows. The same field: terrain, water source, erosion and soil water retention from open data (screen captures of release 1.1.0). Panels: (a) Contours every metre, flow arrows and channels, and the ranked water sources. (b) GNDVI and the line to the best water source, the Galma River (288 m). (c) Soil loss (RUSLE), field mean 8.0 t ha^{-1} yr^{-1}, and the months of risk. (d) Available water from the retention curves mapped from SoilGrids (11.2 % by volume) and the days to the trigger.

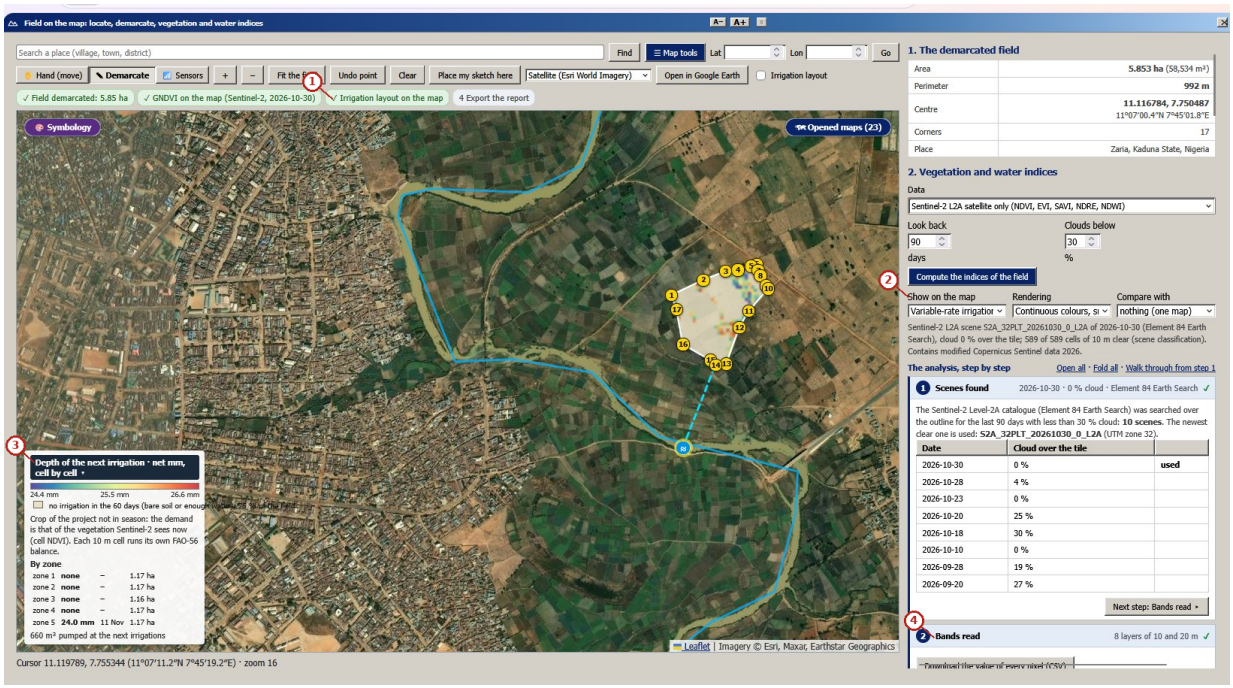
How to read the labels, symbols and colours. Screens: contours and flow, the best water source, soil loss and available water.

What affects it, and why it happens. Water flows down the steepest slope to the channels; soil loss is largest on steep, bare and erodible cells; available water follows soil texture. Water flows down the contours to channels; soil loss is highest on bare, steep, erosive cells; available

water follows the texture mapped from SoilGrids.

How it is made. Screen capture of SWAFL Desktop 1.1.0 (not regenerated).

S8.8 Reading Figure 4 of the article



What the numbered labels point to.

1. "Irrigation layout on the map": a heading or control of this screen; the caption above says which layer it belongs to.
2. "Show on the map": a heading or control of this screen; the caption above says which layer it belongs to.
3. "Depth of the next": a heading or control of this screen; the caption above says which layer it belongs to.
4. "Bands read": satellite indices of the vegetation computed from the Sentinel-2 bands.

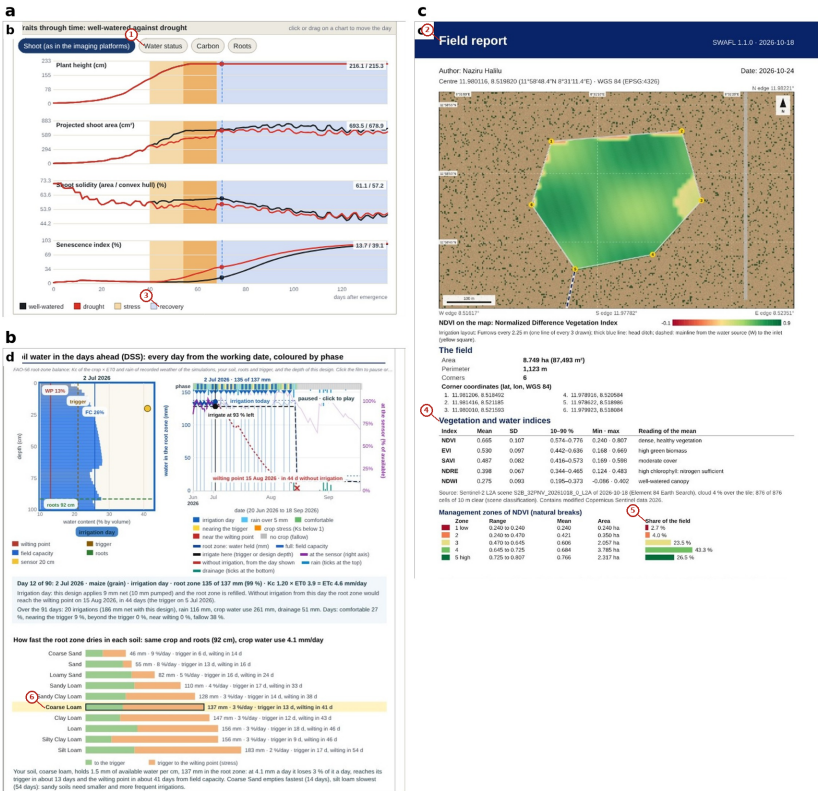
What it shows. Depth of the next irrigation on the Zaria field, cell by cell, from the FAO-56 balance of each 10 m cell (real screenshot).

How to read the labels, symbols and colours. Map of the depth of the next irrigation, cell by cell, with a colour scale in mm.

What affects it, and why it happens. Cells with more green cover use more water and need deeper irrigation sooner. Cells with more vegetation (higher demand) or less available water need deeper irrigation sooner.

How it is made. Screen capture of SWAFL Desktop 1.1.0 (not regenerated).

S8.9 Reading Figure 5 of the article



What the numbered labels point to.

1. "Water status Carbon Roots": the roots: depth, density and uptake.
2. "Field report": export of the field report and its maps.
3. "recovery": a heading or control of this screen; the caption above says which layer it belongs to.
4. "Vegetation and water indices": satellite indices of the vegetation computed from the Sentinel-2 bands.
5. "Share of the field": a heading or control of this screen; the caption above says which layer it belongs to.
6. "Coarse Loam": the soil of the project, which every layer uses.

What it shows. (a) Traits through time, well-watered against drought, with the phases of the stress marked. (b) Soil water in the days ahead with the irrigation decision of the design. (c) The field report (page 1).

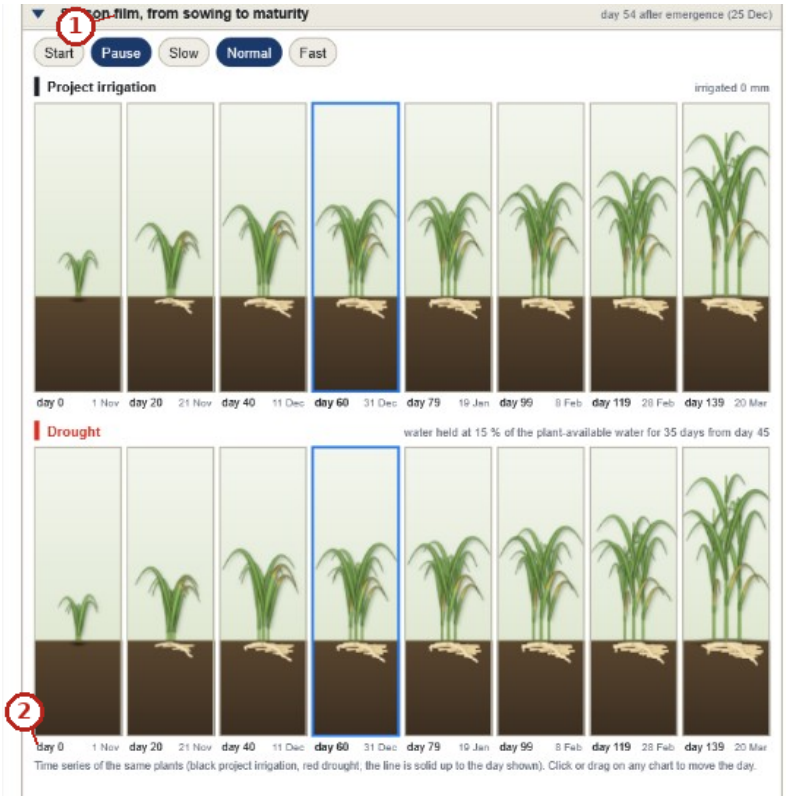
How to read the labels, symbols and colours. (a) Traits through time, well-watered against

drought; (b) soil water ahead with the decision; (c) the field report.

What affects it, and why it happens. Under drought the plant takes less water, closes stomata and grows less, so traits fall behind the well-watered plant. Traits fall behind under drought because uptake and growth fall with soil water.

How it is made. Screen capture of SWAFL Desktop 1.1.0 (not regenerated).

S8.10 Reading Figure 6 of the article



What the numbered labels point to.

1. "film, from sowing to maturity": a heading or control of this screen; the caption above says which layer it belongs to.
2. "day 0": a heading or control of this screen; the caption above says which layer it belongs to.

What it shows. The season film of the phenotyping layer: the same plants from sowing to maturity every 20 days, under the irrigation of the project (top) and under a drought that holds the water at 15 % of the plant-available water for 35 days from day 45 (bottom), with shoots and roots drawn from the state of the model (screen capture of release 1.1.0).

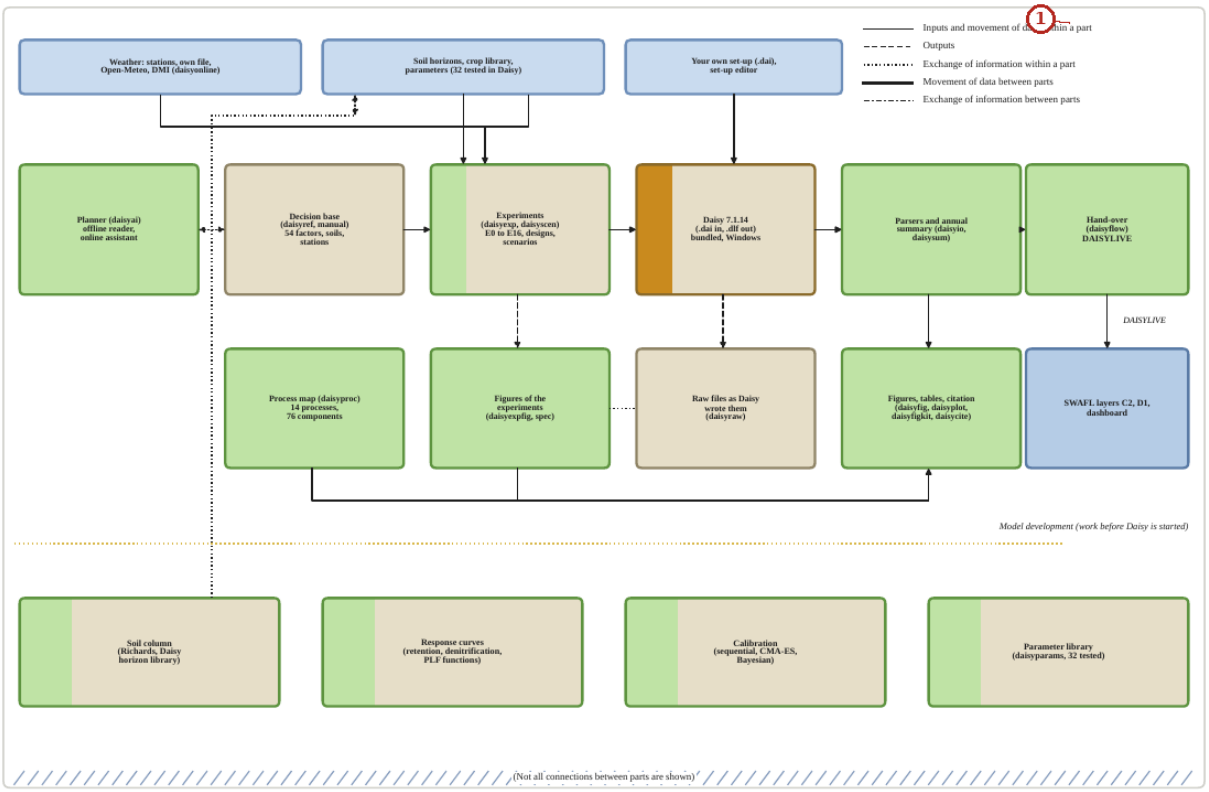
How to read the labels, symbols and colours. The same plants every 20 days under the

project's irrigation (top) and a drought (bottom).

What affects it, and why it happens. Water stress reduces growth and sends roots deeper, so the drought plants are smaller with deeper roots. Under drought the plants grow less and roots go deeper, as the model's water stress reduces growth.

How it is made. Screen capture of SWAFL Desktop 1.1.0 (not regenerated).

S8.11 Reading Figure 7 of the article



What the numbered labels point to.

(1) Panel (a): see the caption above.

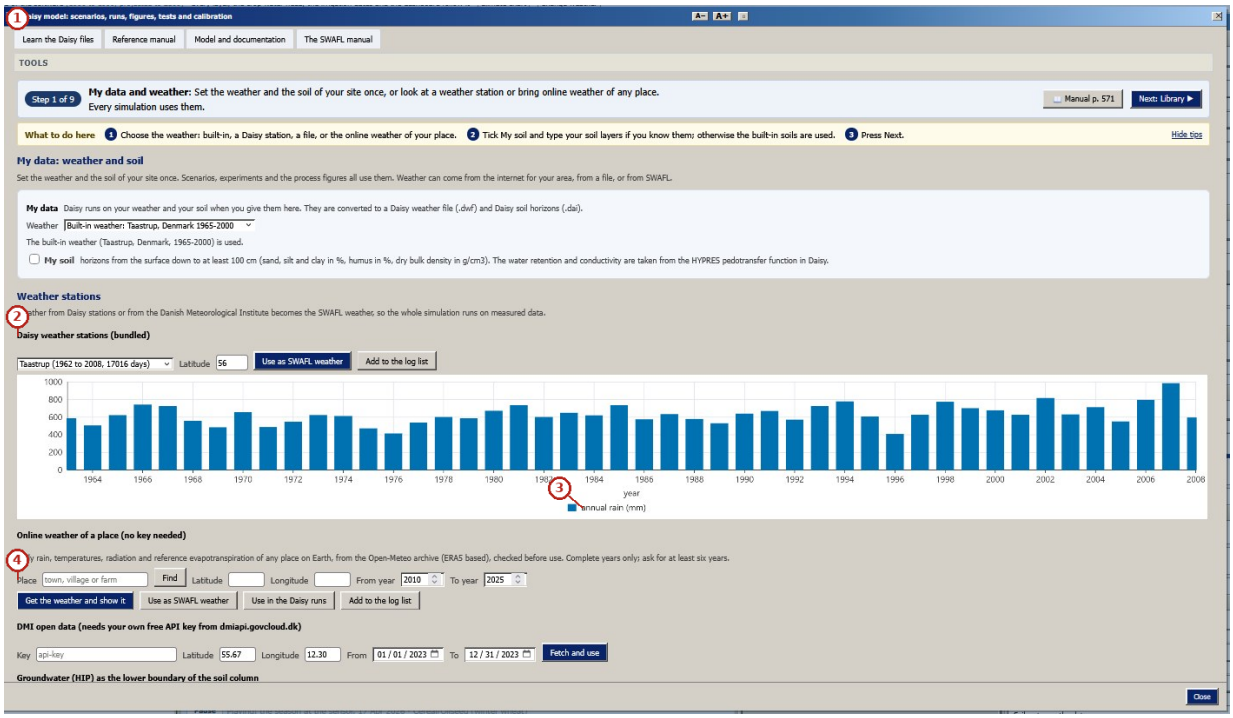
What it shows. Modules of the Daisy workbench and the hand-over, in the style of the process map of Daisy. The chain runs from the planner and the decision base through the experiments, Daisy and the parsers to the hand-over; the modules below the dotted line are used before Daisy is started.

How to read the labels, symbols and colours. Modules of the workbench and the hand-over; the dotted line separates what runs before Daisy.

What affects it, and why it happens. A question becomes a checked design, Daisy runs, the logs are parsed and handed to the layers.

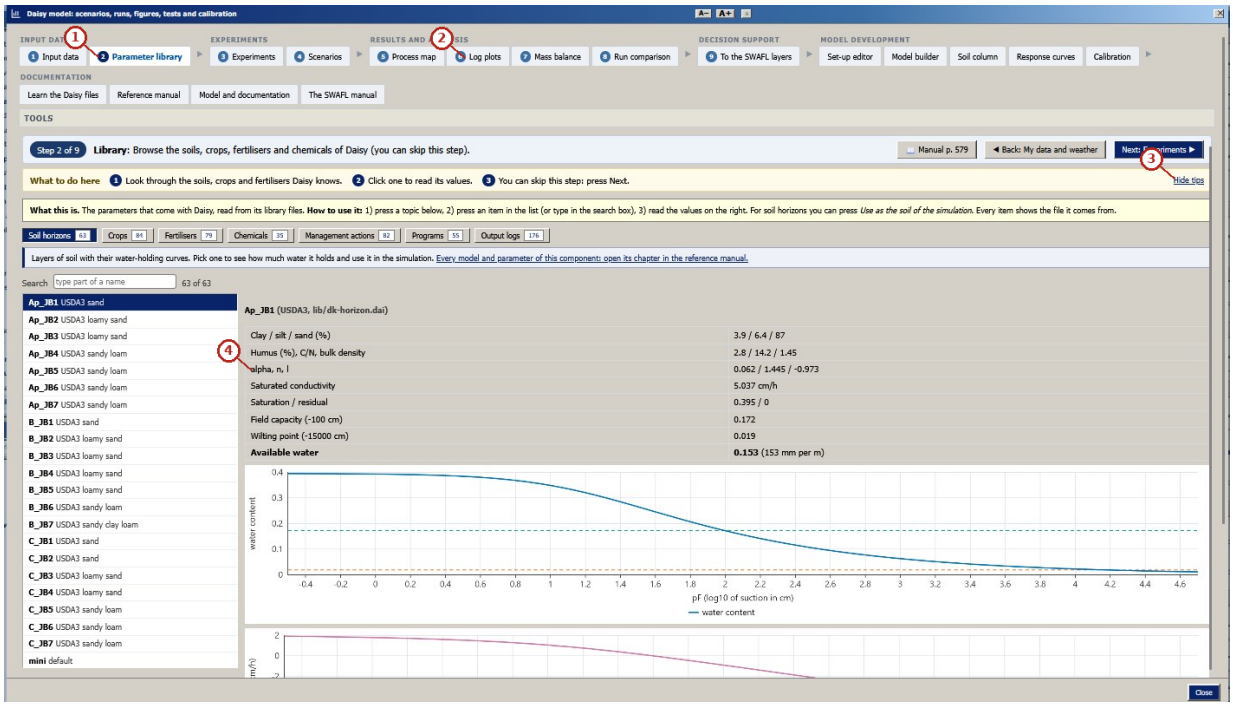
How it is made. Figure drawn by the script from the input data. Script: `Manuscripts/reproducibility/p456/p456_figs_architecture.py`. Input: `Data/daisy_ensemble.csv.gz`.

S8.12 Reading Figure 10 of the article



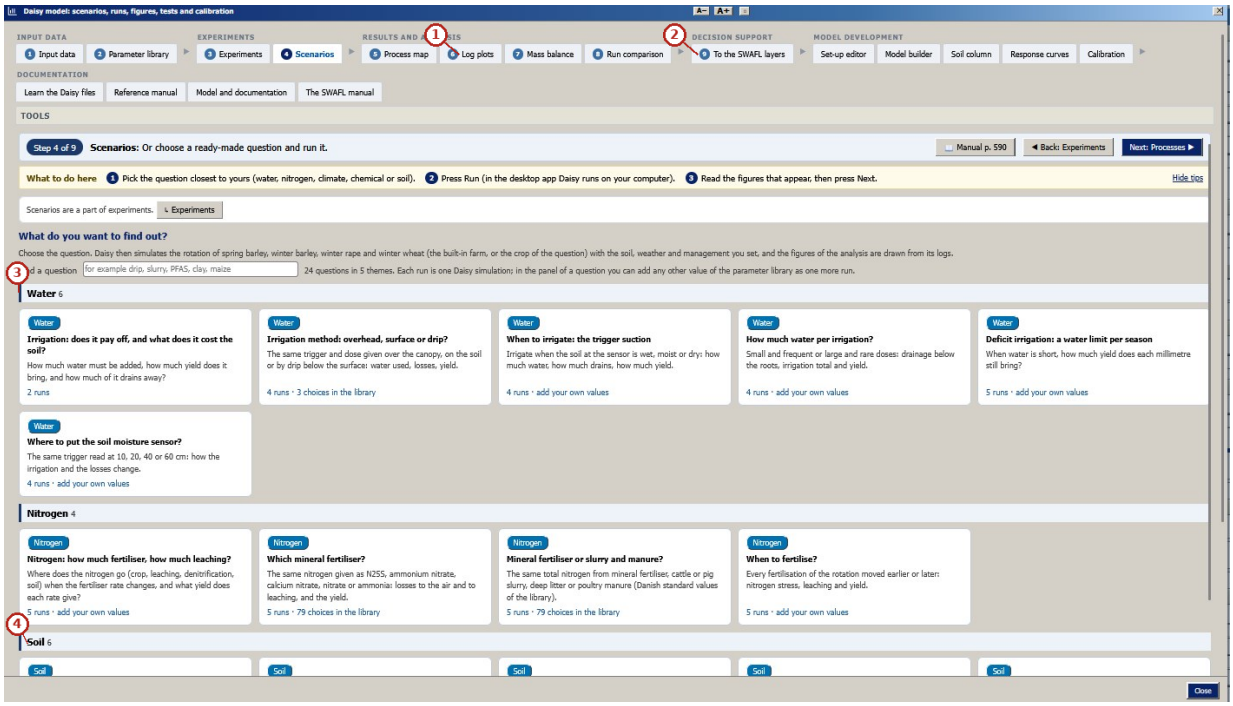
What the numbered labels point to (picture 1 of 8: Step 1: my data and weather (manual p. 760).).

1. "Daisy runs, figures, tests and calibration": the Daisy model and its workbench.
2. "Daisy weather stations (bundled)": the Daisy model and its workbench.
3. "annual rain (mm)": rain, from the weather of the project.
4. "Place": a heading or control of this screen; the caption above says which layer it belongs to.



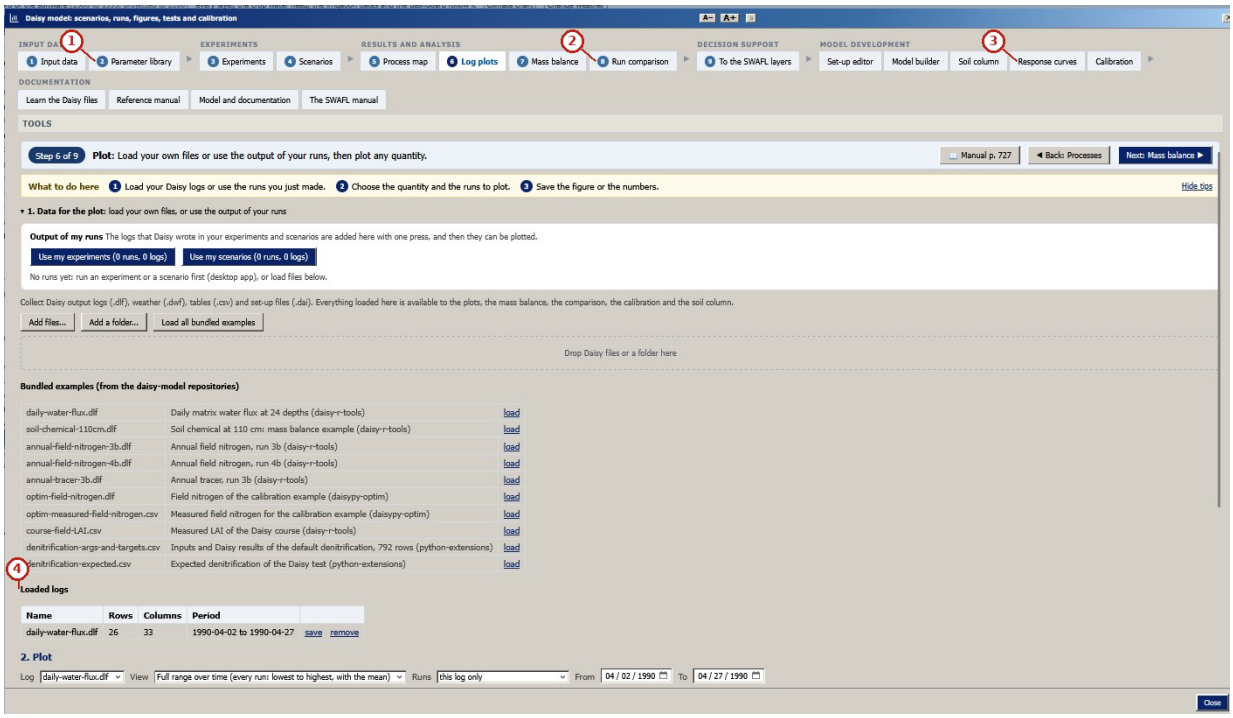
What the numbered labels point to (picture 2 of 8: Step 2: library of soils, crops and fertilisers (p. 583).).

1. "Parameter library": a heading or control of this screen; the caption above says which layer it belongs to.
2. "Leg plots": a heading or control of this screen; the caption above says which layer it belongs to.
3. "Hide tos": a heading or control of this screen; the caption above says which layer it belongs to.
4. "alpha, n": a heading or control of this screen; the caption above says which layer it belongs to.



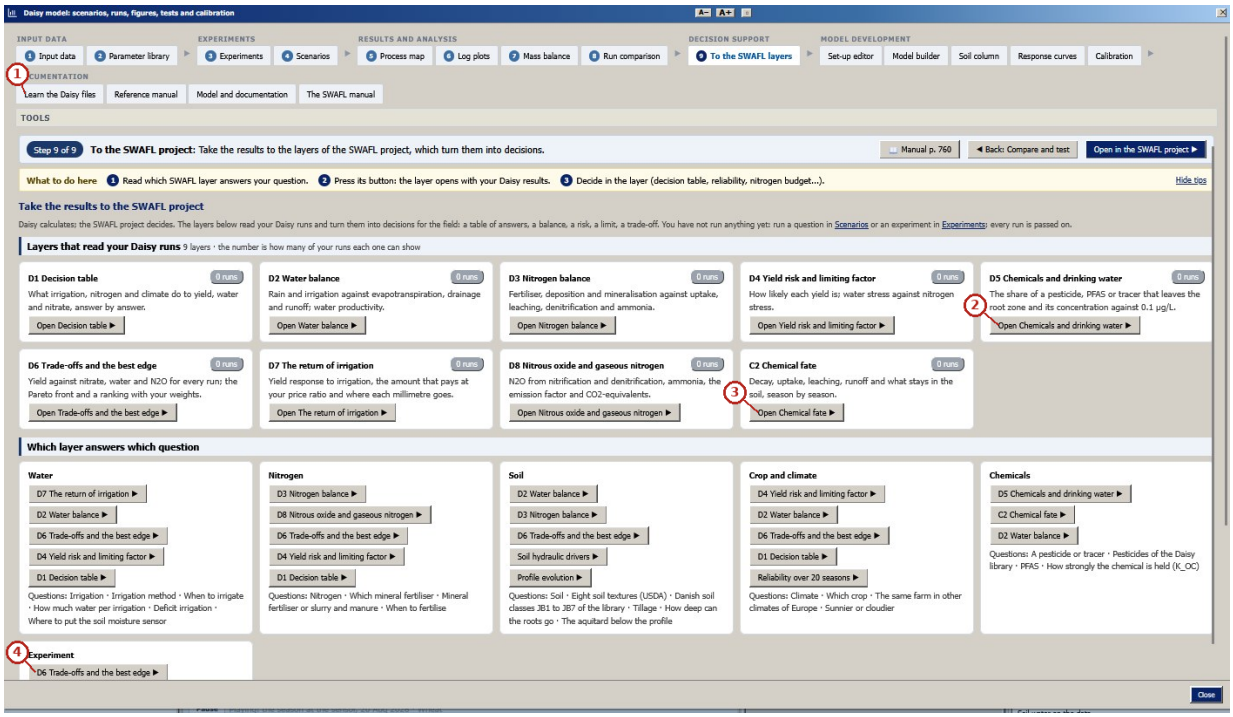
What the numbered labels point to (picture 3 of 8: Step 4: scenarios, a question and its runs (p. 762).).

1. "Log plots": a heading or control of this screen; the caption above says which layer it belongs to.
2. "SWAFL layers": the menu bar of the program (file, view, run, Daisy, help).
3. "water": a heading or control of this screen; the caption above says which layer it belongs to.
4. "Soil 6": the soil of the project, which every layer uses.



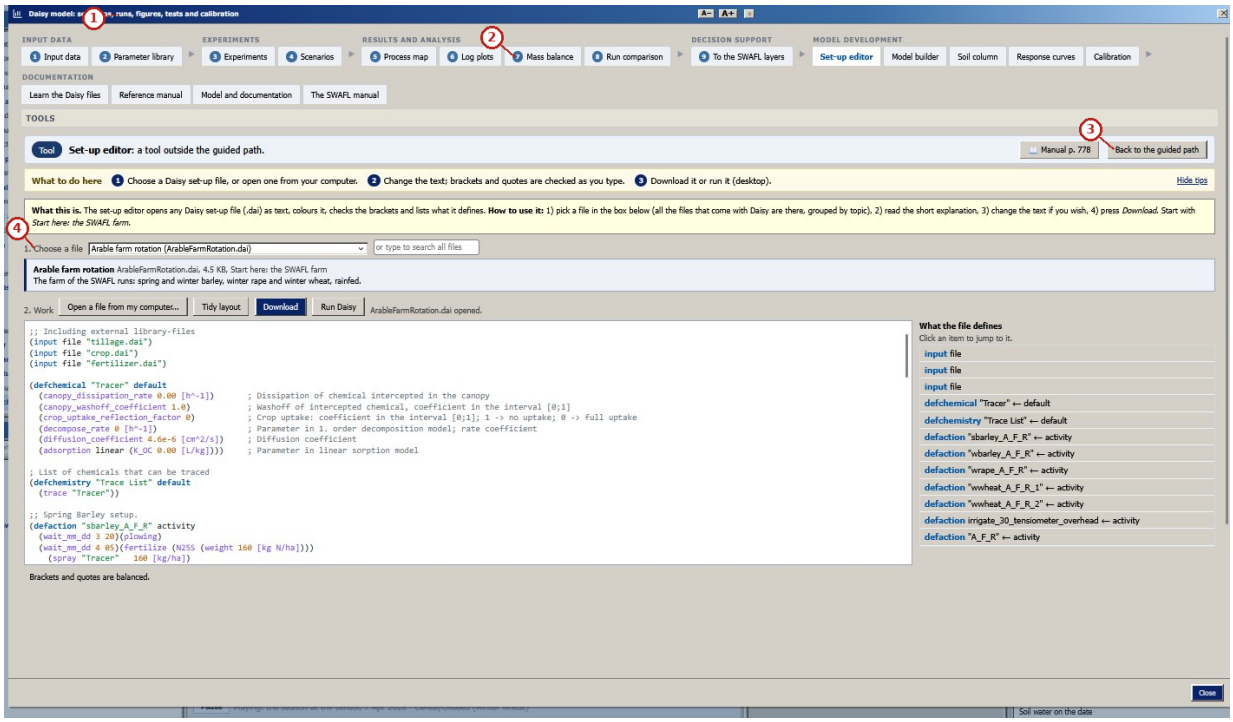
What the numbered labels point to (picture 4 of 8: Step 6: log plots of the runs (p. 763).).

1. "Parameter library": a heading or control of this screen; the caption above says which layer it belongs to.
2. "Run comparison": a heading or control of this screen; the caption above says which layer it belongs to.
3. "Response curves": a heading or control of this screen; the caption above says which layer it belongs to.
4. "Loaded logs": a heading or control of this screen; the caption above says which layer it belongs to.



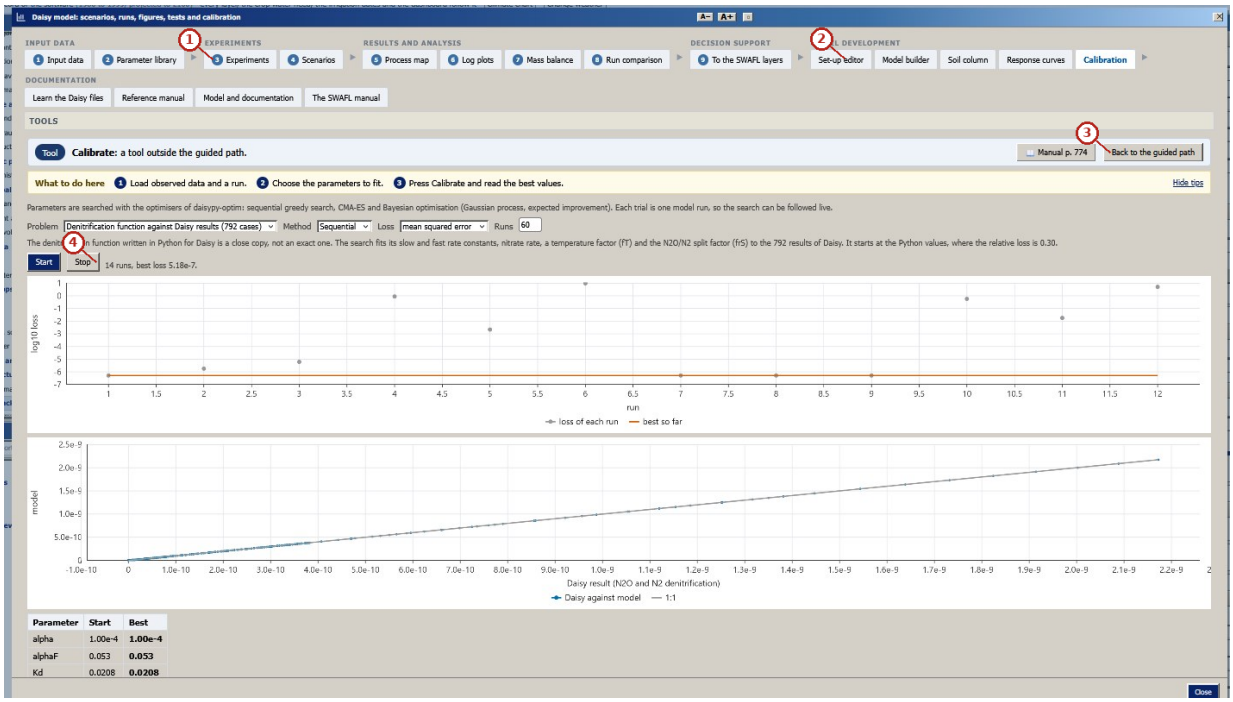
What the numbered labels point to (picture 5 of 8: Step 9: to the SWAFL layers, which layer answers which question (p. 764).).

1. "Learn the Daisy files": the Daisy model and its workbench.
2. "Open Chemicals and drinking": a heading or control of this screen; the caption above says which layer it belongs to.
3. "Open Chemical": a heading or control of this screen; the caption above says which layer it belongs to.
4. "D6 and the best edge": a heading or control of this screen; the caption above says which layer it belongs to.



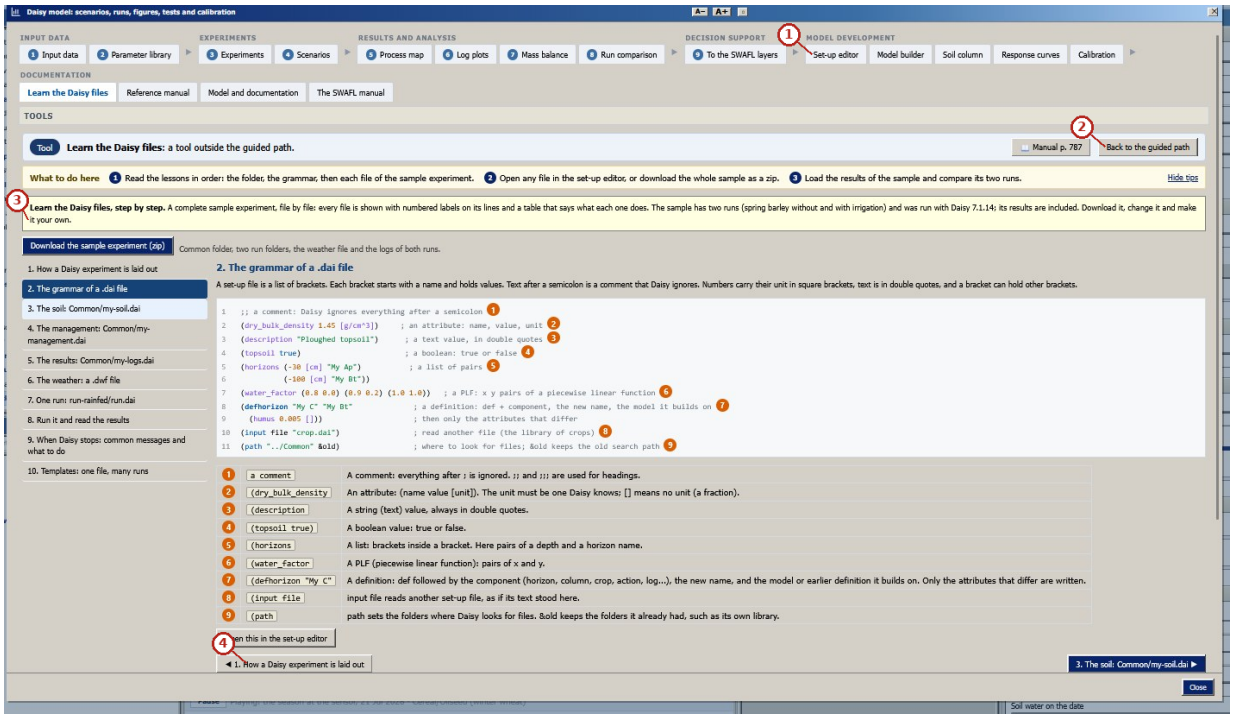
What the numbered labels point to (picture 6 of 8: Tool: set-up editor with bracket check (p. 783)).).

1. "runs, figures, tests and cal": a heading or control of this screen; the caption above says which layer it belongs to.
2. "Mass balance": a heading or control of this screen; the caption above says which layer it belongs to.
3. "Back to the guided path": moves to the next or previous step of the guided path.
4. "Choose file farm rotation": a heading or control of this screen; the caption above says which layer it belongs to.



What the numbered labels point to (picture 7 of 8: Tool: calibration (daisypy-optim) (p. 779).).

1. "Experiments Scenarios": a heading or control of this screen; the caption above says which layer it belongs to.
2. "editor builder": a heading or control of this screen; the caption above says which layer it belongs to.
3. "Back to the guided path": moves to the next or previous step of the guided path.
4. "14 runs, best loss": a heading or control of this screen; the caption above says which layer it belongs to.



What the numbered labels point to (picture 8 of 8: Tool: learn the Daisy files, the grammar of a .dai file (p. 792).).

1. "Setup editor": a heading or control of this screen; the caption above says which layer it belongs to.
2. "Back to the guided path": moves to the next or previous step of the guided path.
3. "it your own": a heading or control of this screen; the caption above says which layer it belongs to.
4. "How a Daisy out": the Daisy model and its workbench.

What it shows. The guided path of the Daisy workbench and three of its tools (real screenshots of release 1.1.0). Each step carries its step number, a one-line "what to do here", the manual page and Back and Next buttons. Panels: (a) Step 1: my data and weather (manual p. 760). (b) Step 2: library of soils, crops and fertilisers (p. 583). (c) Step 4: scenarios, a question and its runs (p. 762). (d) Step 6: log plots of the runs (p. 763). (e) Step 9: to the SWAFL layers, which layer answers which question (p. 764). (f) Tool: set-up editor with bracket check (p. 783). (g) Tool: calibration (daisypy-optim) (p. 779). (h) Tool: learn the Daisy files, the grammar of a .dai file (p. 792).

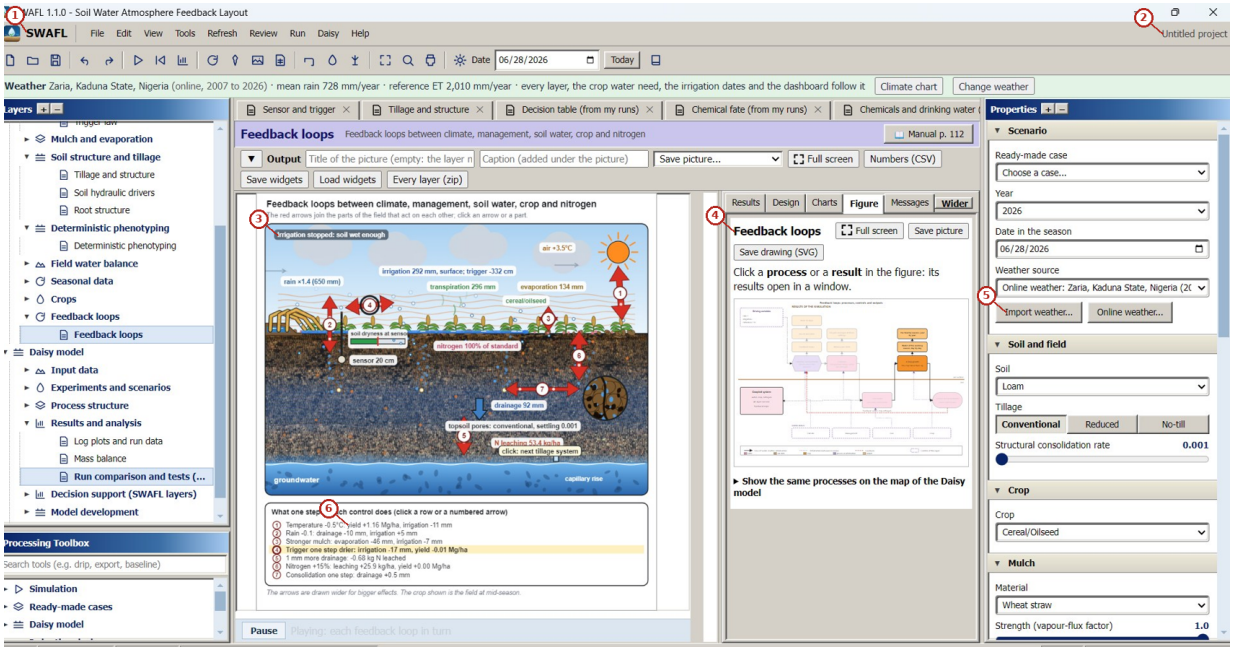
How to read the labels, symbols and colours. Eight screens of the guided path and tools, each with its manual page.

What affects it, and why it happens. The guided path orders the work from data to decision;

each step opens the manual page that documents it. The path leads a user from data to decision in nine steps.

How it is made. Screen capture of SWAFL Desktop 1.1.0 (not regenerated).

S8.13 Reading Figure 11 of the article



What the numbered labels point to.

1. "SWAFL Tools Run Daisy Help": the menu bar of the program (file, view, run, Daisy, help).
2. "Untitled project": a heading or control of this screen; the caption above says which layer it belongs to.
3. "Irrigation stopped: soil wet enough": the soil of the project, which every layer uses.
4. "Feedback loops Full screen": the feedback loops between climate, management, soil water, crop and nitrogen.
5. "Import weather... Online weather": the weather of the project: built-in record, a file of the user or online weather of any place.
6. "yield irrigation -11 mm": a heading or control of this screen; the caption above says which layer it belongs to.

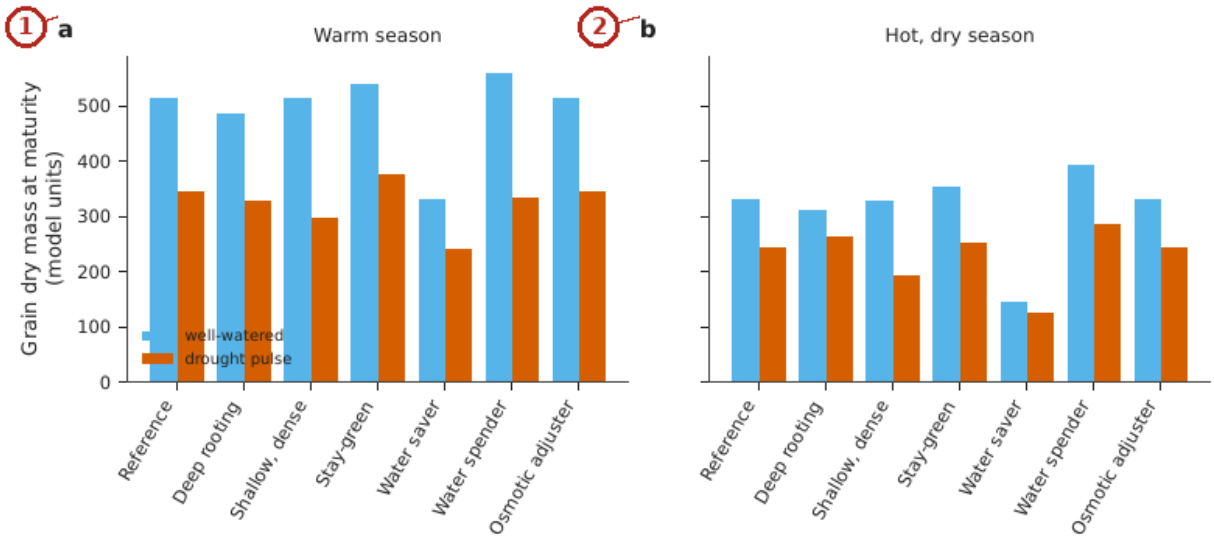
What it shows. The main window with the online weather of Zaria, Nigeria (2007–2026): Layers tree, the feedback-loops layer with the controls of the season, the Properties panel and the Processing toolbox (real screenshot).

How to read the labels, symbols and colours. Main window: Layers tree, feedback layer, Properties panel, toolbox.

What affects it, and why it happens. The Properties panel sets the soil, crop, mulch and weather that every layer uses, so changing it redraws the layer. Controls in the Properties panel drive every layer at once.

How it is made. Screen capture of SWAFL Desktop 1.1.0 (not regenerated).

S8.14 Reading Figure 21 of the article



What the numbered labels point to.

1. Panel (a): and a hot, dry. *Why:* Drought shortens grain filling, so grain mass falls; genotypes differ in how much.
2. Panel (b): season. *Why:* Heat and dryness add to the drought effect, so all genotypes lose more.

What it shows. Grain dry mass at maturity of seven genotypes, well-watered and under a drought pulse, in a warm (a) and a hot, dry (b) season.

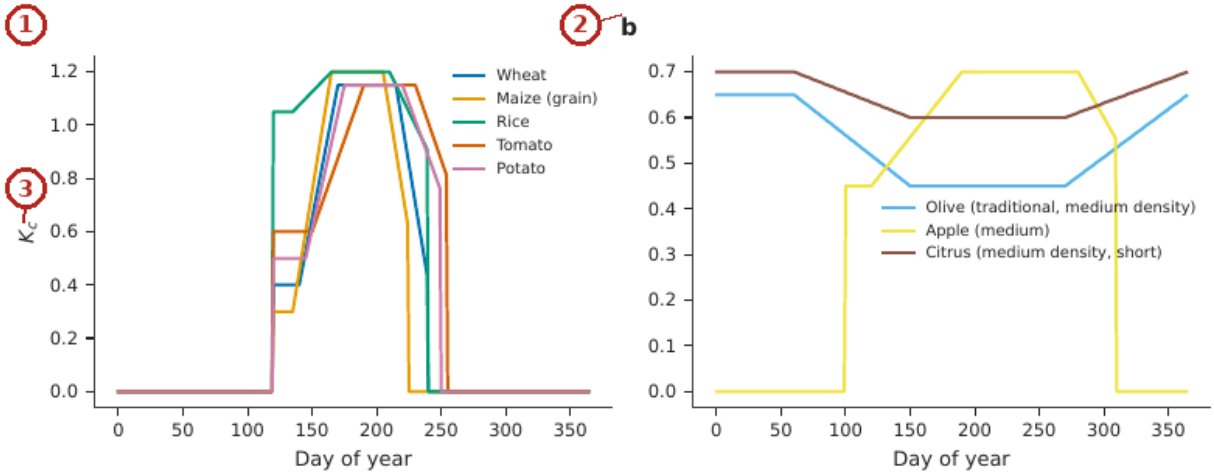
How to read the labels, symbols and colours. Grain mass of seven genotypes, well-watered against drought, in a warm (a) and a hot, dry (b) season.

What affects it, and why it happens. Drought and heat shorten the grain-filling period and reduce assimilation, so all genotypes lose grain, by different amounts.

How it is made. Figure drawn by the script from the input data. Script: `Manuscripts/repr`

oducibility/p456/p456_figs_data.py. Input: Data/daisy_ensemble.csv.gz.

S8.15 Reading Figure 22 of the article



What the numbered labels point to.

(1) Panel (a): five annual crops. *Why:* Coefficients rise with canopy cover, stay at the mid-season value and fall with senescence.

(2) Panel (b): three perennial crops (medium density). *Why:* Perennial crops keep a canopy most of the year, so the curves stay high longer.

(3) Panel (c): see the caption above.

What it shows. Daily crop coefficient curves of the library (Eq. (2) of the article) at the default sowing date, (a) five annual crops, (b) three perennial crops (medium density).

How to read the labels, symbols and colours. Daily crop coefficient curves of the library crops at the default sowing date.

What affects it, and why it happens. Coefficients rise with canopy development to a mid-season plateau and fall with senescence (FAO-56).

How it is made. Figure drawn by the script from the input data. Script: Manuscripts/repr oducibility/p456/p456_figs_data.py. Input: Data/daisy_ensemble.csv.gz.

S8.16 Derivation: Normalised spectral indices

Green leaves absorb red light for photosynthesis and scatter near-infrared light in the leaf mesophyll. The normalised difference NDVI $= (\rho_N - \rho_R)/(\rho_N + \rho_R)$ is a contrast of the two

that is bounded in $[-1, 1]$ and cancels a common multiplicative illumination factor k: $(k\rho_N - k\rho_R)/(k\rho_N + k\rho_R)$ equals NDVI. SAVI adds a soil term L to the denominator to reduce the influence of bright soil, $(1 + L)(\rho_N - \rho_R)/(\rho_N + \rho_R + L)$; EVI corrects for the atmosphere with the blue band and for the canopy background; the water indices use the shortwave-infrared band, which liquid water absorbs.

S8.17 Derivation: Crop coefficient through the season

The FAO-56 curve is piecewise linear in time: constant K_{ini} during the initial stage L_1, a linear rise to K_{mid} over the development stage L_2, constant over the mid-season L_3 and a linear fall to K_{end} over the late stage L_4 (Allen et al., 1998). The crop evapotranspiration is $ET_c = K_c\, ET_0$.

S8.18 Derivation: Daily root-zone balance of each map cell

For each cell, the root-zone depletion follows $D_{r,i} = D_{r,i-1} - (P - \mathrm{RO})_i - I_i + ET_{c,i} + \mathrm{DP}_i$, bounded by $0 \le D_r \le TAW$; deep percolation occurs only when the depletion would fall below zero. Irrigation is required when D_r exceeds the readily available water $p\,TAW$; the net depth to apply is D_r (Allen et al., 1998).

S8.19 Derivation: Terrain, flow and wetness

On a filled elevation grid each cell drains to the steepest of its eight neighbours (D8). The upslope area A of a cell is the sum of the areas that drain through it, and the specific area is $a = A/w$ with w the cell width. The topographic wetness index $\ln(a/\tan\beta)$ is large where much water arrives and the slope β is small, which marks where the soil stays wet.

S8.20 Derivation: Soil loss

The revised universal soil loss equation multiplies the rain erosivity R, the soil erodibility K, the slope length and steepness LS, the cover C and the practice P: $A = R\,K\,LS\,C\,P$ in $\mathrm{t\,ha^{-1}\,yr^{-1}}$ (Renard and Freimund, 1994).

S9. Every term of the author's equations

Rule of origin, show(v) $\iff$ **source**(v) $\in$ **Daisy runs**

Term	What it means	Unit	Where the value comes from	When it rises
v	a value drawn in a Daisy layer	–	bridge	–
source(v)	log file and line the value was read from	–	bridge	the value is drawn or not

Run budget, $N = \min(\prod n_i, B)$

Term	What it means	Unit	Where the value comes from	When it rises
n_i	number of levels of factor i	–	design	more combinations
B	budget of runs set by the user	runs	user	more runs, up to the factorial
N	runs made	runs	computed	–

Soil-hydraulic factors

Term	What it means	Unit	Where the value comes from	When it rises
f_k	multiplier of K_s of every horizon	–	experiment design	faster drainage
f_a	multiplier of α	–	experiment design	earlier air entry
f_n	multiplier of $n-1$	–	experiment design	more uniform pores
f_t, $\delta\ell$	multiplier of θ_s, shift of ℓ	–	experiment design	more storage; faster conductivity decline

S10. How every constant is obtained, what it depends on, and how sensitive the results are

S10.1 How every constant of the article is obtained

Constant	Value	How it was obtained	What it depends on
EVI coefficients	2.5, 6, 7.5, 1	standard values of the enhanced vegetation index (Huete et al., 2002)	the sensor; fixed for Sentinel-2
SAVI soil factor L	0.5	standard value for intermediate cover (Huete, 1988)	the cover
Crop coefficients and stage lengths	per crop	FAO-56 tables and a review	the crop, the climate and the sowing date
RUSLE factors R, K, LS, C, P	per cell	R from monthly rainfall (Renard and Freimund, 1994), K from SoilGrids texture, LS from the elevation model, C from NDVI, $P = 1$	the data of the field
Cloud limit of a scene	30 %	chosen: scenes with more cloud over the field are not used	the season
Run budget B	set by the user	–	time available
Scaling factors f_k, f_a, f_n, f_t, $\delta\ell$	set by the user	levels of the experiment	the question asked
N_2:N_2O ratio	from the Daisy function	nitrate, carbon-dioxide and water factors of Daisy	the soil state

S11. How each table of the article is produced

Table 1 of the article. The 22 decision layers of release 1.1.0 by group, and the two layers of Daisy output.

How it is made. Descriptive table (definitions, design, parameters or tests); no computation.

Table 2 of the article. Questions of the workbench and the layers that answer them.

How it is made. Descriptive table (definitions, design, parameters or tests); no computation.

Table 3 of the article. Soils used for the test of the soil column (van Genuchten parameters, $\ell = 0.5$).

How it is made. Table typed from the result file (13 of its numbers found there). Script: `Manuscripts/reproducibility/p3_js/run_drip.js`. Input: `Data/daisy_ensemble.csv.gz`.

Table 4 of the article. The 17 experiments of the study and their prepared designs in the workbench. Hours are the sum of the Daisy wall-clock run times of the study (`run_seconds`) and the estimate for the compact design (runs × the median run time of the experiment). The experiments marked − are not offered as designs; their published runs are bundled.

How it is made. Table typed in the manuscript from the result files of the script. Script: `p4_design_sizes.js`. Input: `Data/daisy_ensemble.csv.gz`.

Table 5 of the article. The Richards solver of the workbench against an independent reference (method of lines, scipy BDF, 0.5 cm nodes). A 100 cm column, initial suction 300 cm, 200 mm of rain on day 1, five dry days, free drainage. D is the drainage below 1 m and S the final storage in mm, for the reference (ref) and the workbench (wb, 0.5 cm nodes); the spread (mm) is the range over node spacings of 4, 2, 1 and 0.5 cm.

How it is made. Table typed from the result file (5 of its numbers found there). Script: `Manuscripts/reproducibility/run_all.sh`. Input: `Data/daisy_ensemble.csv.gz`.

Table 6 of the article. Test questions of the planner and the factors that the offline reader returned (all of the expected factors, station and soil were present in every case).

How it is made. Descriptive table (definitions, design, parameters or tests); no computation.

Table 7 of the article. Mean seasonal nitrogen budget of the two bundled annual nitrogen logs as reported by the bridge (kg N ha^{-1}, ten seasons).

How it is made. Table typed in the manuscript from the result files of the script. Script: `(seethereproducibilityappendixofthepaper)`. Input: `Data/daisy_ensemble.csv.gz`.

Table 8 of the article. The three routes from Daisy to a decision.

How it is made. Descriptive table (definitions, design, parameters or tests); no computation.

Table 9 of the article. Deterministic phenotyping layer: maize in a warm and in a hot, dry season, seven genotypes, well-watered bench against a drought pulse (onset day 40, 28 days, 30 % of the plant-available water, rain-out shelter). Grain dry mass in model units; the penalty is the loss by the drought.

How it is made. Table typed in the manuscript from the result files of the script. Script: `(seethereproducibilityappendixofthepaper)`. Input: `Data/daisy_ensemble.csv.gz`.

Table 10 of the article. What this article adds, item by item: what was established before, what is new here, and the evidence in this article.

How it is made. Descriptive table (definitions, design, parameters or tests); no computation.

S12. From the data to every result: experiments, runs and computations

Where every number starts. All simulated numbers come from one set of 10 190 Daisy runs, organised in seventeen experiments (E0 to E16), each of which varies a few factors on the same ten soils (S01 to S10) under the same weather series (Taastrup, Denmark). Every run writes 20-year mean annual fluxes (rain, irrigation, evapotranspiration and its parts, drainage at 1 m, yield, nitrogen); one row per run is stored in `Data/Comprehensive_Summary/00_summary_all_simulations.csv.gz` (sheet `All_runs_summary` of the workbook) and, for the ensemble analyses, in `Data/daisy_ensemble.csv.gz` (sheet `Ensemble_runs`). A run is used only when its status is `ok` (10 041 runs); the 149 runs whose solver failed are kept in the files but never enter a result. The workbook repeats the per-run algebra as live Excel formulas (the green columns at the end of `All_runs_summary`: other evaporation, balance residual, differences from the rainfed run of the same soil, water productivity), and the sheet `Column_Equations` lists the equation behind every calculated column. The rainfed run of each soil (experiment E0) is the reference against which every irrigated or treated run is compared (sheet `Baselines_E0`). This article is mostly about software; its numbers come from checks of the software against references rather than from a statistical analysis of the runs.

Result	Data	How it is computed	Script, file, workbook
Rule of origin	every Daisy layer of the suite	the application is opened without runs and every layer is checked to show no number	`p5_empty_state.js`
Run budget	any design	$N = \min(\prod_i n_i, B)$, Latin hypercube	live formula `EQ_P3`!B32; chart `CH_P3`
Run time of a Daisy run	the 10 190 runs of the ensemble	distribution of `run_seconds`	`p4_design_fidelity.py` → `out_p4_runtime.csv`

Result	Data	How it is computed	Script, file, workbook
Failed runs and their causes	the 149 runs with a solver failure	Daisy messages matched to four classes	the diagnosis layer; column `status` of `All_runs_summary`
Hydraulic factors for every horizon	one loam run with Daisy 7.1.14, with and without the factors	percolation of the season (177 against 222 mm)	the column tool of the suite
Denitrification split	792 stored cases of Daisy's tools	the function evaluated on the stored inputs and compared case by case with the stored outputs	`p5_bridge_checks.js` → `out_p5_bridge_js.json`
Crop coefficient, water balance and soil loss of a map cell	satellite indices and the field's soil	FAO-56 piecewise K_c, daily root-zone balance, RUSLE	live formulas in `EQ_P3`; chart `CH_P3`
Surface irrigation against the textbook	Example 6.1 of the reference	the shipped solver run with the book's inputs; $\lvert\text{SWAFL} - \text{book}\rvert/\text{book}$	sheet `R_p3_ws61` (live `err_pct`)

S13. The data

Table S7 lists the 17 experiments of the data folder and Fig. S1 shows their outputs. The tables that follow are the result files of the analyses of the paper, printed in full or up to 220 rows; each is also in the workbook `Data/Manuscript_Data/SWAFL_Manuscripts_Data.xlsx`, and every run is in `Data/Comprehensive_Summary`.

Table S7: The data of this study: the 17 Daisy experiments of `Data/Comprehensive_Summary` (10190 runs, 10041 valid; 20 years, 1980–1999, of Taastrup weather). Means of the valid runs; Y yield (Mg ha^{-1}), I irrigation, D drainage at 1 m and ET (mm yr^{-1}), N_L nitrate leached (kg N ha^{-1} yr^{-1}), t run time of Daisy (s). Every row is one file of the folder, linked.

Exp.	File	Runs	Valid	Factors varied	Y	I	D	ET	N_L	t
E0	`Baselines`	10	10	soil	2.19	0	55	418	158	231
E1	`Doses`	200	189	soil, rate, dose	5.13	198	98	564	79	195
E2	`TriggerPressure`	200	197	soil, trigger	4.70	169	80	559	90	139
E3	`SensorDepth`	200	200	soil, sensor depth	4.94	199	86	583	86	146
E4	`Methods`	100	96	soil, method	5.01	164	88	542	83	147
E5	`SHP Dynamics`	300	300	soil, method, tillage	4.13	152	43	579	84	189
E6	`Mediterranean Pollution`	210	193	soil, method, trigger, N rate	3.64	84	239	620	62	118
E7	`Mulching`	200	200	soil, mulch	5.29	183	96	555	74	161
E8	`Rain Intensity Moisture`	1000	1000	initial suction, rain scale	0.68	0	-1	282	-0	6
E9	`Soil Moisture Rain`	1000	987	soil, initial suction, rain scale	0.75	0	19	247	33	7
E10	`Amount Timing Soil`	1000	954	soil, rate, dose, window	4.42	139	76	532	97	146
E11	`Temp Evap Rain`	1000	1000	rain scale, warming, demand	3.81	324	18	754	66	151
E12	`Sensor Trigger Rain`	1000	1000	trigger, sensor depth, rain scale	5.88	260	35	670	32	155
E13	`N Water Soil`	1000	975	soil, method, rate, dose, trigger, N rate	4.00	162	83	547	79	207
E14	`Mulch Rain`	1000	1000	rain scale, mulch	6.15	235	43	634	25	173
E15	`Method Depth Amount`	1000	1000	method, rate, dose, root depth	6.02	217	61	619	37	157
E16	`Edge Cases`	770	740	soil, method, rate, dose, trigger, sensor depth, rain scale, warming, N rate, demand	2.17	344	185	600	93	224

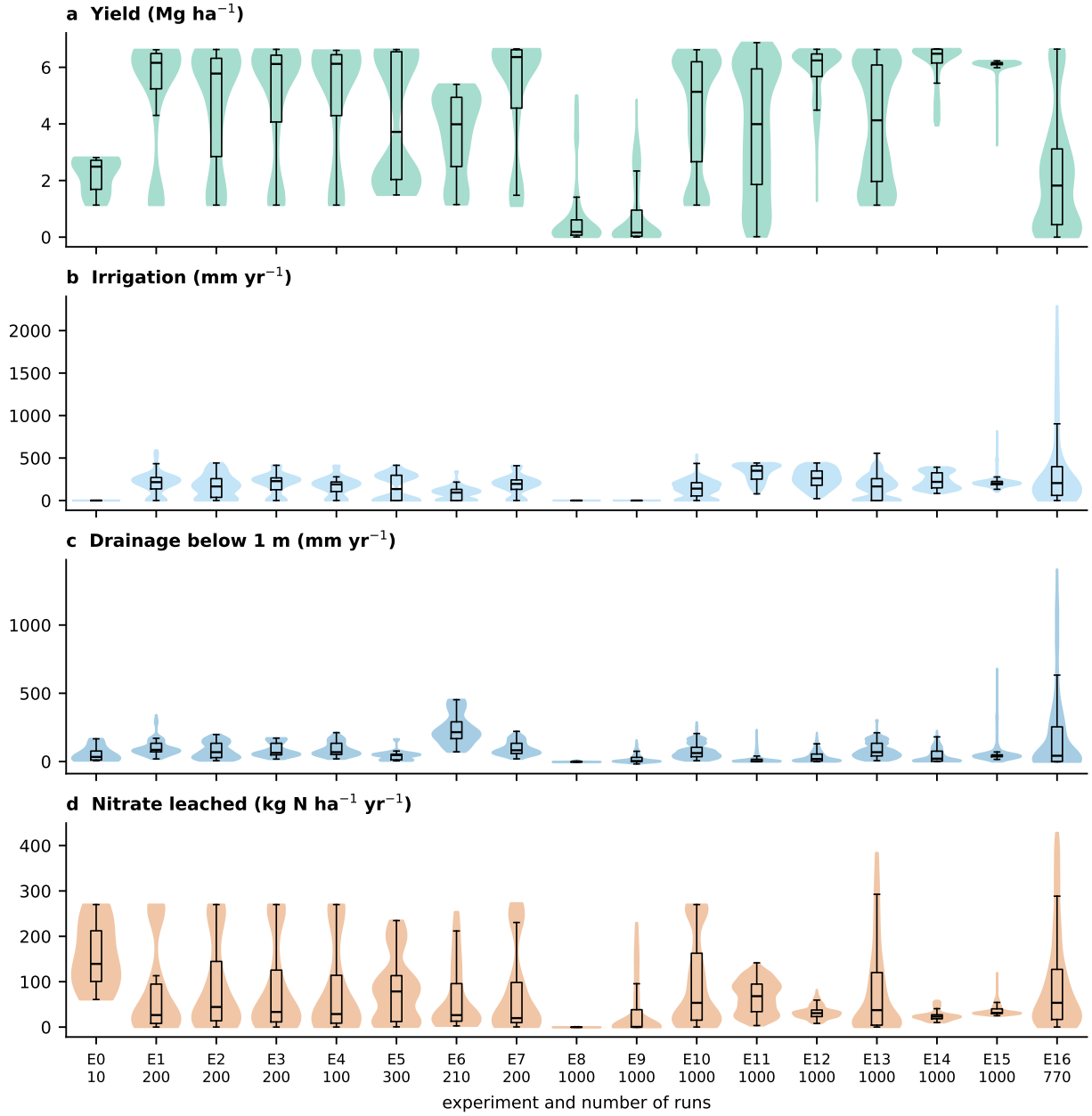


Figure S1: Yield, irrigation, drainage and nitrate leaching of the valid runs, by experiment.

out_p4_calibration.json

Source: `Manuscripts/reproducibility/p456/out_p4_calibration.json`; 140 values.

quantity	value
truth.alphaF	0.05303
truth.Kd	0.02083
truth.alpha	0.0001
loss0	1.409e-31
start	0.204
runs[0].method	sequential
runs[0].seed	1

quantity	value
runs[0].evals	16
runs[0].loss	0.0006235
runs[0].x.alphaF	0.05
runs[0].x.Kd	0.06
runs[0].x.alpha	0.0003
runs[0].ms	20
runs[0].err	0.02553, 0.4594, 0.4771
runs[1].method	sequential
runs[1].seed	2
runs[1].evals	16
runs[1].loss	0.0006235
runs[1].x.alphaF	0.05
runs[1].x.Kd	0.06
runs[1].x.alpha	0.0003
runs[1].ms	4
runs[1].err	0.02553, 0.4594, 0.4771
runs[2].method	sequential
runs[2].seed	3
runs[2].evals	16
runs[2].loss	0.0006235
runs[2].x.alphaF	0.05
runs[2].x.Kd	0.06
runs[2].x.alpha	0.0003
runs[2].ms	4
runs[2].err	0.02553, 0.4594, 0.4771
runs[3].method	sequential
runs[3].seed	4
runs[3].evals	16
runs[3].loss	0.0006235
runs[3].x.alphaF	0.05
runs[3].x.Kd	0.06
runs[3].x.alpha	0.0003
runs[3].ms	4
runs[3].err	0.02553, 0.4594, 0.4771
runs[4].method	sequential
runs[4].seed	5
runs[4].evals	16
runs[4].loss	0.0006235
runs[4].x.alphaF	0.05
runs[4].x.Kd	0.06
runs[4].x.alpha	0.0003
runs[4].ms	4
runs[4].err	0.02553, 0.4594, 0.4771
runs[5].method	cma
runs[5].seed	1
runs[5].evals	154
runs[5].loss	4.312e-06
runs[5].x.alphaF	0.0533

quantity	value
runs[5].x.Kd	0.004069
runs[5].x.alpha	2.777e-05
runs[5].ms	39
runs[5].err	0.002251, 0.7093, 0.5564
runs[6].method	cma
runs[6].seed	2
runs[6].evals	154
runs[6].loss	4.144e-06
runs[6].x.alphaF	0.05313
runs[6].x.Kd	0.006077
runs[6].x.alpha	0.000431
runs[6].ms	36
runs[6].err	0.0008651, 0.535, 0.6345
runs[7].method	cma
runs[7].seed	3
runs[7].evals	154
runs[7].loss	2.003e-06
runs[7].x.alphaF	0.05286
runs[7].x.Kd	0.05183
runs[7].x.alpha	9.474e-05
runs[7].ms	38
runs[7].err	0.001402, 0.3958, 0.02349
runs[8].method	cma
runs[8].seed	4
runs[8].evals	154
runs[8].loss	6.869e-06
runs[8].x.alphaF	0.05337
runs[8].x.Kd	0.02493
runs[8].x.alpha	2.533e-05
runs[8].ms	36
runs[8].err	0.002804, 0.07801, 0.5964
runs[9].method	cma
runs[9].seed	5
runs[9].evals	154
runs[9].loss	9.754e-09
runs[9].x.alphaF	0.05301
runs[9].x.Kd	0.003502
runs[9].x.alpha	0.0001228
runs[9].ms	37
runs[9].err	0.0001234, 0.7744, 0.08912
runs[10].method	bayes
runs[10].seed	1
runs[10].evals	150
runs[10].loss	6.888e-08
runs[10].x.alphaF	0.05305
runs[10].x.Kd	0.04165
runs[10].x.alpha	2.105e-05
runs[10].ms	1779

quantity	value
runs[10].err	0.000177, 0.3008, 0.6768
runs[11].method	bayes
runs[11].seed	2
runs[11].evals	150
runs[11].loss	7.706e-08
runs[11].x.alphaF	0.05307
runs[11].x.Kd	0.09905
runs[11].x.alpha	6.412e-05
runs[11].ms	1671
runs[11].err	0.0003446, 0.6771, 0.193
runs[12].method	bayes
runs[12].seed	3
runs[12].evals	150
runs[12].loss	8.24e-08
runs[12].x.alphaF	0.05306
runs[12].x.Kd	0.009166
runs[12].x.alpha	1.795e-05
runs[12].ms	1638
runs[12].err	0.0003083, 0.3566, 0.746
runs[13].method	bayes
runs[13].seed	4
runs[13].evals	150
runs[13].loss	1.321e-07
runs[13].x.alphaF	0.05302
runs[13].x.Kd	0.003837
runs[13].x.alpha	2.652e-05
runs[13].ms	1586
runs[13].err	7.897e-05, 0.7348, 0.5764
runs[14].method	bayes
runs[14].seed	5
runs[14].evals	150
runs[14].loss	2.571e-07
runs[14].x.alphaF	0.053
runs[14].x.Kd	0.05913
runs[14].x.alpha	1.386e-05
runs[14].ms	1666
runs[14].err	0.000195, 0.453, 0.8582

out_p4_design.json

Source: `Manuscripts/reproducibility/p456/out_p4_design.json`; 96 values.

quantity	value
[0].id	E0
[0].name	Baselines
[0].gui	9

quantity	value
[0].study	10
[0].guiFactorCombos	9
[0].ensCombos	0
[1].id	E1
[1].name	Dose and duration
[1].gui	17
[1].study	200
[1].guiFactorCombos	16
[1].ensCombos	1
[2].id	E2
[2].name	Trigger pressure
[2].gui	21
[2].study	200
[2].guiFactorCombos	18
[2].ensCombos	200
[3].id	E3
[3].name	Sensor depth
[3].gui	21
[3].study	200
[3].guiFactorCombos	18
[3].ensCombos	200
[4].id	E4
[4].name	Methods
[4].gui	6
[4].study	100
[4].guiFactorCombos	4
[4].ensCombos	10
[5].id	E5
[5].name	Soil structure dynamics (SHP)
[5].gui	72
[5].study	300
[5].guiFactorCombos	72
[5].ensCombos	10
[6].id	E6
[6].name	Mediterranean pollution
[6].gui	30
[6].study	210
[6].guiFactorCombos	30
[6].ensCombos	10
[7].id	E7
[7].name	Mulching
[7].unsupported	True
[7].study	200
[8].id	E8
[8].name	Rain intensity and moisture
[8].unsupported	True
[8].study	1000
[9].id	E9

quantity	value
[9].name	Soil moisture and rain
[9].gui	40
[9].study	1000
[9].guiFactorCombos	32
[9].ensCombos	1000
[10].id	E10
[10].name	Amount, timing and soil
[10].gui	20
[10].study	1000
[10].guiFactorCombos	18
[10].ensCombos	100
[11].id	E11
[11].name	Temperature, evaporation, rain
[11].gui	16
[11].study	1000
[11].guiFactorCombos	16
[11].ensCombos	100
[12].id	E12
[12].name	Sensor, trigger, rain
[12].gui	30
[12].study	1000
[12].guiFactorCombos	27
[12].ensCombos	1000
[13].id	E13
[13].name	N, water, soil
[13].gui	32
[13].study	1000
[13].guiFactorCombos	24
[13].ensCombos	10
[14].id	E14
[14].name	Mulch and rain
[14].unsupported	True
[14].study	1000
[15].id	E15
[15].name	Method, depth, amount
[15].gui	30
[15].study	1000
[15].guiFactorCombos	27
[15].ensCombos	100
[16].id	E16
[16].name	Edge cases
[16].gui	24
[16].study	770
[16].guiFactorCombos	16
[16].ensCombos	5

out_p4_fidelity.csv

Source: `Manuscripts/reproducibility/p456/out_p4_fidelity.csv`; 51 rows. Workbook sheet (`CSV file`).

exp	factor	resp	levels_study	levels_compact	rmse_pct	max_pct
E2	trig	yield_Mg_ha	20	6	7.317	26.9
E2	trig	irrigation_mm	20	6	6.443	22.11
E2	trig	drainage_1m_mm	20	6	8.957	36.94
E2	trig	IE_kg_m3	20	6	8.133	28.55
E2	trig	N_leached_kg_ha	20	6	7.183	23.64
E3	z	yield_Mg_ha	20	6	4.638	11.63
E3	z	irrigation_mm	20	6	11.23	24.55
E3	z	drainage_1m_mm	20	6	4.563	9.679
E3	z	IE_kg_m3	20	6	6.511	13.93
E3	z	N_leached_kg_ha	20	6	4.849	13
E9	trig	yield_Mg_ha	10	4	11.68	33.78
E9	trig	drainage_1m_mm	10	4	26.19	79.45
E9	trig	N_leached_kg_ha	10	4	16.2	49.6
E9	P	yield_Mg_ha	10	4	10.03	27.69
E9	P	drainage_1m_mm	10	4	12.32	35
E9	P	N_leached_kg_ha	10	4	7.935	20.58
E10	amt_mm	yield_Mg_ha	10	3	32.48	96.29
E10	amt_mm	irrigation_mm	10	3	29.85	88.58
E10	amt_mm	drainage_1m_mm	10	3	36.08	88.19
E10	amt_mm	IE_kg_m3	10	3	27.91	85.16
E10	amt_mm	N_leached_kg_ha	10	3	30.38	90.76
E11	dT	yield_Mg_ha	10	4	8.061	19.75
E11	dT	irrigation_mm	10	4	11.16	29.98
E11	dT	drainage_1m_mm	10	4	14.95	42.6
E11	dT	IE_kg_m3	10	4	10.98	30.61
E11	dT	N_leached_kg_ha	10	4	13.5	34.91
E11	P	yield_Mg_ha	10	4	6.598	14.63
E11	P	irrigation_mm	10	4	9.053	24.73
E11	P	drainage_1m_mm	10	4	16.9	48.6
E11	P	IE_kg_m3	10	4	10.36	28.78
E11	P	N_leached_kg_ha	10	4	9.729	21.93
E12	z	yield_Mg_ha	10	3	11.16	23.65
E12	z	irrigation_mm	10	3	13.24	24.2
E12	z	drainage_1m_mm	10	3	27.73	65.95
E12	z	IE_kg_m3	10	3	21.02	50.32
E12	z	N_leached_kg_ha	10	3	29.31	64.31
E12	trig	yield_Mg_ha	10	3	15.58	42.43
E12	trig	irrigation_mm	10	3	13.2	28.82
E12	trig	drainage_1m_mm	10	3	17.55	52.43
E12	trig	IE_kg_m3	10	3	18.63	54.2
E12	trig	N_leached_kg_ha	10	3	16.13	44.94
E12	P	yield_Mg_ha	10	3	21.89	64.7

exp	factor	resp	levels_study	levels_compact	rmse_pct	max_pct
E12	P	irrigation_mm	10	3	6.649	16.43
E12	P	drainage_1m_mm	10	3	8.067	23.93
E12	P	IE_kg_m3	10	3	10.71	31.48
E12	P	N_leached_kg_ha	10	3	30.16	89.24
E15	root	yield_Mg_ha	10	3	18.15	56.06
E15	root	irrigation_mm	10	3	31.64	100
E15	root	drainage_1m_mm	10	3	27.96	88.29
E15	root	IE_kg_m3	10	3	15.45	47.71
E15	root	N_leached_kg_ha	10	3	14.68	44.85

out_p4_richards_js.json

Source: `Manuscripts/reproducibility/p456/out_p4_richards_js.json`; 153 values.

quantity	value
SOILS.sand.alpha	0.145
SOILS.sand.n	2.68
SOILS.sand.Ks	29.7
SOILS.sand.Ths	0.43
SOILS.sand.Thr	0.045
SOILS.sand.l	0.5
SOILS.loam.alpha	0.036
SOILS.loam.n	1.56
SOILS.loam.Ks	1.04
SOILS.loam.Ths	0.43
SOILS.loam.Thr	0.078
SOILS.loam.l	0.5
SOILS.siltclayloam.alpha	0.01
SOILS.siltclayloam.n	1.23
SOILS.siltclayloam.Ks	0.07
SOILS.siltclayloam.Ths	0.5
SOILS.siltclayloam.Thr	0.089
SOILS.siltclayloam.l	0.5
init.sand	45.68
init.loam	170.1
init.siltclayloam	394.8
out.sand.1.infil	200
out.sand.1.drain	152.1
out.sand.1.runoff	0
out.sand.1.store0	None
out.sand.1.storeEnd	93.56
out.sand.1.maxErr	0
out.sand.1.sumErr	0
out.sand.1.iters	9964
out.sand.1.ms	1586
out.sand.1.infilD	200, 0, 0, 0, 0, 0

quantity	value
out.sand.1.drainD	18.79, 96.26, 18.37, 8.962, 5.692, 4.046
out.sand.1.store	226.9, 130.6, 112.3, 103.3, 97.6, 93.56
out.sand.2.infil	200
out.sand.2.drain	152.1
out.sand.2.runoff	0
out.sand.2.store0	None
out.sand.2.storeEnd	93.56
out.sand.2.maxErr	0
out.sand.2.sumErr	0
out.sand.2.iters	6306
out.sand.2.ms	654
out.sand.2.infilD	200, 0, 0, 0, 0, 0
out.sand.2.drainD	18.79, 96.23, 18.37, 8.982, 5.7, 4.051
out.sand.2.store	226.9, 130.7, 112.3, 103.3, 97.61, 93.56
out.sand.4.infil	200
out.sand.4.drain	152.2
out.sand.4.runoff	0
out.sand.4.store0	None
out.sand.4.storeEnd	93.53
out.sand.4.maxErr	0
out.sand.4.sumErr	0
out.sand.4.iters	4192
out.sand.4.ms	298
out.sand.4.infilD	200, 0, 0, 0, 0, 0
out.sand.4.drainD	18.79, 96.16, 18.45, 9.001, 5.703, 4.054
out.sand.4.store	226.9, 130.7, 112.3, 103.3, 97.58, 93.53
out.sand.0.5.infil	200
out.sand.0.5.drain	152.1
out.sand.0.5.runoff	0
out.sand.0.5.store0	None
out.sand.0.5.storeEnd	93.54
out.sand.0.5.maxErr	0
out.sand.0.5.sumErr	0
out.sand.0.5.iters	15784
out.sand.0.5.ms	4958
out.sand.0.5.infilD	200, 0, 0, 0, 0, 0
out.sand.0.5.drainD	18.79, 96.27, 18.37, 8.99, 5.676, 4.047
out.sand.0.5.store	226.9, 130.6, 112.2, 103.3, 97.59, 93.54
out.loam.1.infil	200
out.loam.1.drain	54.59
out.loam.1.runoff	0
out.loam.1.store0	None
out.loam.1.storeEnd	315.5
out.loam.1.maxErr	0
out.loam.1.sumErr	0
out.loam.1.iters	10486
out.loam.1.ms	1599
out.loam.1.infilD	200, 0, 0, 0, 0, 0

quantity	value
out.loam.1.drainD	0.009, 5.772, 19.08, 12.88, 9.458, 7.396
out.loam.1.store	370.1, 364.3, 345.2, 332.3, 322.9, 315.5
out.loam.2.infil	200
out.loam.2.drain	54.63
out.loam.2.runoff	0
out.loam.2.store0	None
out.loam.2.storeEnd	315.4
out.loam.2.maxErr	0
out.loam.2.sumErr	0
out.loam.2.iters	8549
out.loam.2.ms	834
out.loam.2.infilD	200, 0, 0, 0, 0, 0
out.loam.2.drainD	0.009, 5.784, 19.04, 12.94, 9.454, 7.397
out.loam.2.store	370.1, 364.3, 345.2, 332.3, 322.8, 315.4
out.loam.4.infil	200
out.loam.4.drain	54.58
out.loam.4.runoff	0
out.loam.4.store0	None
out.loam.4.storeEnd	315.5
out.loam.4.maxErr	0
out.loam.4.sumErr	0
out.loam.4.iters	7345
out.loam.4.ms	477
out.loam.4.infilD	200, 0, 0, 0, 0, 0
out.loam.4.drainD	0.009, 5.825, 18.99, 12.88, 9.466, 7.402
out.loam.4.store	370.1, 364.2, 345.2, 332.4, 322.9, 315.5
out.loam.0.5.infil	200
out.loam.0.5.drain	54.62
out.loam.0.5.runoff	0
out.loam.0.5.store0	None
out.loam.0.5.storeEnd	315.4
out.loam.0.5.maxErr	0
out.loam.0.5.sumErr	0
out.loam.0.5.iters	14781
out.loam.0.5.ms	4257
out.loam.0.5.infilD	200, 0, 0, 0, 0, 0
out.loam.0.5.drainD	0.009, 5.773, 19.09, 12.9, 9.463, 7.389
out.loam.0.5.store	370.1, 364.3, 345.2, 332.3, 322.8, 315.4
out.siltclayloam.1.infil	21.77
out.siltclayloam.1.drain	0.156
out.siltclayloam.1.runoff	178.2
out.siltclayloam.1.store0	None
out.siltclayloam.1.storeEnd	416.4
out.siltclayloam.1.maxErr	0.05
out.siltclayloam.1.sumErr	0.05
out.siltclayloam.1.iters	1345673
out.siltclayloam.1.ms	182464
out.siltclayloam.1.infilD	21.77, 0, 0, 0, 0, 0

quantity	value
out.siltclayloam.1.drainD	0.026, 0.026, 0.026, 0.026, 0.026, 0.026
out.siltclayloam.1.store	416.6, 416.6, 416.5, 416.5, 416.5, 416.4
out.siltclayloam.2.infil	22.54
out.siltclayloam.2.drain	0.156
out.siltclayloam.2.runoff	177.5
out.siltclayloam.2.store0	None
out.siltclayloam.2.storeEnd	417.2
out.siltclayloam.2.maxErr	0.074
out.siltclayloam.2.sumErr	0.074
out.siltclayloam.2.iters	266393
out.siltclayloam.2.ms	20684
out.siltclayloam.2.infilD	22.54, 0, 0, 0, 0, 0
out.siltclayloam.2.drainD	0.026, 0.026, 0.026, 0.026, 0.026, 0.026
out.siltclayloam.2.store	417.4, 417.3, 417.3, 417.3, 417.3, 417.2
out.siltclayloam.4.infil	24.38
out.siltclayloam.4.drain	0.156
out.siltclayloam.4.runoff	175.6
out.siltclayloam.4.store0	None
out.siltclayloam.4.storeEnd	419
out.siltclayloam.4.maxErr	0.007
out.siltclayloam.4.sumErr	-0.007
out.siltclayloam.4.iters	385896
out.siltclayloam.4.ms	21266
out.siltclayloam.4.infilD	24.38, 0, 0, 0, 0, 0
out.siltclayloam.4.drainD	0.026, 0.026, 0.026, 0.026, 0.026, 0.026
out.siltclayloam.4.store	419.1, 419.1, 419.1, 419, 419, 419

out_p4_richards_ref.json

Source: `Manuscripts/reproducibility/p456/out_p4_richards_ref.json`; 8 values.

quantity	value
sand.infil	200
sand.drain	152.6
sand.storeEnd	93.12
sand.S0	45.68
loam.infil	200
loam.drain	55.12
loam.storeEnd	314.9
loam.S0	170.1

out_p4_runtime.csv

Source: `Manuscripts/reproducibility/p456/out_p4_runtime.csv`; 17 rows. Workbook sheet (`CSV file`).

exp	name	study	study_hours	gui	gui_hours
E0	Baselines	10	0.6411	9	0.5925
E1	Dose and duration	200	10.99	17	0.8878
E2	Trigger pressure	200	7.731	21	0.805
E3	Sensor depth	200	8.099	21	0.84
E4	Methods	100	4.1	6	0.2408
E5	Soil structure dynamics (SHP)	300	15.77	72	3.45
E6	Mediterranean pollution	210	6.937	30	0.7833
E7	Mulching	200	8.949	nan	nan
E8	Rain intensity and moisture	1000	1.727	nan	nan
E9	Soil moisture and rain	1000	1.957	40	0.07778
E10	Amount, timing and soil	1000	70.39	20	0.7111
E11	Temperature, evaporation, rain	1000	41.98	16	0.6578
E12	Sensor, trigger, rain	1000	42.92	30	1.225
E13	N, water, soil	1000	57.44	32	1.929
E14	Mulch and rain	1000	48.01	nan	nan
E15	Method, depth, amount	1000	43.58	30	1.275
E16	Edge cases	770	48.22	24	1.21

out_p5_bridge.json

Source: `Manuscripts/reproducibility/p456/out_p5_bridge.json`; 20 values.

quantity	value
parser[0]	annual-field-nitrogen-3b.dlf, 11, 11, 2.246e-16
parser[1]	annual-field-nitrogen-4b.dlf, 11, 11, 1.703e-16
parser[2]	annual-tracer-3b.dlf, 11, 11, 7.006e-34
parser[3]	daily-water-flux.dlf, 26, 26, 3.277e-16
parser[4]	optim-field-nitrogen.dlf, 1, 1, 0
parser[5]	soil-chemical-110cm.dlf, 11665, 11665, 2.918e-15
n.annual-field-nitrogen-3b.dlf.Nfert	0
n.annual-field-nitrogen-3b.dlf.Nup	0.045
n.annual-field-nitrogen-3b.dlf.NL	0.00478
n.annual-field-nitrogen-3b.dlf.Nmin	0.049
n.annual-field-nitrogen-3b.dlf.Nden	0.00473

quantity	value
n.annual-field-nitrogen-3b.dlf.maxabs_Error_kgN	0
n.annual-field-nitrogen-3b.dlf.seasons	10
n.annual-field-nitrogen-4b.dlf.Nfert	0
n.annual-field-nitrogen-4b.dlf.Nup	0.049
n.annual-field-nitrogen-4b.dlf.NL	0.0041
n.annual-field-nitrogen-4b.dlf.Nmin	0.048
n.annual-field-nitrogen-4b.dlf.Nden	0.00442
n.annual-field-nitrogen-4b.dlf.maxabs_Error_kgN	0
n.annual-field-nitrogen-4b.dlf.seasons	10

out_p5_bridge_js.json

Source: `Manuscripts/reproducibility/p456/out_p5_bridge_js.json`; 832 values, first 220 shown.

quantity	value
deniArr[0]	3.206e-11, 3.206e-11, 1.88e-11, 4.709e-11, 4.709e-11, 2.869e-11
deniArr[1]	3.83e-11, 3.83e-11, 2.235e-11, 5.479e-11, 5.479e-11, 3.35e-11
deniArr[2]	4.481e-11, 4.481e-11, 2.599e-11, 6.21e-11, 6.21e-11, 3.816e-11
deniArr[3]	3.869e-11, 3.869e-11, 2.239e-11, 5.305e-11, 5.305e-11, 3.265e-11
deniArr[4]	2.408e-11, 2.408e-11, 1.396e-11, 3.312e-11, 3.312e-11, 2.037e-11
deniArr[5]	5.238e-12, 5.238e-12, 3.111e-12, 1.273e-11, 1.273e-11, 7.671e-12
deniArr[6]	2.994e-12, 2.994e-12, 1.785e-12, 7.394e-12, 7.394e-12, 4.447e-12
deniArr[7]	1.63e-12, 1.63e-12, 9.747e-13, 4.065e-12, 4.065e-12, 2.442e-12
deniArr[8]	8.865e-13, 8.865e-13, 5.309e-13, 2.222e-12, 2.222e-12, 1.334e-12
deniArr[9]	4.67e-13, 4.67e-13, 2.799e-13, 1.174e-12, 1.174e-12, 7.048e-13
deniArr[10]	2.463e-13, 2.463e-13, 1.477e-13, 6.204e-13, 6.204e-13, 3.723e-13
deniArr[11]	1.221e-13, 1.221e-13, 7.327e-14, 3.079e-13, 3.079e-13, 1.848e-13
deniArr[12]	2.007e-13, 2.007e-13, 1.204e-13, 2.665e-13, 2.665e-13, 1.599e-13
deniArr[13]	9.756e-14, 9.756e-14, 5.853e-14, 1.295e-13, 1.295e-13, 7.772e-14
deniArr[14]	0, 0, 0, 0, 0, 0
deniArr[15]	0, 0, 0, 0, 0, 0
deniArr[16]	0, 0, 0, 0, 0, 0
deniArr[17]	0, 0, 0, 0, 0, 0
deniArr[18]	0, 0, 0, 0, 0, 0
deniArr[19]	0, 0, 0, 0, 0, 0
deniArr[20]	0, 0, 0, 0, 0, 0
deniArr[21]	0, 0, 0, 0, 0, 0
deniArr[22]	0, 0, 0, 0, 0, 0
deniArr[23]	0, 0, 0, 0, 0, 0
deniArr[24]	0, 0, 0, 0, 0, 0
deniArr[25]	0, 0, 0, 0, 0, 0
deniArr[26]	0, 0, 0, 0, 0, 0
deniArr[27]	0, 0, 0, 0, 0, 0
deniArr[28]	0, 0, 0, 0, 0, 0
deniArr[29]	0, 0, 0, 0, 0, 0
deniArr[30]	0, 0, 0, 0, 0, 0

quantity	value
deniArr[31]	0, 0, 0, 0, 0, 0
deniArr[32]	0, 0, 0, 0, 0, 0
deniArr[33]	1.124e-10, 1.124e-10, 6.329e-11, 1.363e-10, 1.363e-10, 8.598e-11
deniArr[34]	7.183e-11, 7.183e-11, 4.113e-11, 9.431e-11, 9.431e-11, 5.856e-11
deniArr[35]	5.099e-11, 5.099e-11, 2.946e-11, 6.949e-11, 6.949e-11, 4.283e-11
deniArr[36]	3.823e-11, 3.823e-11, 2.213e-11, 5.249e-11, 5.249e-11, 3.23e-11
deniArr[37]	2.373e-11, 2.373e-11, 1.376e-11, 3.264e-11, 3.264e-11, 2.006e-11
deniArr[38]	5.254e-12, 5.254e-12, 3.12e-12, 1.276e-11, 1.276e-11, 7.687e-12
deniArr[39]	2.973e-12, 2.973e-12, 1.773e-12, 7.343e-12, 7.343e-12, 4.417e-12
deniArr[40]	1.634e-12, 1.634e-12, 9.773e-13, 4.075e-12, 4.075e-12, 2.448e-12
deniArr[41]	8.821e-13, 8.821e-13, 5.283e-13, 2.211e-12, 2.211e-12, 1.328e-12
deniArr[42]	4.679e-13, 4.679e-13, 2.805e-13, 1.177e-12, 1.177e-12, 7.062e-13
deniArr[43]	2.457e-13, 2.457e-13, 1.474e-13, 6.189e-13, 6.189e-13, 3.714e-13
deniArr[44]	1.221e-13, 1.221e-13, 7.323e-14, 3.079e-13, 3.079e-13, 1.848e-13
deniArr[45]	2.007e-13, 2.007e-13, 1.204e-13, 2.666e-13, 2.666e-13, 1.6e-13
deniArr[46]	9.747e-14, 9.747e-14, 5.848e-14, 1.294e-13, 1.294e-13, 7.765e-14
deniArr[47]	0, 0, 0, 0, 0, 0
deniArr[48]	0, 0, 0, 0, 0, 0
deniArr[49]	0, 0, 0, 0, 0, 0
deniArr[50]	0, 0, 0, 0, 0, 0
deniArr[51]	0, 0, 0, 0, 0, 0
deniArr[52]	0, 0, 0, 0, 0, 0
deniArr[53]	0, 0, 0, 0, 0, 0
deniArr[54]	0, 0, 0, 0, 0, 0
deniArr[55]	0, 0, 0, 0, 0, 0
deniArr[56]	0, 0, 0, 0, 0, 0
deniArr[57]	0, 0, 0, 0, 0, 0
deniArr[58]	0, 0, 0, 0, 0, 0
deniArr[59]	0, 0, 0, 0, 0, 0
deniArr[60]	0, 0, 0, 0, 0, 0
deniArr[61]	0, 0, 0, 0, 0, 0
deniArr[62]	0, 0, 0, 0, 0, 0
deniArr[63]	0, 0, 0, 0, 0, 0
deniArr[64]	0, 0, 0, 0, 0, 0
deniArr[65]	0, 0, 0, 0, 0, 0
deniArr[66]	1.964e-10, 1.964e-10, 1.074e-10, 2.027e-10, 2.027e-10, 1.32e-10
deniArr[67]	1.222e-10, 1.222e-10, 6.842e-11, 1.432e-10, 1.432e-10, 9.084e-11
deniArr[68]	6.988e-11, 6.988e-11, 3.993e-11, 9.051e-11, 9.051e-11, 5.63e-11
deniArr[69]	4.024e-11, 4.024e-11, 2.326e-11, 5.486e-11, 5.486e-11, 3.379e-11
deniArr[70]	2.371e-11, 2.371e-11, 1.376e-11, 3.255e-11, 3.255e-11, 2e-11
deniArr[71]	5.244e-12, 5.244e-12, 3.114e-12, 1.272e-11, 1.272e-11, 7.665e-12
deniArr[72]	2.963e-12, 2.963e-12, 1.767e-12, 7.321e-12, 7.321e-12, 4.403e-12
deniArr[73]	1.632e-12, 1.632e-12, 9.761e-13, 4.07e-12, 4.07e-12, 2.445e-12
deniArr[74]	8.806e-13, 8.806e-13, 5.274e-13, 2.208e-12, 2.208e-12, 1.325e-12
deniArr[75]	4.677e-13, 4.677e-13, 2.803e-13, 1.176e-12, 1.176e-12, 7.058e-13
deniArr[76]	2.455e-13, 2.455e-13, 1.472e-13, 6.183e-13, 6.183e-13, 3.711e-13
deniArr[77]	1.22e-13, 1.22e-13, 7.318e-14, 3.077e-13, 3.077e-13, 1.847e-13
deniArr[78]	2.006e-13, 2.006e-13, 1.204e-13, 2.666e-13, 2.666e-13, 1.6e-13

quantity	value
deniArr[79]	9.741e-14, 9.741e-14, 5.844e-14, 1.293e-13, 1.293e-13, 7.76e-14
deniArr[80]	0, 0, 0, 0, 0, 0
deniArr[81]	0, 0, 0, 0, 0, 0
deniArr[82]	0, 0, 0, 0, 0, 0
deniArr[83]	0, 0, 0, 0, 0, 0
deniArr[84]	0, 0, 0, 0, 0, 0
deniArr[85]	0, 0, 0, 0, 0, 0
deniArr[86]	0, 0, 0, 0, 0, 0
deniArr[87]	0, 0, 0, 0, 0, 0
deniArr[88]	0, 0, 0, 0, 0, 0
deniArr[89]	0, 0, 0, 0, 0, 0
deniArr[90]	0, 0, 0, 0, 0, 0
deniArr[91]	0, 0, 0, 0, 0, 0
deniArr[92]	0, 0, 0, 0, 0, 0
deniArr[93]	0, 0, 0, 0, 0, 0
deniArr[94]	0, 0, 0, 0, 0, 0
deniArr[95]	0, 0, 0, 0, 0, 0
deniArr[96]	0, 0, 0, 0, 0, 0
deniArr[97]	0, 0, 0, 0, 0, 0
deniArr[98]	0, 0, 0, 0, 0, 0
deniArr[99]	2.86e-10, 2.86e-10, 1.53e-10, 2.548e-10, 2.548e-10, 1.715e-10
deniArr[100]	1.737e-10, 1.737e-10, 9.543e-11, 1.836e-10, 1.836e-10, 1.189e-10
deniArr[101]	9.365e-11, 9.365e-11, 5.287e-11, 1.142e-10, 1.142e-10, 7.186e-11
deniArr[102]	4.498e-11, 4.498e-11, 2.591e-11, 6.029e-11, 6.029e-11, 3.725e-11
deniArr[103]	2.415e-11, 2.415e-11, 1.4e-11, 3.3e-11, 3.3e-11, 2.029e-11
deniArr[104]	5.245e-12, 5.245e-12, 3.115e-12, 1.271e-11, 1.271e-11, 7.659e-12
deniArr[105]	2.956e-12, 2.956e-12, 1.763e-12, 7.303e-12, 7.303e-12, 4.392e-12
deniArr[106]	1.629e-12, 1.629e-12, 9.744e-13, 4.063e-12, 4.063e-12, 2.441e-12
deniArr[107]	8.794e-13, 8.794e-13, 5.267e-13, 2.205e-12, 2.205e-12, 1.324e-12
deniArr[108]	4.673e-13, 4.673e-13, 2.801e-13, 1.175e-12, 1.175e-12, 7.052e-13
deniArr[109]	2.454e-13, 2.454e-13, 1.471e-13, 6.179e-13, 6.179e-13, 3.708e-13
deniArr[110]	1.219e-13, 1.219e-13, 7.312e-14, 3.076e-13, 3.076e-13, 1.846e-13
deniArr[111]	2.004e-13, 2.004e-13, 1.203e-13, 2.664e-13, 2.664e-13, 1.599e-13
deniArr[112]	9.735e-14, 9.735e-14, 5.841e-14, 1.293e-13, 1.293e-13, 7.756e-14
deniArr[113]	0, 0, 0, 0, 0, 0
deniArr[114]	0, 0, 0, 0, 0, 0
deniArr[115]	0, 0, 0, 0, 0, 0
deniArr[116]	0, 0, 0, 0, 0, 0
deniArr[117]	0, 0, 0, 0, 0, 0
deniArr[118]	0, 0, 0, 0, 0, 0
deniArr[119]	0, 0, 0, 0, 0, 0
deniArr[120]	0, 0, 0, 0, 0, 0
deniArr[121]	0, 0, 0, 0, 0, 0
deniArr[122]	0, 0, 0, 0, 0, 0
deniArr[123]	0, 0, 0, 0, 0, 0
deniArr[124]	0, 0, 0, 0, 0, 0
deniArr[125]	0, 0, 0, 0, 0, 0
deniArr[126]	0, 0, 0, 0, 0, 0

quantity	value
deniArr[127]	0, 0, 0, 0, 0, 0
deniArr[128]	0, 0, 0, 0, 0, 0
deniArr[129]	0, 0, 0, 0, 0, 0
deniArr[130]	0, 0, 0, 0, 0, 0
deniArr[131]	0, 0, 0, 0, 0, 0
deniArr[132]	3.764e-10, 3.764e-10, 1.979e-10, 2.94e-10, 2.94e-10, 2.043e-10
deniArr[133]	2.296e-10, 2.296e-10, 1.242e-10, 2.196e-10, 2.196e-10, 1.453e-10
deniArr[134]	1.196e-10, 1.196e-10, 6.675e-11, 1.373e-10, 1.373e-10, 8.741e-11
deniArr[135]	5.156e-11, 5.156e-11, 2.956e-11, 6.756e-11, 6.756e-11, 4.191e-11
deniArr[136]	2.508e-11, 2.508e-11, 1.453e-11, 3.405e-11, 3.405e-11, 2.095e-11
deniArr[137]	5.274e-12, 5.274e-12, 3.132e-12, 1.277e-11, 1.277e-11, 7.693e-12
deniArr[138]	2.952e-12, 2.952e-12, 1.761e-12, 7.293e-12, 7.293e-12, 4.386e-12
deniArr[139]	1.627e-12, 1.627e-12, 9.728e-13, 4.057e-12, 4.057e-12, 2.437e-12
deniArr[140]	8.783e-13, 8.783e-13, 5.26e-13, 2.202e-12, 2.202e-12, 1.322e-12
deniArr[141]	4.669e-13, 4.669e-13, 2.799e-13, 1.174e-12, 1.174e-12, 7.046e-13
deniArr[142]	2.452e-13, 2.452e-13, 1.47e-13, 6.175e-13, 6.175e-13, 3.706e-13
deniArr[143]	1.218e-13, 1.218e-13, 7.306e-14, 3.074e-13, 3.074e-13, 1.845e-13
deniArr[144]	2.003e-13, 2.003e-13, 1.201e-13, 2.663e-13, 2.663e-13, 1.598e-13
deniArr[145]	9.729e-14, 9.729e-14, 5.837e-14, 1.292e-13, 1.292e-13, 7.751e-14
deniArr[146]	0, 0, 0, 0, 0, 0
deniArr[147]	0, 0, 0, 0, 0, 0
deniArr[148]	0, 0, 0, 0, 0, 0
deniArr[149]	0, 0, 0, 0, 0, 0
deniArr[150]	0, 0, 0, 0, 0, 0
deniArr[151]	0, 0, 0, 0, 0, 0
deniArr[152]	0, 0, 0, 0, 0, 0
deniArr[153]	0, 0, 0, 0, 0, 0
deniArr[154]	0, 0, 0, 0, 0, 0
deniArr[155]	0, 0, 0, 0, 0, 0
deniArr[156]	0, 0, 0, 0, 0, 0
deniArr[157]	0, 0, 0, 0, 0, 0
deniArr[158]	0, 0, 0, 0, 0, 0
deniArr[159]	0, 0, 0, 0, 0, 0
deniArr[160]	0, 0, 0, 0, 0, 0
deniArr[161]	0, 0, 0, 0, 0, 0
deniArr[162]	0, 0, 0, 0, 0, 0
deniArr[163]	0, 0, 0, 0, 0, 0
deniArr[164]	0, 0, 0, 0, 0, 0
deniArr[165]	4.711e-10, 4.711e-10, 2.444e-10, 3.245e-10, 3.245e-10, 2.33e-10
deniArr[166]	2.868e-10, 2.868e-10, 1.53e-10, 2.497e-10, 2.497e-10, 1.689e-10
deniArr[167]	1.476e-10, 1.476e-10, 8.148e-11, 1.594e-10, 1.594e-10, 1.027e-10
deniArr[168]	5.941e-11, 5.941e-11, 3.389e-11, 7.585e-11, 7.585e-11, 4.727e-11
deniArr[169]	2.648e-11, 2.648e-11, 1.532e-11, 3.562e-11, 3.562e-11, 2.194e-11
deniArr[170]	5.342e-12, 5.342e-12, 3.172e-12, 1.291e-11, 1.291e-11, 7.781e-12
deniArr[171]	2.954e-12, 2.954e-12, 1.762e-12, 7.297e-12, 7.297e-12, 4.389e-12
deniArr[172]	1.625e-12, 1.625e-12, 9.717e-13, 4.052e-12, 4.052e-12, 2.434e-12
deniArr[173]	8.773e-13, 8.773e-13, 5.254e-13, 2.199e-12, 2.199e-12, 1.321e-12
deniArr[174]	4.665e-13, 4.665e-13, 2.796e-13, 1.173e-12, 1.173e-12, 7.04e-13

quantity	value
deniArr[175]	2.45e-13, 2.45e-13, 1.47e-13, 6.171e-13, 6.171e-13, 3.703e-13
deniArr[176]	1.217e-13, 1.217e-13, 7.299e-14, 3.072e-13, 3.072e-13, 1.843e-13
deniArr[177]	2.001e-13, 2.001e-13, 1.2e-13, 2.661e-13, 2.661e-13, 1.597e-13
deniArr[178]	9.722e-14, 9.722e-14, 5.833e-14, 1.291e-13, 1.291e-13, 7.746e-14
deniArr[179]	0, 0, 0, 0, 0, 0
deniArr[180]	0, 0, 0, 0, 0, 0
deniArr[181]	0, 0, 0, 0, 0, 0
deniArr[182]	0, 0, 0, 0, 0, 0
deniArr[183]	0, 0, 0, 0, 0, 0
deniArr[184]	0, 0, 0, 0, 0, 0
deniArr[185]	0, 0, 0, 0, 0, 0
deniArr[186]	0, 0, 0, 0, 0, 0
deniArr[187]	0, 0, 0, 0, 0, 0
deniArr[188]	0, 0, 0, 0, 0, 0
deniArr[189]	0, 0, 0, 0, 0, 0
deniArr[190]	0, 0, 0, 0, 0, 0
deniArr[191]	0, 0, 0, 0, 0, 0
deniArr[192]	0, 0, 0, 0, 0, 0
deniArr[193]	0, 0, 0, 0, 0, 0
deniArr[194]	0, 0, 0, 0, 0, 0
deniArr[195]	0, 0, 0, 0, 0, 0
deniArr[196]	0, 0, 0, 0, 0, 0
deniArr[197]	0, 0, 0, 0, 0, 0
deniArr[198]	5.655e-10, 5.655e-10, 2.901e-10, 3.469e-10, 3.469e-10, 2.573e-10
deniArr[199]	3.463e-10, 3.463e-10, 1.826e-10, 2.755e-10, 2.755e-10, 1.904e-10
deniArr[200]	1.769e-10, 1.769e-10, 9.67e-11, 1.8e-10, 1.8e-10, 1.175e-10
deniArr[201]	6.822e-11, 6.822e-11, 3.87e-11, 8.468e-11, 8.468e-11, 5.304e-11
deniArr[202]	2.827e-11, 2.827e-11, 1.632e-11, 3.762e-11, 3.762e-11, 2.321e-11
deniArr[203]	5.45e-12, 5.45e-12, 3.236e-12, 1.315e-11, 1.315e-11, 7.925e-12
deniArr[204]	2.964e-12, 2.964e-12, 1.768e-12, 7.322e-12, 7.322e-12, 4.404e-12
deniArr[205]	1.624e-12, 1.624e-12, 9.713e-13, 4.05e-12, 4.05e-12, 2.433e-12
deniArr[206]	8.764e-13, 8.764e-13, 5.249e-13, 2.197e-12, 2.197e-12, 1.319e-12
deniArr[207]	4.661e-13, 4.661e-13, 2.794e-13, 1.172e-12, 1.172e-12, 7.034e-13
deniArr[208]	2.449e-13, 2.449e-13, 1.469e-13, 6.167e-13, 6.167e-13, 3.701e-13
deniArr[209]	1.216e-13, 1.216e-13, 7.292e-14, 3.07e-13, 3.07e-13, 1.842e-13
deniArr[210]	1.998e-13, 1.998e-13, 1.199e-13, 2.659e-13, 2.659e-13, 1.596e-13
deniArr[211]	9.715e-14, 9.715e-14, 5.829e-14, 1.29e-13, 1.29e-13, 7.741e-14
deniArr[212]	0, 0, 0, 0, 0, 0
deniArr[213]	0, 0, 0, 0, 0, 0
deniArr[214]	0, 0, 0, 0, 0, 0
deniArr[215]	0, 0, 0, 0, 0, 0
deniArr[216]	0, 0, 0, 0, 0, 0
deniArr[217]	0, 0, 0, 0, 0, 0
deniArr[218]	0, 0, 0, 0, 0, 0
deniArr[219]	0, 0, 0, 0, 0, 0

out_p5_chem.json

Source: `Manuscripts/reproducibility/p456/out_p5_chem.json`; 6 values.

quantity	value
resid	-0.1097, 0.1082, 0.0011, -3.553e-15, 0.0007, -0.1228, 0.1221, 0, -0.0002, 0.001
maxAbs	0.1228
sumAbs	0.4658
applied	2290
leachTotal	393
uptakeTotal	1934

out_p6_crops.json

Source: `Manuscripts/reproducibility/p456/out_p6_crops.json`; 1495 values, first 220 shown.

quantity	value
n	48
groups.field	19
groups.perennial	29
habits.annual	19
habits.deciduous	18
habits.evergreen	11
combos	110
issues[0][0]	citrus
issues[0][1]	Young
issues[0][2]	Kc mid < Kc ini
issues[1][0]	citrus
issues[1][1]	Low density
issues[1][2]	Kc mid < Kc ini
issues[2][0]	citrus
issues[2][1]	Medium density, short
issues[2][2]	Kc mid < Kc ini
issues[3][0]	citrus
issues[3][1]	Medium density, tall
issues[3][2]	Kc mid < Kc ini
issues[4][0]	citrus
issues[4][1]	High density, short
issues[4][2]	Kc mid < Kc ini
issues[5][0]	citrus
issues[5][1]	High density, tall
issues[5][2]	Kc mid < Kc ini
issues[6][0]	olive
issues[6][1]	Young
issues[6][2]	Kc mid < Kc ini, Kc mid 0.25

quantity	value
issues[7][0]	olive
issues[7][1]	Traditional, low density
issues[7][2]	Kc mid < Kc ini, Kc mid 0.3
issues[8][0]	olive
issues[8][1]	Traditional, medium density
issues[8][2]	Kc mid < Kc ini
issues[9][0]	olive
issues[9][1]	Intensive (hedge pruned)
issues[9][2]	Kc mid < Kc ini
issues[10][0]	olive
issues[10][1]	Super-intensive, medium
issues[10][2]	Kc mid < Kc ini
issues[11][0]	olive
issues[11][1]	Super-intensive, high
issues[11][2]	Kc mid < Kc ini
issues[12][0]	datepalm
issues[12][1]	Young, low
issues[12][2]	Kc mid < Kc ini
rows[0].key	rice
rows[0].dens	None
rows[0].group	field
rows[0].habit	annual
rows[0].kcIni	1.05
rows[0].kcMid	1.2
rows[0].kcEnd	0.9
rows[0].len	120
rows[0].root	75
rows[0].p	0.2
rows[0].ky	None
rows[0].kcSum	135.1
rows[0].iss	
rows[1].key	maize
rows[1].dens	None
rows[1].group	field
rows[1].habit	annual
rows[1].kcIni	0.3
rows[1].kcMid	1.2
rows[1].kcEnd	0.6
rows[1].len	105
rows[1].root	135
rows[1].p	0.55
rows[1].ky	1.25
rows[1].kcSum	92.85
rows[1].iss	
rows[2].key	pearlmillet
rows[2].dens	None
rows[2].group	field
rows[2].habit	annual

quantity	value
rows[2].kcIni	0.3
rows[2].kcMid	1
rows[2].kcEnd	0.3
rows[2].len	105
rows[2].root	150
rows[2].p	0.55
rows[2].ky	None
rows[2].kcSum	77
rows[2].iss	
rows[3].key	sorghum
rows[3].dens	None
rows[3].group	field
rows[3].habit	annual
rows[3].kcIni	0.3
rows[3].kcMid	1
rows[3].kcEnd	0.55
rows[3].len	125
rows[3].root	150
rows[3].p	0.55
rows[3].ky	0.9
rows[3].kcSum	91.88
rows[3].iss	
rows[4].key	cassava
rows[4].dens	None
rows[4].group	field
rows[4].habit	annual
rows[4].kcIni	0.3
rows[4].kcMid	0.8
rows[4].kcEnd	0.3
rows[4].len	210
rows[4].root	65
rows[4].p	0.35
rows[4].ky	None
rows[4].kcSum	133
rows[4].iss	
rows[5].key	yam
rows[5].dens	None
rows[5].group	field
rows[5].habit	annual
rows[5].kcIni	0.5
rows[5].kcMid	1.1
rows[5].kcEnd	0.85
rows[5].len	240
rows[5].root	60
rows[5].p	0.4
rows[5].ky	None
rows[5].kcSum	220.3
rows[5].iss	

quantity	value
rows[6].key	cowpea
rows[6].dens	None
rows[6].group	field
rows[6].habit	annual
rows[6].kcIni	0.4
rows[6].kcMid	1.05
rows[6].kcEnd	0.5
rows[6].len	100
rows[6].root	80
rows[6].p	0.45
rows[6].ky	None
rows[6].kcSum	76.7
rows[6].iss	
rows[7].key	greengram
rows[7].dens	None
rows[7].group	field
rows[7].habit	annual
rows[7].kcIni	0.4
rows[7].kcMid	0.96
rows[7].kcEnd	0.51
rows[7].len	90
rows[7].root	75
rows[7].p	0.45
rows[7].ky	None
rows[7].kcSum	66.78
rows[7].iss	
rows[8].key	soybean
rows[8].dens	None
rows[8].group	field
rows[8].habit	annual
rows[8].kcIni	0.4
rows[8].kcMid	1.15
rows[8].kcEnd	0.5
rows[8].len	120
rows[8].root	95
rows[8].p	0.5
rows[8].ky	0.85
rows[8].kcSum	103.6
rows[8].iss	
rows[9].key	groundnut
rows[9].dens	None
rows[9].group	field
rows[9].habit	annual
rows[9].kcIni	0.4
rows[9].kcMid	1.15
rows[9].kcEnd	0.6
rows[9].len	120
rows[9].root	75

quantity	value
rows[9].p	0.5
rows[9].ky	0.7
rows[9].kcSum	102.9
rows[9].iss	
rows[10].key	cotton
rows[10].dens	None
rows[10].group	field
rows[10].habit	annual
rows[10].kcIni	0.35
rows[10].kcMid	1.15
rows[10].kcEnd	0.5
rows[10].len	190
rows[10].root	135
rows[10].p	0.65
rows[10].ky	0.85
rows[10].kcSum	160.2
rows[10].iss	
rows[11].key	wheat
rows[11].dens	None
rows[11].group	field
rows[11].habit	annual
rows[11].kcIni	0.4
rows[11].kcMid	1.15
rows[11].kcEnd	0.41
rows[11].len	120
rows[11].root	165
rows[11].p	0.55
rows[11].ky	1.05
rows[11].kcSum	102.5
rows[11].iss	
rows[12].key	chickpea
rows[12].dens	None
rows[12].group	field
rows[12].habit	annual
rows[12].kcIni	0.4
rows[12].kcMid	1
rows[12].kcEnd	0.4
rows[12].len	120
rows[12].root	80
rows[12].p	0.5
rows[12].ky	None
rows[12].kcSum	93
rows[12].iss	
rows[13].key	mustard
rows[13].dens	None
rows[13].group	field
rows[13].habit	annual

quantity	value
rows[13].kcIni	0.35

out_p6_pheno.json

Source: `Manuscripts/reproducibility/p456/out_p6_pheno.json`; 560 values, first 220 shown.

quantity	value
[0].clim	warm
[0].geno	Reference
[0].tr	ww
[0].closure_mm	-6.724e-07
[0].det	0
[0].P	0
[0].I	589.1
[0].T	330
[0].Es	252.3
[0].drain	0
[0].grain	515.1
[0].shoot	542.3
[0].root	172.6
[0].rdepth	119.5
[0].cwsiMax	0.2467
[0].ctdMax	1.066
[0].psiMin	-1.213
[0].lai	2.632
[0].height	213.7
[0].stage	3.657
[1].clim	warm
[1].geno	Reference
[1].tr	dr
[1].closure_mm	2.013e-05
[1].det	0
[1].P	0
[1].I	519.7
[1].T	253.9
[1].Es	259.1
[1].drain	1.907e-06
[1].grain	345.4
[1].shoot	478.6
[1].root	166.1
[1].rdepth	119.5
[1].cwsiMax	0.3367
[1].ctdMax	1.076
[1].psiMin	-1.285
[1].lai	2.429
[1].height	213

quantity	value
[1].stage	3.657
[2].clim	warm
[2].geno	Deep rooting
[2].tr	ww
[2].closure_mm	-2.057e-05
[2].det	0
[2].P	0
[2].I	584.2
[2].T	305.8
[2].Es	271.6
[2].drain	0
[2].grain	485.8
[2].shoot	481.6
[2].root	186.1
[2].rdepth	119.5
[2].cwsiMax	0.2172
[2].ctdMax	1.078
[2].psiMin	-1.203
[2].lai	2.326
[2].height	213.7
[2].stage	3.657
[3].clim	warm
[3].geno	Deep rooting
[3].tr	dr
[3].closure_mm	-3.866e-05
[3].det	0
[3].P	0
[3].I	518.4
[3].T	239.2
[3].Es	272.4
[3].drain	0
[3].grain	328.9
[3].shoot	432.8
[3].root	179.2
[3].rdepth	119.5
[3].cwsiMax	0.346
[3].ctdMax	1.085
[3].psiMin	-1.282
[3].lai	2.192
[3].height	213.1
[3].stage	3.657
[4].clim	warm
[4].geno	Shallow, dense
[4].tr	ww
[4].closure_mm	-1.514e-05
[4].det	0
[4].P	0
[4].I	589.6

quantity	value
[4].T	331.3
[4].Es	251.6
[4].drain	0
[4].grain	516.2
[4].shoot	544
[4].root	172.7
[4].rdepth	109
[4].cwsiMax	0.2477
[4].ctdMax	1.066
[4].psiMin	-1.214
[4].lai	2.644
[4].height	213.6
[4].stage	3.657
[5].clim	warm
[5].geno	Shallow, dense
[5].tr	dr
[5].closure_mm	2.381e-05
[5].det	0
[5].P	0
[5].I	501.3
[5].T	228.3
[5].Es	266.3
[5].drain	1.907e-06
[5].grain	297.7
[5].shoot	429.1
[5].root	158.4
[5].rdepth	111.5
[5].cwsiMax	0.3839
[5].ctdMax	1.082
[5].psiMin	-1.27
[5].lai	2.219
[5].height	212.7
[5].stage	3.657
[6].clim	warm
[6].geno	Stay-green
[6].tr	ww
[6].closure_mm	1.436e-05
[6].det	0
[6].P	0
[6].I	611.2
[6].T	395
[6].Es	209.6
[6].drain	0
[6].grain	541.7
[6].shoot	552.9
[6].root	176.6
[6].rdepth	119.5
[6].cwsiMax	0.2492

quantity	value
[6].ctdMax	1.113
[6].psiMin	-1.211
[6].lai	2.738
[6].height	213.7
[6].stage	3.657
[7].clim	warm
[7].geno	Stay-green
[7].tr	dr
[7].closure_mm	4.894e-06
[7].det	0
[7].P	0
[7].I	543.4
[7].T	319
[7].Es	217.8
[7].drain	0
[7].grain	376.9
[7].shoot	488.7
[7].root	170.7
[7].rdepth	119.5
[7].cwsiMax	0.3559
[7].ctdMax	1.11
[7].psiMin	-1.265
[7].lai	2.576
[7].height	213
[7].stage	3.657
[8].clim	warm
[8].geno	Water saver
[8].tr	ww
[8].closure_mm	-1.134e-05
[8].det	0
[8].P	0
[8].I	506.2
[8].T	164.5
[8].Es	334.8
[8].drain	0
[8].grain	330.9
[8].shoot	336
[8].root	143.3
[8].rdepth	119.5
[8].cwsiMax	0.3778
[8].ctdMax	0.9156
[8].psiMin	-0.8846
[8].lai	1.48
[8].height	213.7
[8].stage	3.657
[9].clim	warm
[9].geno	Water saver
[9].tr	dr

quantity	value
[9].closure_mm	-6.134e-06
[9].det	0
[9].P	0
[9].I	453.8
[9].T	135
[9].Es	311.9
[9].drain	0
[9].grain	241.9
[9].shoot	300.1
[9].root	135.6
[9].rdepth	119.5
[9].cwsiMax	0.4086
[9].ctdMax	0.9363
[9].psiMin	-0.9405
[9].lai	1.364
[9].height	213.2
[9].stage	3.657
[10].clim	warm
[10].geno	Water spender
[10].tr	ww
[10].closure_mm	1.024e-05
[10].det	0
[10].P	0
[10].I	615.1
[10].T	377
[10].Es	231.5
[10].drain	0
[10].grain	561.5
[10].shoot	608
[10].root	180.6
[10].rdepth	119.5
[10].cwsiMax	0.1435
[10].ctdMax	1.114
[10].psiMin	-1.59
[10].lai	3.01
[10].height	213.5
[10].stage	3.657

out_p6_sweep.json

Source: `Manuscripts/reproducibility/p456/out_p6_sweep.json`; 286 values, first 220 shown.

quantity	value
[0].id	sid
[0].ran	False
[0].engineInputs	19

quantity	value
[0].ms	6655
[0].tabs	Results, Design, Charts, Figure, Messages
[0].inputs	0
[0].chartsTab	8
[0].errors	
[0].kpi	62
[0].msgs	0
[0].charts	4
[0].textLen	2259
[0].canvas	None
[1].id	drip
[1].ran	False
[1].engineInputs	19
[1].ms	6639
[1].tabs	Results, Design, Charts, Figure, Messages
[1].inputs	0
[1].chartsTab	8
[1].errors	
[1].kpi	62
[1].msgs	2
[1].charts	4
[1].textLen	2500
[1].canvas	None
[2].id	spr
[2].ran	False
[2].engineInputs	19
[2].ms	6839
[2].tabs	Results, Design, Charts, Figure, Messages
[2].inputs	0
[2].chartsTab	8
[2].errors	
[2].kpi	62
[2].msgs	4
[2].charts	4
[2].textLen	2860
[2].canvas	None
[3].id	v1
[3].ran	True
[3].engineInputs	19
[3].ms	6431
[3].tabs	Results, Design, Charts, Figure, Messages
[3].inputs	0
[3].chartsTab	4
[3].errors	
[3].kpi	62
[3].msgs	4
[3].charts	4
[3].textLen	1672

quantity	value
[3].canvas	None
[4].id	n1
[4].ran	True
[4].engineInputs	19
[4].ms	6350
[4].tabs	Results, Design, Charts, Figure, Messages
[4].inputs	0
[4].chartsTab	4
[4].errors	
[4].kpi	62
[4].msgs	6
[4].charts	6
[4].textLen	3419
[4].canvas	606, 460
[5].id	v4
[5].ran	True
[5].engineInputs	19
[5].ms	6388
[5].tabs	Results, Design, Charts, Figure, Messages
[5].inputs	0
[5].chartsTab	4
[5].errors	
[5].kpi	62
[5].msgs	4
[5].charts	4
[5].textLen	1584
[5].canvas	None
[6].id	v6
[6].ran	True
[6].engineInputs	19
[6].ms	6371
[6].tabs	Results, Design, Charts, Figure, Messages
[6].inputs	0
[6].chartsTab	4
[6].errors	
[6].kpi	62
[6].msgs	2
[6].charts	4
[6].textLen	1503
[6].canvas	None
[7].id	e1
[7].ran	True
[7].engineInputs	19
[7].ms	6574
[7].tabs	Results, Design, Charts, Figure, Messages
[7].inputs	0
[7].chartsTab	4
[7].errors	

quantity	value
[7].kpi	62
[7].msgs	3
[7].charts	4
[7].textLen	1802
[7].canvas	None
[8].id	n2
[8].ran	True
[8].engineInputs	19
[8].ms	6405
[8].tabs	Results, Design, Charts, Figure, Messages
[8].inputs	0
[8].chartsTab	4
[8].errors	
[8].kpi	62
[8].msgs	2
[8].charts	4
[8].textLen	1674
[8].canvas	None
[9].id	n3
[9].ran	True
[9].engineInputs	19
[9].ms	6370
[9].tabs	Results, Design, Charts, Figure, Messages
[9].inputs	0
[9].chartsTab	4
[9].errors	
[9].kpi	62
[9].msgs	5
[9].charts	4
[9].textLen	1680
[9].canvas	None
[10].id	v3
[10].ran	True
[10].engineInputs	19
[10].ms	6457
[10].tabs	Results, Design, Charts, Figure, Messages
[10].inputs	0
[10].chartsTab	4
[10].errors	
[10].kpi	62
[10].msgs	3
[10].charts	4
[10].textLen	1661
[10].canvas	None
[11].id	v7
[11].ran	True
[11].engineInputs	19
[11].ms	6461

quantity	value
[11].tabs	Results, Design, Charts, Figure, Messages
[11].inputs	0
[11].chartsTab	4
[11].errors	
[11].kpi	62
[11].msgs	0
[11].charts	6
[11].textLen	3237
[11].canvas	606, 440
[12].id	r1
[12].ran	True
[12].engineInputs	19
[12].ms	6429
[12].tabs	Results, Design, Charts, Figure, Messages
[12].inputs	0
[12].chartsTab	4
[12].errors	
[12].kpi	62
[12].msgs	3
[12].charts	4
[12].textLen	1987
[12].canvas	None
[13].id	dph
[13].ran	False
[13].engineInputs	19
[13].ms	6328
[13].tabs	Results, Design, Charts, Figure, Messages
[13].inputs	0
[13].chartsTab	0
[13].errors	
[13].kpi	0
[13].msgs	0
[13].charts	0
[13].textLen	292
[13].canvas	None
[14].id	field
[14].ran	True
[14].engineInputs	19
[14].ms	6716
[14].tabs	Results, Design, Charts, Figure, Messages
[14].inputs	0
[14].chartsTab	4
[14].errors	
[14].kpi	62
[14].msgs	5
[14].charts	4
[14].textLen	2084
[14].canvas	None

quantity	value
[15].id	spac
[15].ran	True
[15].engineInputs	19
[15].ms	6562
[15].tabs	Results, Design, Charts, Figure, Messages
[15].inputs	0
[15].chartsTab	4
[15].errors	
[15].kpi	62
[15].msgs	0
[15].charts	4
[15].textLen	1470
[15].canvas	None
[16].id	v2
[16].ran	True
[16].engineInputs	19
[16].ms	6490
[16].tabs	Results, Design, Charts, Figure, Messages
[16].inputs	0
[16].chartsTab	4
[16].errors	
[16].kpi	62
[16].msgs	4
[16].charts	4
[16].textLen	1695

out_p7_planner.json

Source: `Manuscripts/reproducibility/p456/out_p7_planner.json`; 57 values.

quantity	value
n_questions	8
passed	8
rows[0].q	How does the irrigation trigger of -100 to -800 cm change drainage on a sandy loam in Bologna over 10 years?
rows[0].expected	trig
rows[0].got	irrig, trig
rows[0].station	it-bologna
rows[0].soil	usda:sandyloam
rows[0].pass	True
rows[1].q	What happens to nitrate leaching if I cut the nitrogen dose by 20 % in Foulum?
rows[1].expected	nfac
rows[1].got	nfac, mm
rows[1].station	dk-foulum
rows[1].soil	None

quantity	value
rows[1].pass	True
rows[2].q	Compare 20 % less rain and 20 % more rain at Jyndevad on a loamy sand
rows[2].expected	precip
rows[2].got	precip
rows[2].station	dk-jyndevad
rows[2].soil	usda:loamysand
rows[2].pass	True
rows[3].q	Warming of 2 degrees and 4 degrees in Poland, Jadwisin
rows[3].expected	temp
rows[3].got	temp
rows[3].station	pl-jadwisin
rows[3].soil	None
rows[3].pass	True
rows[4].q	Is drip irrigation better than rainfed on a silt loam in the Czech Republic?
rows[4].expected	irrig
rows[4].got	irrig, method
rows[4].station	cz-valecov
rows[4].soil	usda:siltloam
rows[4].pass	True
rows[5].q	Effect of the sensor depth 10, 30 and 60 cm on the irrigation trigger
rows[5].expected	z
rows[5].got	irrig, trig, z
rows[5].station	None
rows[5].soil	None
rows[5].pass	True
rows[6].q	Pesticide half-life of 20 and 60 days and sorption koc of 50 and 200 l/kg
rows[6].expected	half, koc
rows[6].got	koc, half
rows[6].station	None
rows[6].soil	None
rows[6].pass	True
rows[7].q	What if the rooting depth is 60 and 120 cm in Bratislava?
rows[7].expected	root
rows[7].got	root
rows[7].station	sk-bratislava
rows[7].soil	None
rows[7].pass	True
catalogue_factors	54
catalogue_soils	10
catalogue_stations	8
validate.factors[0].id	precip
validate.factors[0].levels	0.4, 1, 1.6
validate.responses	yield
validate.notes	unknown factor ”not/_a/_factor” left out

S14. Where every table and figure comes from

Item	How it is made	Script(s)	Input and output
Figure 1	Figure drawn by the script from the input data	Manuscripts/reproducibility/p456/p456_figs_architecture.py	Data/daisy_ensemble.csv.gz (10 190 runs); Manuscripts/figures/p6/fig01_system.pdf
Table 1	Descriptive table (definitions, design, parameters or tests); no computation	-	
Figure 2	Screen capture of SWAFL Desktop 1.1.0 (not regenerated)	-	Manuscripts/figures/shots_jpg/zaria_ndvi.jpg; Manuscripts/figures/shots_jpg/zaria_scenes.jpg; Manuscripts/figures/shots_jpg/real_sketch.jpg
Figure 3	Screen capture of SWAFL Desktop 1.1.0 (not regenerated)	-	Manuscripts/figures/shots_jpg/zaria_terrain.jpg; Manuscripts/figures/shots_jpg/zaria_gndvi_water.jpg; Manuscripts/figures/shots_jpg/zaria_rusle.jpg; Manuscripts/figures/shots_jpg/zaria_available_water.jpg
Figure 4	Screen capture of SWAFL Desktop 1.1.0 (not regenerated)	-	Manuscripts/figures/shots_jpg/zaria_vri_depth.jpg
Figure 5	Screen capture of SWAFL Desktop 1.1.0 (not regenerated)	-	Manuscripts/figures/p6/fig03_pheno_figures.jpg
Figure 6	Screen capture of SWAFL Desktop 1.1.0 (not regenerated)	-	Manuscripts/figures/shots_jpg/real_season_film.jpg
Figure 7	Figure drawn by the script from the input data	Manuscripts/reproducibility/p456/p456_figs_architecture.py	Data/daisy_ensemble.csv.gz (10 190 runs); Manuscripts/figures/p4/fig01_workbench.pdf
Figure 8	Screen capture of SWAFL Desktop 1.1.0 (not regenerated)	-	Manuscripts/figures/p4/fig02a_process_map.png
Figure 9	Screen capture of SWAFL Desktop 1.1.0 (not regenerated)	-	Manuscripts/figures/p4/fig02b_soil_column.png
Figure 10	Screen capture of SWAFL Desktop 1.1.0 (not regenerated)	-	Manuscripts/figures/shots_jpg/wb_step1_input.jpg; Manuscripts/figures/shots_jpg/wb_step2_library.jpg; Manuscripts/figures/shots_jpg/wb_step4_scenarios.jpg; Manuscripts/figures/shots_jpg/wb_step6_logplots.jpg; Manuscripts/figures/shots_jpg/wb_step9_layers.jpg; Manuscripts/figures/shots_jpg/wb_tool_setup_editor.jpg; Manuscripts/figures/shots_jpg/wb_tool_calibrate.jpg; Manuscripts/figures/shots_jpg/wb_tool_learn_files.jpg

Item	How it is made	Script(s)	Input and output
Figure 11	Screen capture of SWAFL Desktop 1.1.0 (not regenerated)	-	Manuscripts/figures/shots_jpg/zaria_main_feedback.jpg
Figure 12	Figure drawn by the script from the input data	Manuscripts/reproducibility/p456/p456_figs_schematics.py	Data/daisy_ensemble.csv.gz (10 190 runs); Manuscripts/figures/p5/fig01_bridge.pdf
Table 2	Descriptive table (definitions, design, parameters or tests); no computation	-	
Table 3	Table typed from the result file (13 of its numbers found there)	Manuscripts/reproducibility/p3_js/run_drip.js; Manuscripts/reproducibility/analysis/p3_lit_figs.py; Manuscripts/reproducibility/analysis/make_supplement_data.py	Data/daisy_ensemble.csv.gz (10 190 runs); Manuscripts/results/drip_out.json
Table 4	Table typed in the manuscript from the result files of the script	p4_design_sizes.js; out_p4_design.js	Data/daisy_ensemble.csv.gz (10 190 runs)
Figure 13	Figure drawn by the script from the input data	Manuscripts/reproducibility/p456/p456_figs_data.py; p4_design_fidelity.py	Data/daisy_ensemble.csv.gz (10 190 runs); Manuscripts/figures/p4/fig03_fidelity.pdf
Table 5	Table typed from the result file (5 of its numbers found there)	Manuscripts/reproducibility/run_all.sh; Manuscripts/reproducibility/make_e12.py; Manuscripts/reproducibility/audit_numbers.py; p4_richards.js; p4_richards_ref.py	Data/daisy_ensemble.csv.gz (10 190 runs); Manuscripts/results/e12.json
Figure 14	Figure drawn by the script from the input data	Manuscripts/reproducibility/p456/p456_figs_data.py; p4_richards.js; p4_richards_ref.py	Data/daisy_ensemble.csv.gz (10 190 runs); Manuscripts/figures/p4/fig04_richards.pdf
Figure 15	Figure drawn by the script from the input data	Manuscripts/reproducibility/p456/p456_figs_data.py; p5_bridge_checks.js	Data/daisy_ensemble.csv.gz (10 190 runs); Manuscripts/figures/p4/fig05_denitrification.pdf
Table 6	Descriptive table (definitions, design, parameters or tests); no computation	-	
Table 7	Table typed in the manuscript from the result files of the script	(see the reproducibility appendix of the paper)	Data/daisy_ensemble.csv.gz (10 190 runs)
Table 8	Descriptive table (definitions, design, parameters or tests); no computation	-	
Figure 16	Figure drawn by the script from the input data	Manuscripts/reproducibility/p456/p456_figs_data.py	Data/daisy_ensemble.csv.gz (10 190 runs); Manuscripts/figures/p5/fig04_economics.pdf
Figure 17	Screen capture of SWAFL Desktop 1.1.0 (not regenerated)	-	Manuscripts/figures/p5/fig02_flowfigures.jpg
Figure 18	Screen capture of SWAFL Desktop 1.1.0 (not regenerated)	-	Manuscripts/figures/shots_jpg/ui_layers_tree.jpg; Manuscripts/figures/shots_jpg/ui_field_panel1.jpg; Manuscripts/figures/shots_jpg/ui_field_panel2.jpg

Item	How it is made	Script(s)	Input and output
Figure 19	Screen capture of SWAFL Desktop 1.1.0 (not regenerated)	-	Manuscripts/figures/shots_jpg/wb_summary_effects.jpg; Manuscripts/figures/shots_jpg/wb_every_log_lai.jpg
Figure 20	Screen capture of SWAFL Desktop 1.1.0 (not regenerated)	-	Manuscripts/figures/shots_jpg/wb_why_failed.jpg; Manuscripts/figures/shots_jpg/wb_all_outputs.jpg
Table 9	Table typed in the manuscript from the result files of the script	(see the reproducibility appendix of the paper)	Data/daisy_ensemble.csv.gz (10 190 runs)
Figure 21	Figure drawn by the script from the input data	Manuscripts/reproducibility/p456/p456_figs_data.py	Data/daisy_ensemble.csv.gz (10 190 runs); Manuscripts/figures/p6/fig04_pheno.pdf
Figure 22	Figure drawn by the script from the input data	Manuscripts/reproducibility/p456/p456_figs_data.py	Data/daisy_ensemble.csv.gz (10 190 runs); Manuscripts/figures/p6/fig05_crops.pdf
Table 10	Descriptive table (definitions, design, parameters or tests); no computation	-	

References

Richard G. Allen, Luis S. Pereira, Dirk Raes, and Martin Smith. Crop evapotranspiration: Guidelines for computing crop water requirements. Technical Report FAO Irrigation and Drainage Paper 56, FAO, Rome, 1998. URL https://www.fao.org/4/x0490e/x0490e00.htm.

Matthias Drusch, Umberto Del Bello, Sebastien Carlier, et al. Sentinel-2: ESA's optical high-resolution mission for GMES operational services. *Remote Sensing of Environment*, 120:25–36, 2012.

Naziru Halilu. *SWAFL Desktop 1.1.0: Technical Manual (1155 pages)*, 2026. Distributed with the software and at https://github.com/halilunaziru73-creator/Coupled-Water-Fluxes-Pore-Scale-Dynamics-Precision-Irrigation-Mulching-Feedback-Loops/tree/main/Software/v1.1.0/Manual.

Søren Hansen, Per Abrahamsen, Christian T. Petersen, and Merete Styczen. Daisy: model use, calibration, and validation. *Transactions of the ASABE*, 55:1317–1335, 2012. doi: 10.13031/2013.42244.

Alfredo Huete, Kamel Didan, Tomoaki Miura, E. Patricia Rodriguez, Xiang Gao, and Laerte G. Ferreira. Overview of the radiometric and biophysical performance of the MODIS vegetation indices. *Remote Sensing of Environment*, 83:195–213, 2002.

Alfredo R. Huete. A soil-adjusted vegetation index (SAVI). *Remote Sensing of Environment*, 25:295–309, 1988.

Laura Poggio, Luis M. de Sousa, Niels H. Batjes, Gerard B. M. Heuvelink, Bas Kempen, Eloi Ribeiro, and David Rossiter. SoilGrids 2.0: producing soil information for the globe with quantified spatial uncertainty. *SOIL*, 7:217–240, 2021.

K. G. Renard and J. R. Freimund. Using monthly precipitation data to estimate the R-factor in the revised USLE. *Journal of Hydrology*, 157:287–306, 1994.

Jiří Šimůnek, Martinus Th. van Genuchten, and Miroslav Šejna. Recent developments and applications of the HYDRUS computer software packages. *Vadose Zone Journal*, 15(7), 2016. doi: 10.2136/vzj2016.04.0033.